Introduction to

Linear Algebra



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Chapter 1

Systems of linear equations

1.1 A motivating problem: linear economic models

The economy of a country may for simplicity be subdivided into five sectors: agriculture, construction, energy, manufacturing, and services. In order to function, each industry needs to use some of the commodities produced by other industries: for instance the services industry certainly needs electricity produced by the energy sector to function normally. We encode this interdependence in the following table, where each column represents the percentage of the production of a given industry that is consumed by each industry:

	agriculture	construction	energy	manufacturing	services
agriculture	10%	20%	20%	10%	10%
construction	20%	20%	30%	20%	20%
energy	20%	20%	10%	10%	10%
manufacturing	30%	20%	20%	20%	30%
services	20%	20%	20%	40%	30%

For instance, we read from this table 20% of the production of the energy industry is consumed by the services industry, or that 10% of the production of the manufacturing industry is consumed by the agriculture industry.
For simplicity, this example does not take external demand into account, and we will assume that the economy is in equilibrium, that is, the total output (production) equates the total input (internal demand). To determine suitable prices $p_1, \ldots, p_5$ for the output of each industry, we thus need to solve the following system of equations:

$$\begin{cases} p_1 &= 0.1\ p_1 + 0.2\ p_2 + 0.2\ p_3 + 0.1\ p_4 + 0.1\ p_5 \\ p_2 &= 0.2\ p_1 + 0.2\ p_2 + 0.3\ p_3 + 0.2\ p_4 + 0.2\ p_5 \\ p_3 &= 0.2\ p_1 + 0.2\ p_2 + 0.1\ p_3 + 0.1\ p_4 + 0.1\ p_5 \\ p_4 &= 0.3\ p_1 + 0.2\ p_2 + 0.2\ p_3 + 0.2\ p_4 + 0.3\ p_5 \\ p_5 &= 0.2\ p_1 + 0.2\ p_2 + 0.2\ p_3 + 0.4\ p_4 + 0.3\ p_5, \end{cases}$$

Such a model for the economy of a country is called the *Leontief Input-Output (closed) model.*

The goal of this first chapter is to study an algorithm (*Gaussian elimination*) to solve such systems of equations in full generality, and develop a conceptual understanding allowing us to discuss the existence or the number of solutions of such systems.

1.2 Systems of linear equations

Definition 1.1. A *linear equation* in n variables $x_1, \ldots, x_n$ is an equation which can be expressed in the form

$$a_1x_1 + a_2x_2 + \ldots + a_nx_n = b$$

with $a_1, \ldots, a_n, b \in \mathbb{R}$. A *system of linear equations* consists of a a finite number of linear equations:

$$\begin{cases} a_{11}x_1 + a_{12}x_2 + \ldots + a_{1n}x_n &= b_1 \\ a_{21}x_1 + a_{22}x_2 + \ldots + a_{2n}x_n &= b_2 \\ \vdots & \vdots \\ a_{m1}x_1 + a_{m2}x_2 + \ldots + a_{mn}x_n &= b_m \end{cases}$$

A *solution* of a system of linear equations is a tuple $(x_1, \ldots, x_n)$ that satisfy each equation of the system. We say that this system of linear equations is

(i) *consistent*, if it has at least one solution, and

(ii) *inconsistent*, if it has no solution.

Example 1.2. (i) The system:

$$\begin{cases} x_1 + x_2 &= 2 \\ x_1 - x_2 &= 0 \end{cases}$$

is consistent and has exactly one solution: the second equation yields $x_1 = x_2$ and the first equation yields $x_1 = x_2 = 1$.

(ii) The system

$$\begin{cases} x_1 + x_2 &= 2 \\ 2x_1 + 2x_2 &= 4 \end{cases}$$

is consistent and has infinitely many solutions: the second equation is a multiple of the first one, so all couples of the form $(x, 2-x)$, for $x \in \mathbb{R}$ are solutions of the system.

(iii) The system

$$\begin{cases} x_1 + x_2 &= 2 \\ 3x_1 + 3x_2 &= 7 \end{cases}$$

is inconsistent: the first equation yields $3x_1 + 3x_2 = 6$, which is incompatible with the second equation.

(iv) The system

$$\begin{cases} x_1 + x_2 &= 2 \\ x_1 - x_2 &= 0 \\ 3x_1 + 2x_2 &= 4 \end{cases}$$

is inconsistent: the first two equations yield $x_1 = x_2 = 1$, which does not solve the third equation.

Remark: systems of linear equations with complex coefficients. In this course, we will focus mostly on systems of equations with coefficients in $\mathbb{R}$. However, everything explained in this chapter generalises without any difficulty to systems of equations over $\mathbb{C}$.

1.3 Gaussian elimination

In this section, we describe an algorithm to efficiently solve any system of linear equations. We start by describing systems of a particularly simple nature.

Definition 1.3. We say that a system of m linear equations is in *(upper) triangular form* if, for every $2 \leq i \leq m$, the i-th equation only does not contain the unknowns $x_1, \ldots, x_{i-1}$.

A system in triangular form is much simpler to solve, by using substitution, starting from the last equation and going up.

Example 1.4. Consider the following system in triangular form

$$\begin{cases} x_1 & & +x_3 & = 5 \\ & x_2 & +2x_3 & = 4 \\ & & 3x_3 & = 3 \end{cases}$$

Starting from the bottom, we get $x_3 = 1$. The second equation then gives $x_2 = 2$, and finally the first equation gives $x_1 = 4$. Thus, the unique solution of the system is $(4, 2, 1)$.

Matrices associated to systems of linear equations. We will explain how to reduce a general system of linear equations to a triangular one. We first introduce a short way to encode a system of linear equations.

Definition 1.5. An $m \times n$ *matrix (with real coefficients)* is a rectangular table of real numbers containing m rows and n columns. The (i, j)*-entry* of the matrix A is the entry in row i and column j, and is denoted a_{ij} (or $(A)_{ij}$ if there is a possible confusion). The set of all $m \times n$ matrices with real coefficients is denoted $M_{m,n}(\mathbb{R})$ (or simply $M_n(\mathbb{R})$ if $m = n$).
To a system of m linear equations in n variables, we associate a matrix as follows:

$$\begin{cases} a_{11}x_1 + a_{12}x_2 + \ldots + a_{1n}x_n &= b_1 \\ a_{21}x_1 + a_{22}x_2 + \ldots + a_{2n}x_n &= b_2 \\ \vdots & \vdots \\ a_{m1}x_1 + a_{m2}x_2 + \ldots + a_{mn}x_n &= b_m. \end{cases} \rightsquigarrow \begin{pmatrix} a_{11} & a_{12} & \ldots & a_{1n} \\ a_{21} & a_{22} & \ldots & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{m1} & a_{m2} & \ldots & a_{mn} \end{pmatrix}.$$

The *augmented matrix* associated to this system of linear equations is the following $m \times (n+1)$ matrix (the vertical line is only here for readability):

$$\left(\begin{array}{cccc|c} a_{11} & a_{12} & \ldots & a_{1n} & b_1 \\ a_{21} & a_{22} & \ldots & a_{2n} & b_2 \\ \vdots & \vdots & & \vdots & \\ a_{m1} & a_{m2} & \ldots & a_{mn} & b_m \end{array}\right).$$

Elementary row operations. Consider the system of linear equations

$$\begin{cases} a_{11}x_1 + a_{12}x_2 + \ldots + a_{1n}x_n &= b_1 \\ a_{21}x_1 + a_{22}x_2 + \ldots + a_{2n}x_n &= b_2 \\ \vdots & \vdots \\ a_{m1}x_1 + a_{m2}x_2 + \ldots + a_{mn}x_n &= b_m. \end{cases}$$

We do not change the set of solutions of the system if we perform one of the following operations on the equations:

(a) row-switching transformation: $R_i \leftrightarrow R_j$.

(b) row-multiplying transformation: $R_i \to aR_i$ for $a \neq 0$.

(c) row-addition transformation: $R_i \to R_i + aR_j$ for another $j \neq i$ and $a \in \mathbb{R}$.

Equivalently, these operations can be performed on the augmented matrix associated to the system. We will now explain how to use the elementary operations to reduce a system of equations to a system in triangular form.

Method: Gaussian elimination.

The following algorithm allows you to solve a system of linear equations by reducing it to an upper triangular system:

(1) *Placing the pivot.* Consider the associated augmented matrix. Go to the first column of the matrix, say j, which is not entirely zero. Choose a row which has a non-zero entry in this column and move it to the top. This non-zero term will be called a *pivot*. The row containing this pivot will be refered to as R_p for the next step.

(2) *Killing off the coefficients below the pivot.* For each row R_i below the first one, replace R_i by a combination $aR_i + bR_p$, with the constants $a \neq 0$ et b such that the entry in the column j vanishes.

(3) *Induction step.* Consider the augmented matrix obtained by forgetting the first row. If this augmented matrix has only one line, or if all the columns of the associated matrix are zero, the algorithm stops. Otherwise, apply the previous algorithm, starting from (1), to this new augmented matrix. (When applying this algorithm, we will still write the rows that are being forgotten, but we will consider them as somehow frozen)

Example 1.6. Here is a simple example

$$\begin{cases} x_1 + 2x_2 + x_3 &= 1 \\ x_1 + 3x_2 + 4x_3 &= 3 \\ 2x_1 + 5x_2 + 6x_3 &= 5 \end{cases} \quad \text{We first rewrite it as} \quad \left(\begin{array}{ccc|c} 1 & 2 & 1 & 1 \\ 1 & 3 & 4 & 3 \\ 2 & 5 & 6 & 5 \end{array}\right).$$

Now, apply elementary row operations:

$$\overset{\substack{R_2 \to R_2 - R_1 \\ R_3 \to R_3 - 2R_1}}{\rightsquigarrow} \left(\begin{array}{ccc|c} 1 & 2 & 1 & 1 \\ 0 & 1 & 3 & 2 \\ 0 & 1 & 4 & 3 \end{array}\right) \overset{R_3 \to R_3 - R_2}{\rightsquigarrow} \left(\begin{array}{ccc|c} 1 & 2 & 1 & 1 \\ 0 & 1 & 3 & 2 \\ 0 & 0 & 1 & 1 \end{array}\right)$$

We are thus back to a triangular system:

$$\begin{cases} x_1 + 2x_2 + x_3 &= 1 \\ x_2 + 3x_3 &= 2 \\ x_3 &= 1 \end{cases}$$

Here, we can apply the standard substitution technique, starting from the bottom row, to com-

pletely solve the system:

$$\begin{aligned} x_1 &= 1 - 2x_2 - x_3 = 2 \\ x_2 &= 2 - 3x_3 = -1 \\ x_3 &= 1. \end{aligned}$$

Echelon form. After applying the Gaussian elimination, the augmented matrix we end up with is said to be in (row) *echelon form*, as defined below:

Definition 1.7. An augmented matrix is in *(row) echelon form* if the first non-zero entry in a row (the *pivot* of that row) appears strictly to the right of the pivots of the previous rows.

Example 1.8. The following augmented matrices are in echelon form:

$$\left(\begin{array}{cccc|c} 1 & 5 & 3 & -1 & 2 \\ 0 & 0 & 2 & 3 & 0 \\ 0 & 0 & 0 & -1 & 1 \end{array}\right), \quad \left(\begin{array}{ccc|c} 3 & 1 & 4 & 2 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right), \quad \left(\begin{array}{ccc|c} 1 & 0 & 3 & 1 \\ 0 & 4 & 0 & 2 \\ 0 & 0 & 3 & 9 \end{array}\right).$$

The following augmented matrices are not in echelon form:

$$\left(\begin{array}{ccc|c} 1 & 5 & 3 & -1 \\ 1 & 0 & 1 & 3 \\ 0 & 0 & 0 & 1 \end{array}\right), \quad \left(\begin{array}{ccc|c} 1 & 8 & 4 & 2 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{array}\right).$$

Consistency. From the row echelon form, it is straightforward to see whether a system of linear equations is consistent.

Theorem 1.9. A system of linear equations has either 0, 1, or infinitely many solutions.
It is consistent if and only if the echelon form of the associated augmented matrix does not contain a row of the form

$$(\,0 \;\ldots\; 0 \mid \alpha\,), \quad \alpha \neq 0.$$

A consistent system of linear equation has exactly one solution if every column of the associated matrix contains a pivot, and infinitely many solutions otherwise.

Example 1.10.

unique solution:

$$\left(\begin{array}{ccc|c} 1 & \times & \times & \times \\ 0 & 1 & \times & \times \\ 0 & 0 & 1 & \times \end{array}\right)$$

(Substitution yields the unique solution.)

infinitely many solutions:

$$\left(\begin{array}{ccc|c} 1 & \times & \times & \times \\ 0 & 1 & \times & \times \\ 0 & 0 & 0 & 0 \end{array}\right)$$

(x_3 can be chosen arbitrarily.)

no solution:

$$\left(\begin{array}{ccc|c} 1 & \times & \times & \times \\ 0 & 1 & \times & \times \\ 0 & 0 & 0 & 1 \end{array}\right)$$

(Last row: $0x_1 + 0x_2 + 0x_3 = 1$ yields a contradiction.)

General solution. Once a system of linear equations is put in echelon form, it is possible to find all the solutions to that system.

Definition 1.11. In an augmented matrix in row echelon form, the variables corresponding to columns containing a pivot are called *pivot variables*. The other variables are called *free variables*.

Method: Finding the general solution to a system of linear equations.

The following algorithm allows you to find the general solution to a system of linear equations:

(1) Bring the augmented matrix to row echelon form via Gaussian elimination.

(2) Free variables are set as arbitrary constants $\alpha, \beta, \gamma, \ldots \in \mathbb{R}$.

(3) The values of the pivot variables are then determined from the free variables by substitution, starting from the bottom row.

The result is called the *general solution* of the system.

Example 1.12. Here is an example of a system of linear equations whose augmented matrix has already been put in row echelon form:

$$\left(\begin{array}{cccc|c} 1 & 5 & 3 & 2 & 3 \\ 0 & 0 & 3 & 4 & 8 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right)$$

In this system, x_1 and x_3 are pivot variables, while x_2 and x_4 are free variables. We thus set $x_4 = \alpha, x_2 = \beta$, and we obtain by substitution, starting from the bottom,

$$3x_3 + 4x_4 = 8 \rightsquigarrow x_3 = \frac{1}{3}(8 - 4x_4) = \frac{1}{3}(8 - 4\alpha)$$

and

$$x_1 + 5x_2 + 3x_3 + 2x_4 = 3 \rightsquigarrow x_1 = 3 - 5x_2 - 3x_3 - 2x_4 = 3 - 5\beta - 3 \times \frac{1}{3}(8 - 4\alpha) - 2\alpha = -5 + 2\alpha - 5\beta.$$

The general solution is of the form $(-5+2\alpha-5\beta, \beta, \frac{1}{3}(8-4\alpha), \alpha)$ for $\alpha, \beta \in \mathbb{R}$. For instance, for $\alpha = -1$ and $\beta = 0$, we find the solution $(-7, 0, 4, -1)$. For $\alpha = 2, \beta = 1$, we find the solution $(-6, 1, 0, 2)$.

1.4 Homogeneous and inhomogeneous systems of linear equations

Homogeneous systems of linear equations.

Definition 1.13. A system of linear equations

$$(E) \begin{cases} a_{11}x_1 + a_{12}x_2 + \ldots + a_{1n}x_n &= b_1 \\ a_{21}x_1 + a_{22}x_2 + \ldots + a_{2n}x_n &= b_2 \\ \vdots & \vdots \\ a_{m1}x_1 + a_{m2}x_2 + \ldots + a_{mn}x_n &= b_m \end{cases}$$

is said to be *homogeneous* if and only if $b_1 = \ldots = b_m = 0$. Otherwise, it is said to be *inhomoge-*

neous. The homogeneous system of linear equations *associated to* (E) is the following system (E_0):

$$(E_0)\begin{cases} a_{11}x_1 + a_{12}x_2 + \ldots + a_{1n}x_n &= 0 \\ a_{21}x_1 + a_{22}x_2 + \ldots + a_{2n}x_n &= 0 \\ \vdots & \vdots \\ a_{m1}x_1 + a_{m2}x_2 + \ldots + a_{mn}x_n &= 0. \end{cases}$$

Remark. While we have seen examples of systems of linear equations with no solution, a homogeneous system is always consistent because it has at least the solution $x_1 = \ldots = x_n = 0$.
Here is a simple case where a homogeneous system of linear equations has infinitely many solutions:

Theorem 1.14. A homogeneous system of linear equations with more unknowns than equations has infinitely many solutions.

Proof. Suppose a homogeneous system of linear equations with m equations in n unknowns, $m < n$. Using Gaussian elimination on the corresponding augmented matrix with m rows and $n+1$ columns, we obtain a matrix in row echelon form with $k \leq m < n$ rows with non-zero entries. Hence at least one of the variables is a free variable, which implies that there are infinitely many solutions. □

Example 1.15. The system of equations:

$$\begin{cases} x + y + z &= 0 \\ x + 2y + 3z &= 0. \end{cases}$$

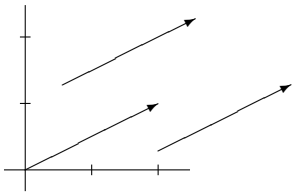
admits infinitely many solutions, as it is a system of two equations in three variables.

Chapter 2

Vectors

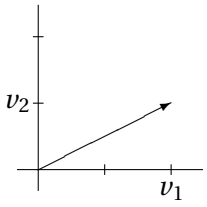
2.1 Vectors in $\mathbb{R}^2$ and $\mathbb{R}^3$.

Vectors in $\mathbb{R}^2$. We briefly recall some facts about vectors in $\mathbb{R}^2$. Intuitively, a *vector* of $\mathbb{R}^2$ can be thought as an arrow encoding a length and a direction. A vector however does not have an origin, so one may draw many arrows in the plane corresponding to the same vector:



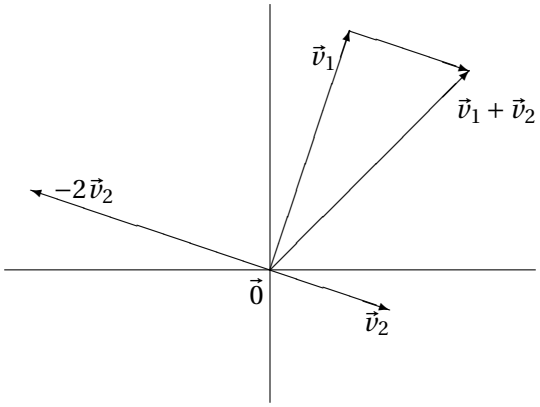
One may identify the set of all vectors in $\mathbb{R}^2$ with the set of points of the plane $\mathbb{R}^2$, by associating to each point of the plane the vector from the origin to that point. Under this identification, the origin corresponds to a vector called the *null vector* and denoted $\vec{0}$.
Each point and thus each vector $\vec{v} \in \mathbb{R}^2$ can be denoted by a pair of real numbers (thus the notation $\mathbb{R}^2$): the horizontal displacement and the vertical displacement. Such a vector is denoted by the 2×1 matrix

$$\vec{v} = \begin{pmatrix} v_1 \\ v_2 \end{pmatrix}.$$



It is a very useful convention to write vectors vertically, i.e. as 2×1 matrices. For typesetting reasons however, we often write that vector as $\vec{v} = (v_1, v_2)^T$, i.e. as a 1×2 matrix instead. The T, short for *transpose*, will be explained later in the course.
We have the following rules for adding two vectors $\vec{v} = (v_1, v_2)^T$ and $\vec{w} = (w_1, w_2)^T$ and for multiplying a vector by a number $\lambda \in \mathbb{R}$ (called a *scalar*):

$$\vec{v} + \vec{w} = \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} + \begin{pmatrix} w_1 \\ w_2 \end{pmatrix} := \begin{pmatrix} v_1 + w_1 \\ v_2 + w_2 \end{pmatrix} \quad \text{and} \quad \lambda \vec{v} := \begin{pmatrix} \lambda v_1 \\ \lambda v_2 \end{pmatrix}.$$



With our convention, the null vector is the vector with coordinates $(0,0)^T$. Note that if we multiply a vector by 0, we obtain the null vector: $0\vec{v} = \vec{0}$ for all $\vec{v} \in \mathbb{R}^2$.

Vectors in $\mathbb{R}^3$. Most of the notions introduced on $\mathbb{R}^2$ straightforwardly extend to $\mathbb{R}^3$. A point in the Euclidean dimensions encodes a vector $\vec{v} \in \mathbb{R}^3$, i.e. an arrow from the origin $\vec{0}$ to that point. We describe the vector $\vec{v}$ by a 3×1 matrix

$$\vec{v} = \begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix},$$

or sometimes as a 1×3 matrix $\vec{v} = (v_1, v_2, v_3)^T$ for typesetting reasons. We can add vectors and multiply them by scalars as before:

$$\vec{v} + \vec{w} = \begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix} + \begin{pmatrix} w_1 \\ w_2 \\ w_3 \end{pmatrix} = \begin{pmatrix} v_1 + w_1 \\ v_2 + w_2 \\ v_3 + w_3 \end{pmatrix} \quad \text{and} \quad \lambda\vec{v} = \begin{pmatrix} \lambda v_1 \\ \lambda v_2 \\ \lambda v_3 \end{pmatrix}.$$

2.2 Vectors in $\mathbb{R}^n$

The generalisation to $\mathbb{R}^n$ should now be clear.

Definition 2.1. We define:

$$\mathbb{R}^n = \left\{ \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} \;\middle|\; x_i \in \mathbb{R}\,, \quad i = 1, \ldots, n \right\}.$$

An element of $\mathbb{R}^n$ is called a *vector* (of $\mathbb{R}^n$), and is identified with an $n \times 1$ matrix. For vectors $\vec{x}, \vec{y} \in \mathbb{R}^n$ and a scalar $\lambda \in \mathbb{R}$, we define the following operations on vectors:

- addition: $\begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} + \begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix} := \begin{pmatrix} x_1 + y_1 \\ \vdots \\ x_n + y_n \end{pmatrix}$,
- multiplication by a scalar: $\lambda \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} := \begin{pmatrix} \lambda x_1 \\ \vdots \\ \lambda x_n \end{pmatrix}$.

From a given family of vectors, one can construct new vectors using additions and multiplication scalars.

Definition 2.2. Let $\vec{v}_1,\dots,\vec{v}_k \in \mathbb{R}^n$. A *linear combination* of $\vec{v}_1,\dots,\vec{v}_k$ is a vector of the form

$$\lambda_1\vec{v}_1+\dots+\lambda_k\vec{v}_k \qquad \text{for some } \lambda_1,\dots,\lambda_k\in\mathbb{R}.$$

Example 2.3. The vector $\begin{pmatrix}1\\3\end{pmatrix}\in\mathbb{R}^2$ is a linear combination of $\begin{pmatrix}1\\2\end{pmatrix}$ and $\begin{pmatrix}2\\3\end{pmatrix}$, since

$$\begin{pmatrix}1\\3\end{pmatrix}=3\begin{pmatrix}1\\2\end{pmatrix}-\begin{pmatrix}2\\3\end{pmatrix}.$$

Writing a vector as a linear combination of other vectors can be thought of as 'decomposing' that vector. Given a family of vectors $\vec{v}_1,\dots,\vec{v}_k$ of $\mathbb{R}^n$, we will study the following questions:

- Can every vector of $\mathbb{R}^n$ be written as a linear combination of $\vec{v}_1,\dots,\vec{v}_k$?
- If not, *which* vectors of $\mathbb{R}^n$ can be written as a linear combination of $\vec{v}_1,\dots,\vec{v}_k$?
- *In how many ways* can a vector of $\mathbb{R}^n$ be written as a linear combination of $\vec{v}_1,\dots,\vec{v}_k$?

The standard basis of $\mathbb{R}^n$. There is already a standard family of vectors for which these questions have a simple answer.

Definition 2.4. We introduce the following vectors of $\mathbb{R}^n$:

$$\vec{e}_1:=\begin{pmatrix}1\\0\\\vdots\\0\end{pmatrix},\vec{e}_2:=\begin{pmatrix}0\\1\\0\\\vdots\end{pmatrix},\dots,\vec{e}_n:=\begin{pmatrix}0\\\vdots\\0\\1\end{pmatrix}.$$

This family of vectors is generally called the *standard basis* of $\mathbb{R}^n$.

Theorem 2.5. Every vector $\vec{x}=(x_1,\dots,x_n)^T$ of $\mathbb{R}^n$ can be written in a unique way as a linear combination of $\vec{e}_1,\dots,\vec{e}_n$, namely:

$$\vec{x}=x_1\vec{e}_1+\dots+x_n\vec{e}_n.$$

Systems of linear equations in vector form.

Definition 2.6. Consider a system of linear equations whose associated matrix is an $m\times n$ matrix of the form:

$$A:=\begin{pmatrix}a_{11}&a_{12}&\dots&a_{1n}\\a_{21}&a_{22}&\dots&a_{2n}\\\vdots&\vdots&&\vdots\\a_{m1}&a_{m2}&\dots&a_{mn}\end{pmatrix}=\left(\begin{array}{c|c|c|c}\text{col}_1(A)&\cdots&\text{col}_n(A)\end{array}\right).$$

For $\vec{x} = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}$ a vector of $\mathbb{R}^n$, we define :

$$A\vec{x} := x_1 \text{col}_1(A) + \ldots + x_n \text{col}_n(A) \in \mathbb{R}^m.$$

In other words, for $1 \leq i \leq m$, the i-th component of $A\vec{x}$ is $(A\vec{x})_i = \sum_{1 \leq k \leq n} a_{ik} x_k$.

Proposition 2.7. Consider the following system of linear equations:

$$\begin{cases} a_{11}x_1 + a_{12}x_2 + \ldots + a_{1n}x_n &= b_1 \\ a_{21}x_1 + a_{22}x_2 + \ldots + a_{2n}x_n &= b_2 \\ \vdots & \vdots \\ a_{m1}x_1 + a_{m2}x_2 + \ldots + a_{mn}x_n &= b_m \end{cases}$$

Let A be the matrix associated to this system, and let $\vec{b} = \begin{pmatrix} b_1 \\ \vdots \\ b_m \end{pmatrix} \in \mathbb{R}^m$. A vector $\vec{x} = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}$ is solution of that system if and only we have:

$$A\vec{x} = \vec{b}.$$

Solving that system is thus equivalent to the following problem:

$$\textit{Is } \vec{b} \in \mathbb{R}^m \textit{ a linear combination of } \text{col}_1(A), \ldots, \text{col}_n(A)\textit{?}$$

In practice, this result means that any equation between vectors can be solved by reducing it to a system of linear equations. This approach will be used many times in this course. Thus, an $m \times n$ matrix is not 'just a bunch of numbers', but we can use it to associate to a vector $\vec{x} \in \mathbb{R}^n$ a new vector $A\vec{x} \in \mathbb{R}^m$. In other words, we can associate to any $m \times n$ matrix a map from $\mathbb{R}^n$ to $\mathbb{R}^m$.
Such maps have very interesting properties, as one immediately checks the following:

Proposition 2.8. Let A be an $m \times n$ matrix. Then we have for $\vec{x}_1, \vec{x}_2 \in \mathbb{R}^n$ and $\lambda \in \mathbb{R}$:

$$A(\vec{x}_1 + \vec{x}_2) = A\vec{x}_1 + A\vec{x}_2,$$
$$A(\lambda\vec{x}_1) = \lambda A\vec{x}_1.$$

A map $\mathbb{R}^n \to \mathbb{R}^m$ that satisfies similar properties is called a *linear* map. Their properties will be investigated in details in a few chapters.

2.3 Spans of vectors and spanning families in $\mathbb{R}^n$

Given a family $\vec{v}_1, \ldots, \vec{v}_k$ of vectors of $\mathbb{R}^n$, we start by considering the questions: What vectors of $\mathbb{R}^n$ can be obtained as a linear combination of $\vec{v}_1, \ldots, \vec{v}_k$? We first introduce some definition:

Definition 2.9. The *span* of a family $\vec{v}_1, \ldots, \vec{v}_k$ of vectors of $\mathbb{R}^n$ is the set $\operatorname{span}(\vec{v}_1, \ldots, \vec{v}_k)$ of vectors of $\mathbb{R}^n$ that can be written as a linear combination of $\vec{v}_1, \ldots, \vec{v}_k$. In other words,

$$\operatorname{span}(\vec{v}_1, \ldots, \vec{v}_k) = \left\{ \sum_{i=1}^{k} \lambda_i \vec{v}_i \;\middle|\; \lambda_i \in \mathbb{R} \text{ for every } i \right\}.$$

Checking whether a given vector is a linear combination of a family of vectors is checked by solving a system of linear equations. Here is an example:

Example 2.10. Let us determine whether the vector $(3,-4,2)^T$ is a linear combination of $(1,0,2)^T$ and $(1,1,3)^T$. We have to solve the equation

$$\lambda \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix} + \kappa \begin{pmatrix} 1 \\ 1 \\ 3 \end{pmatrix} = \begin{pmatrix} 3 \\ -4 \\ 2 \end{pmatrix}$$

with variables $\lambda, \kappa \in \mathbb{R}$. By Proposition 2.7, we express this as a system of linear equations, and perform Gaussian elimination:

$$\left(\begin{array}{cc|c} 1 & 1 & 3 \\ 0 & 1 & -4 \\ 2 & 3 & 2 \end{array}\right) \overset{R_2 \leftrightarrow R_2 - 2R_1}{\rightsquigarrow} \left(\begin{array}{cc|c} 1 & 1 & 3 \\ 0 & 1 & -4 \\ 0 & 1 & -4 \end{array}\right) \overset{R_3 \to R_3 - R_2}{\rightsquigarrow} \left(\begin{array}{cc|c} 1 & 1 & 3 \\ 0 & 1 & -4 \\ 0 & 0 & 0 \end{array}\right)$$

From this echelon form, we see that the system has exactly one solution (all variables are pivot variables and no inconsistent line). We solve the resulting system by substitution, which yields $\kappa = -4$ and $\lambda = 3 - \kappa = 7$. Thus,

$$7 \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix} - 4 \begin{pmatrix} 1 \\ 1 \\ 3 \end{pmatrix} = \begin{pmatrix} 3 \\ -4 \\ 2 \end{pmatrix}.$$

Remark 2.11. Since checking whether a given vector $\vec{v}$ is a linear combination of vectors $\vec{v}_1, \ldots, \vec{v}_k$ is checked by solving a system of linear equations, it follows from Theorem 1.9 that there is either no way to write $\vec{v}$ as a linear combination $\vec{v}_1, \ldots, \vec{v}_k$ (precisely when $\vec{v}$ is not in the span of $\vec{v}_1, \ldots, \vec{v}_k$), exactly one way, or infinitely many ways.

Proposition 2.12. Let $\vec{u}_1, \ldots, \vec{u}_k$ be a family of vectors of $\mathbb{R}^n$ and consider $V := \operatorname{span}(\vec{u}_1, \ldots, \vec{u}_k)$. We have the following:

- $\vec{0} \in V$,
- For every $\vec{x}, \vec{y} \in V$, we also have $\vec{x} + \vec{y} \in V$.
- For every $\vec{x} \in V$ and $\lambda \in \mathbb{R}$, we also have $\lambda \vec{x} \in V$.

More generally, any linear combination of vectors of V is again in V.

Definition 2.13. A subset of $\mathbb{R}^n$ containing the null vector and stable under linear combinations is called a *subspace* of $\mathbb{R}^n$.

Subspaces will be play an important role in this course and will be investigated in more detail in the next chapter.

Visualising spans of vectors. The algebraic properties of spans of vectors mentioned above have a geometric counterpart: Spans of vectors, seen as subsets of $\mathbb{R}^n$, have a very simple shape: line or plane through the origin in $\mathbb{R}^3$ for instance. More complicated shapes, such as spheres, hyperboloids, etc. can never be spans.
As an illustration, we now list the various possibilities for the span of two vectors of $\mathbb{R}^2$. In particular, we see that such spans are geometrically very simple. This phenomenon will be explained and investigated further in the next chapter.

Proposition 2.14. The span of two vectors of $\mathbb{R}^2$ is either:

(i) $\{\vec{0}\}$,

(ii) a straight line through the origin,

(iiii) all of $\mathbb{R}^2$.

Proof. Consider two arbitrary vectors $\vec{v} = (v_1, v_2)^T$, $\vec{w} = (w_1, w_2)^T$. If $\vec{v} = \vec{w} = \vec{0}$, the span of $\vec{v}$ and $\vec{w}$ is just $\{\vec{0}\}$. If they are collinear and for instance $\vec{v}$ is non-vanishing, then the span of $\vec{v}$ and $\vec{w}$ is the straight line parallel to $\vec{v}$ going through the origin.
If they are not collinear, then we now show that $\vec{v}$ and $\vec{w}$ span $\mathbb{R}^2$. Let $\vec{u} = (u_1, u_2)^T$ be an arbitrary vector of $\mathbb{R}^2$. We want to write $\vec{u}$ as a linear combination of $\vec{v}$ and $\vec{w}$. In other words, we want to find numbers $\lambda, \mu \in \mathbb{R}$ such that $\lambda\vec{v} + \mu\vec{w} = \vec{u}$. By taking coordinates, this yields the following equations:

$$\begin{cases} \lambda v_1 + \mu w_1 &= u_1 \\ \lambda v_2 + \mu w_2 &= u_2 \end{cases}$$

Here, the variables are λ and μ, and the coefficients $u_1, u_2, v_1, v_2, w_1, w_2$ are constants. We thus have a system of linear equations. We perform the following row operations:

$$\begin{matrix} R_1 \to w_2 R_1 - w_1 R_2 \\ R_2 \to v_1 R_2 - v_2 R_1 \end{matrix} \begin{cases} (v_1 w_2 - v_2 w_1)\lambda &= u_1 w_2 - u_2 w_1 \\ (v_2 w_1 - v_1 w_2)\mu &= u_2 v_1 - u_1 v_2 \end{cases}$$

Since $\vec{v}$ and $\vec{w}$ are not collinear, we have $v_1 w_2 - v_2 w_1 \neq 0$, so we can find unique solutions λ and μ in terms of the other constants. Thus, there is a unique solution to our system of equations, so $\vec{u}$ is a linear combination of $\vec{v}$ and $\vec{w}$. □

Spanning families.

Definition 2.15. We say that a family $\vec{v}_1, \ldots, \vec{v}_k$ of vectors of $\mathbb{R}^n$ *spans* $\mathbb{R}^n$ (or is a *spanning family* of $\mathbb{R}^n$, or that $\mathbb{R}^n$ is *spanned by* $\vec{v}_1, \ldots, \vec{v}_k$), if $\text{span}(\vec{v}_1, \ldots, \vec{v}_k) = \mathbb{R}^n$, that is, if every vector of $\mathbb{R}^n$ can be written as a linear combination of $\vec{v}_1, \ldots, \vec{v}_k$.

Proposition 2.16. Let $\vec{v}_1,\dots,\vec{v}_k$ be a spanning family of $\mathbb{R}^n$. Then $k \geq n$. In other words, a spanning family of $\mathbb{R}^n$ contains at least n vectors.

Proof. Given a vector $\vec{b} = (b_1,\dots,b_n)^T$ of $\mathbb{R}^n$, we have to consider the equation

$$\lambda_1\vec{v}_1 + \cdots + \lambda_k\vec{v}_k = \vec{b},$$

where $\lambda_1,\dots,\lambda_k$ are variables. Taking coordinates gives us a system of n equations (one for each coordinate of $\mathbb{R}^n$) with k variables. We can now perform Gaussian elimination to get a system in echelon form. We will now show that each row contains a pivot, which will be enough to conclude that $k \geq n$: Since each pivot must be strictly to the right of the pivots of the previous rows, this means that there must be at least as many columns as rows in the associated matrix, hence $k \geq n$.
If there was a row without a pivot, then the last row of the augmented matrix the echelon form would be of the form

$$\left(\begin{array}{ccc|c} 0 & \cdots & 0 & \beta \end{array}\right),$$

where β is a non-trivial linear combination of $b_1,\dots,b_n$. Note that if $\beta \neq 0$, then we have a forbidden row, and the system has no solution. We can now choose specific values of $b_1,\dots,b_n$ such that $\beta \neq 0$. This corresponds to an equation of the form

$$\lambda_1\vec{v}_1 + \cdots + \lambda_k\vec{v}_k = \begin{pmatrix} b_1 \\ \vdots \\ b_n \end{pmatrix}$$

which also has no solution. Thus, the vector $\vec{b} = (b_1,\dots,b_n)^T$ is not a linear combination of $\vec{v}_1,\dots,\vec{v}_n$, hence these vectors do not form a spanning family. □

2.4 Linearly independent families in $\mathbb{R}^n$

We now study family of vectors $\vec{v}_1,\dots,\vec{v}_k \in \mathbb{R}^n$ for which there is a unique way to write a given in the span as a linear combination of $\vec{v}_1,\dots,\vec{v}_k$.

Definition 2.17. A family of vectors of $\mathbb{R}^n$ is *linearly dependent* if one of the vectors is a linear combination of the other vectors. Equivalently, a family of vectors $\vec{v}_1,\dots,\vec{v}_k$ is linearly dependent if there exist scalars $\lambda_1,\dots,\lambda_k \in \mathbb{R}$ not all zero such that

$$\lambda_1\vec{v}_1 + \ldots + \lambda_k\vec{v}_k = \vec{0}.$$

A family of vectors $\vec{v}_1,\dots,\vec{v}_k$ is *linearly independent* if for every $\lambda_1,\dots,\lambda_k \in \mathbb{R}$, we have

$$\lambda_1\vec{v}_1 + \ldots + \lambda_k\vec{v}_k = \vec{0} \implies \lambda_1 = \cdots = \lambda_k = 0.$$

Proposition 2.18. A family of vectors $\vec{v}_1,\dots,\vec{v}_k$ is linearly independent if and only if there is a unique way to write any vector of the span as a linear combination of $\vec{v}_1,\dots,\vec{v}_k$.

Determining whether vectors are linearly independent amounts to solving a system of linear equations, which we do using Gaussian elimination.

Method: Determining whether vectors are linearly independent

To determine whether vectors $\vec{v}_1,\dots,\vec{v}_k$ are linearly independent, one uses Gaussian elimination to find all solutions to the system

$$c_1\vec{v}_1 + c_2\vec{v}_2 + \dots + c_k\vec{v}_k = \vec{0}\,.$$

If all variables $c_1,\dots,c_k$ are pivot variables, this homogeneous system of linear equations has only the trivial solution $c_1 = c_2 = \dots = c_k = 0$. In this case, $\vec{v}_1,\dots\vec{v}_k$ are linearly independent. Otherwise, nontrivial solutions exist and the vectors are linearly dependent.

Example 2.19. We want to determine whether the vectors $(2,1,1)^T$, $(1,2,1)^T$, and $(1,1,2)^T$ are linearly independent. We need to study the equation

$$\lambda\begin{pmatrix}2\\1\\1\end{pmatrix}+\kappa\begin{pmatrix}1\\2\\1\end{pmatrix}+\mu\begin{pmatrix}1\\1\\2\end{pmatrix}=\begin{pmatrix}0\\0\\0\end{pmatrix}$$

with variables $\lambda,\kappa,\mu\in\mathbb{R}$. Taking coordinates, this leads to the system of linear equations:

$$\begin{cases}2\lambda+\kappa+\mu=0\\ \lambda+2\kappa+\mu=0\\ \lambda+\kappa+2\mu=0\end{cases}$$

We use Gaussian elimination to solve this system. We get

$$\left(\begin{array}{ccc|c}2&1&1&0\\1&2&1&0\\1&1&2&0\end{array}\right)\overset{R_1\leftrightarrow R_2}{\rightsquigarrow}\left(\begin{array}{ccc|c}1&2&1&0\\2&1&1&0\\1&1&2&0\end{array}\right)\overset{\substack{R_2\to R_2-2R_1\\ R_3\to R_3-R_1}}{\rightsquigarrow}\left(\begin{array}{ccc|c}1&2&1&0\\0&-3&-1&0\\0&1&-1&0\end{array}\right)\overset{R_3\leftrightarrow 3R_3+R_1}{\rightsquigarrow}\left(\begin{array}{ccc|c}1&2&1&0\\0&-3&-1&0\\0&0&-4&0\end{array}\right)$$

Thus, all the variables are pivot variables and the system is consistent as it is homogeneous, so the system has exactly one solution. Since the system has at least one solution, namely $\lambda = \kappa = \mu = 0$, this must be the only solution. As a consequence, the vectors $(2,1,1)^T$, $(1,2,1)^T$, and $(1,1,2)^T$ are linearly independent.

Example 2.20. Let us determine whether the vectors $(1,2,3,-1)^T$, $(2,1,3,1)^T$ and $(4,5,9,-1)^T$ are linearly dependent. We must find all solutions (c_1,c_2,c_3) of the homogeneous system:

$$c_1\begin{pmatrix}1\\2\\3\\-1\end{pmatrix}+c_2\begin{pmatrix}2\\1\\3\\1\end{pmatrix}+c_3\begin{pmatrix}4\\5\\9\\-1\end{pmatrix}=\begin{pmatrix}0\\0\\0\\0\end{pmatrix}.$$

We reduce the associated system using Gaussian elimination:

$$\begin{cases}c_1+2c_2+4c_3=0\\ 2c_1+c_2+5c_3=0\\ 3c_1+3c_2+9c_3=0\\ -c_1+c_2-c_3=0\end{cases}\rightsquigarrow\left(\begin{array}{ccc|c}1&2&4&0\\2&1&5&0\\3&3&9&0\\-1&1&-1&0\end{array}\right)$$

$$\overset{\substack{R_2\to R_2-2R_1\\ R_3\to R_3-3R_1\\ R_4\to R_4+R_1}}{\rightsquigarrow}\left(\begin{array}{ccc|c}1&2&4&0\\0&-3&-3&0\\0&-3&-3&0\\0&3&3&0\end{array}\right)\overset{\substack{R_3\to R_3-R_2\\ R_4\to R_4+R_3\\ R_2\to -\frac{1}{3}R_2}}{\rightsquigarrow}\left(\begin{array}{ccc|c}1&2&4&0\\0&1&1&0\\0&0&0&0\\0&0&0&0\end{array}\right).$$

Not all the variables are pivot variables (c_3 is a free variable) and therefore the system has infinitely many solutions, which implies that the vectors are linearly dependent. More precisely, if we rewrite it as a system of linear equations, we get

$$\left\{\begin{array}{rcll}c_1+2c_2+4c_3&=&0&\to c_1=-2c_2-4c_3=-2\alpha\\ c_2+c_3&=&0&\to c_3=\alpha\Rightarrow c_2=-\alpha\end{array}\right.$$

For instance, by setting $\alpha=1$ we obtain

$$-2\begin{pmatrix}1\\2\\3\\-1\end{pmatrix}-\begin{pmatrix}2\\1\\3\\1\end{pmatrix}+\begin{pmatrix}4\\5\\9\\-1\end{pmatrix}=\begin{pmatrix}0\\0\\0\\0\end{pmatrix}.$$

Number of vectors in a linearly independent family.

Proposition 2.21. Let $\vec{v}_1,\dots,\vec{v}_k$ be a linearly independent family of $\mathbb{R}^n$. Then $k\leq n$. In other words, a linearly independent family of $\mathbb{R}^n$ contains at most n vectors.

Proof. We prove the contrapositive, namely that a family of k vectors of $\mathbb{R}^n$, with $k>n$, cannot be linearly independent. Let $\vec{v}_1,\dots,\vec{v}_k$ be such a family. We have to consider the equation

$$\lambda_1\vec{v}_1+\cdots+\lambda_k\vec{v}_k=\vec{0},$$

which is a homogeneous system of n equations (one for each coordinate of $\mathbb{R}^n$) with k variables $\lambda_1,\dots,\lambda_k$. Since by assumption there are more variables than equations, we know from Theorem 1.14 that such a system has infinitely many solutions. In particular, there exists a solution where $\lambda_1,\dots,\lambda_k$ are not all zero, which yields a non-trivial linear combination. □

2.5 Bases of $\mathbb{R}^n$.

We now study families of vectors such that *every* vector of $\mathbb{R}^n$ can be written *in a unique way* as a linear combination of $\vec{v}_1,\dots,\vec{v}_k$. We have already seen one example of such a family, namely the standard basis of $\mathbb{R}^n$:

$$\vec{e}_1=\begin{pmatrix}1\\0\\\vdots\\0\end{pmatrix},\vec{e}_2=\begin{pmatrix}0\\1\\0\\\vdots\end{pmatrix},\dots,\vec{e}_n=\begin{pmatrix}0\\\vdots\\0\\1\end{pmatrix}.$$

This family of vectors is generally referred to as the *standard basis* of $\mathbb{R}^n$. This notion can be generalised as follows:

Definition 2.22. A *basis* of $\mathbb{R}^n$ is a family of vectors $\vec{v}_1,\dots,\vec{v}_k$ such that every vector of $\mathbb{R}^n$ can be written in a unique way as a linear combination of $\vec{v}_1,\dots,\vec{v}_k$. In other words, a family of

vectors is a basis if and only if it is both spanning (existence of a linear combination) and linearly independent (uniqueness of a linear combination).

The following is a direct consequences of Proposition 2.21 and 2.16:

Theorem 2.23. A basis of $\mathbb{R}^n$ contains exactly n vectors.

Characterisation of bases of $\mathbb{R}^n$

Theorem 2.24. Let $\vec{u}_1, \ldots, \vec{u}_n$ be a family of **exactly** n vectors of $\mathbb{R}^n$. The following are equivalent:

(i) $\vec{u}_1, \ldots, \vec{u}_n$ is a basis of $\mathbb{R}^n$.

(ii) $\vec{u}_1, \ldots, \vec{u}_n$ spans $\mathbb{R}^n$.

(iii) $\vec{u}_1, \ldots, \vec{u}_n$ is linearly independent.

Proof. By definition of a basis and Theorem 2.23, we have $(i) \Rightarrow (ii), (iii)$. Let us show that $(ii) \Rightarrow (i)$. We want to show that $\vec{u}_1, \ldots, \vec{u}_n$ is linearly independent. To that end, we try to solve the homogeneous system of equations $\lambda_1 \vec{u}_1 + \ldots + \lambda_n \vec{u}_n = \vec{0}$ by Gaussian elimination. Since we have n equations with n variables, either every variable is a pivot variable, or there exists a free variable. In the latter case, we can mimic proof of Proposition 2.16 to find a vector $\vec{v}$ of $\mathbb{R}^n$ such that reducing the system of equations $\lambda_1 \vec{u}_1 + \ldots + \lambda_n \vec{u}_n = \vec{v}$ would give a bottom row of the form

$$\left(\begin{array}{ccc|c} 0 & \ldots & 0 & 1 \end{array} \right).$$

This would imply that the system associated to the equation $\lambda_1 \vec{u}_1 + \ldots + \lambda_n \vec{u}_n = \vec{v}$ is inconsistent, contradicting the fact that the family spans $\mathbb{R}^n$. Thus, every variable is a pivot variable, and we know from the first chapter that the system has exactly one solution: $(\lambda_1, \ldots, \lambda_n) = (0, \ldots, 0)$.
Let us now show that $(iii) \Rightarrow (i)$, using a similar strategy. We want to show that $\vec{u}_1, \ldots, \vec{u}_n$ spans $\mathbb{R}^n$. To that end, we try to solve the system of equations $\lambda_1 \vec{u}_1 + \ldots + \lambda_n \vec{u}_n = \vec{v}$, for some vector $\vec{v}$ of $\mathbb{R}^n$, by Gaussian elimination. Since we have n equations with n variables, either every variable is a pivot variable, or there exists a free variable. In the latter case, the system of equations $\lambda_1 \vec{u}_1 + \ldots + \lambda_n \vec{u}_n = \vec{0}$ would have infinitely many solutions, contradicting the fact that the family is linearly independent. Thus, every variable is a pivot variable, which implies that the system of equations $\lambda_1 \vec{u}_1 + \ldots + \lambda_n \vec{u}_n = \vec{v}$ admits at least one solution. □

The implication $(iii) \Rightarrow (i)$ gives us a possible method to show that a given set of vectors forms a basis of $\mathbb{R}^n$. However, we will see in the next section an often faster way to show that a given family of vectors forms a basis of $\mathbb{R}^n$.

2.6 Volume and determinant

We have seen in the previous section that a basis of $\mathbb{R}^n$ is a family of exactly n vectors $\vec{u}_1, \ldots, \vec{u}_n \in \mathbb{R}^n$ that spans $\mathbb{R}^n$. A geometric way to visualise this is to imagine the 'parallelepiped' generated by $\vec{u}_1, \ldots, \vec{u}_n$, that is, the set of vectors of the form

$$\lambda_1 \vec{u}_1 + \cdots + \lambda_n \vec{u}_n \text{ with } \lambda_1, \ldots, \lambda_n \in [0, 1].$$

Then $\vec{u}_1, \dots, \vec{u}_n$ should be a basis if and only if that parallelepiped has an 'n-dimensional volume'. But this is just a heuristic and is not rigorous at all, in particular since we do not know what we mean by 'n-dimensional volume' in $\mathbb{R}^n$. In the next paragraph, we will see how to rigorously define this idea of volume (and how it is related to the notion of determinant of matrices you have seen in previous courses), and how it gives a simple criterion to check that a set of n vectors forms a basis of $\mathbb{R}^n$.

Searching for a definition of volume in $\mathbb{R}^n$. Let $\vec{e}_1, \dots, \vec{e}_n$ be the standard basis of $\mathbb{R}^n$. While we do not know a priori how to define volumes in $\mathbb{R}^n$, let us think of the properties such a notion should have.
A good notion of volume in $\mathbb{R}^n$ should be a map

$$\mathrm{vol}_n : \underbrace{\mathbb{R}^n \times \ldots \times \mathbb{R}^n}_{n \text{ times}} \to \mathbb{R}$$

that sends a tuple of vectors $(\vec{x}_1, \dots, \vec{x}_n)$ to a real number $\mathrm{vol}_n(\vec{x}_1, \dots, \vec{x}_n)$, the *algebraic* volume of the parallelepiped generated by these vectors. Such a function should satisfy a certain number of properties that correspond to our intuitive idea of volume in dimension n:

(V1) vol_n is *n-linear*, that is, for every vectors $\vec{u}_1, \dots, \vec{u}_{k-1}, \vec{u}_{k+1}, \dots, \vec{u}_n$, the map

$$T : \mathbb{R}^n \to \mathbb{R}, \quad \vec{x} \mapsto \mathrm{vol}_n(\vec{u}_1, \dots, \vec{u}_{k-1}, \vec{x}, \vec{u}_{k+1}, \dots, \vec{u}_n)$$

is linear, that is, satisfies

1. $T(\vec{x} + \vec{y}) = T(\vec{x}) + T(\vec{y})$ for all $\vec{x}, \vec{y} \in \mathbb{R}^n$
2. $T(\lambda \vec{x}) = \lambda T(\vec{x})$ for all $\vec{x} \in \mathbb{R}^n$ and $\lambda \in \mathbb{R}$.

(V2) If $\vec{x}_1, \dots, \vec{x}_n$ is linearly dependent, then $\mathrm{vol}_n(\vec{x}_1, \dots, \vec{x}_n) = 0$.

(V3) $\mathrm{vol}_n(\vec{e}_1, \dots, \vec{e}_n) = 1$.

Before showing how these properties allow us to compute the volume in many cases, we need the following result:

Proposition 2.25. For a function that satisfies conditions $(V1)$, condition $(V2)$ is equivalent to being *alternate*, that is, exchanging two distinct vectors in $\vec{x}_1, \dots, \vec{x}_n$ changes the sign of the function:

$$\text{For } \vec{x}_1, \dots, \vec{x}_n \in \mathbb{R}^n \text{ and } i \neq j, \ \mathrm{vol}_n(\dots, \vec{x}_i, \dots, \vec{x}_j, \dots) = -\mathrm{vol}_n(\dots, \vec{x}_j, \dots, \vec{x}_i, \dots)$$

Proof. Using n-linearity, we get:

$$\begin{aligned}\mathrm{vol}_n(\dots, \vec{x}_i + \vec{x}_j, \dots, \vec{x}_i + \vec{x}_j, \dots) = \mathrm{vol}_n(\dots, \vec{x}_i, \dots, \vec{x}_i, \dots) + \mathrm{vol}_n(\dots, \vec{x}_j, \dots, \vec{x}_j, \dots) \\ + \mathrm{vol}_n(\dots, \vec{x}_i, \dots, \vec{x}_j, \dots) + \mathrm{vol}_n(\dots, \vec{x}_j, \dots, \vec{x}_i, \dots).\end{aligned}$$

Since vol_n satisfies $(V2)$ it follows that $\mathrm{vol}_n(\dots, \vec{x}_j, \dots, \vec{x}_j, \dots) + \mathrm{vol}_n(\dots, \vec{x}_i, \dots, \vec{x}_j, \dots) = 0$.
Reciprocally, if $\mathrm{vol}_n(\dots, \vec{x}_i, \dots, \vec{x}_j, \dots) = -\mathrm{vol}_n(\dots, \vec{x}_j, \dots, \vec{x}_i, \dots)$ for every $i \neq j$, it follows in particular that if $\vec{x}_i = \vec{x}_j =: \vec{x}$ for $i \neq j$, then $\mathrm{vol}_n(\dots, \vec{x}, \dots, \vec{x}, \dots) = -\mathrm{vol}_n(\dots, \vec{x}, \dots, \vec{x}, \dots)$, hence the function vanishes whenever two components are equal. Now, if $\vec{x}_1, \dots, \vec{x}_n$ is a family of linearly dependent vectors, with for instance $\vec{x}_1 = \sum_{2 \leq i \leq n} a_i \vec{x}_i$, then it follows from n-linearity that

$$\mathrm{vol}_n(\vec{x}_1, \dots, \vec{x}_n) = \mathrm{vol}_n\Big(\sum_{2 \leq i \leq n} a_i \vec{x}_i, \vec{x}_2, \dots, \vec{x}_n\Big) = \sum_{2 \leq i \leq n} a_i \mathrm{vol}_n(\vec{x}_i, \vec{x}_2, \dots, \vec{x}_n) = 0.$$

□

Dimension 2. Let us show how properties $(V1)$-$(V3)$ allow us to completely compute the volume function. Let $\vec{x},\vec{y}$ be two vectors of $\mathbb{R}^2$, which we write in the canonical basis $\vec{e}_1,\vec{e}_2$ of $\mathbb{R}^2$ as $\vec{x} = a\vec{e}_1 + b\vec{e}_2$ and $\vec{y} = c\vec{e}_1 + d\vec{e}_2$.

$$\begin{aligned}\mathrm{vol}_2(\vec{x},\vec{y}) &= \mathrm{vol}_2(a\vec{e}_1 + b\vec{e}_2, c\vec{e}_1 + d\vec{e}_2)\\ &= ac \times \underbrace{\mathrm{vol}_2(\vec{e}_1,\vec{e}_1)}_{=0} + ad \times \mathrm{vol}_2(\vec{e}_1,\vec{e}_2) + bc \times \underbrace{\mathrm{vol}_2(\vec{e}_2,\vec{e}_1)}_{=-\mathrm{vol}_2(\vec{e}_1,\vec{e}_2)} + bd \times \underbrace{\mathrm{vol}_2(\vec{e}_2,\vec{e}_2)}_{=0}\\ &= ad \times \mathrm{vol}_2(\vec{e}_1,\vec{e}_2) - bc \times \mathrm{vol}_2(\vec{e}_1,\vec{e}_2)\\ &= ad - bc.\end{aligned}$$

Dimension 3. Let $\vec{x},\vec{y},\vec{z}$ be three vectors of $\mathbb{R}^3$, which we write in the canonical basis $\vec{e}_1,\vec{e}_2,\vec{e}_3$ of $\mathbb{R}^3$ as $\vec{x} = a\vec{e}_1 + b\vec{e}_2 + c\vec{e}_3$, $\vec{y} = d\vec{e}_1 + e\vec{e}_2 + f\vec{e}_3$, and $\vec{z} = g\vec{e}_1 + h\vec{e}_2 + i\vec{e}_3$
By 3-linearity, we get:

$$\begin{aligned}\mathrm{vol}_3(\vec{x},\vec{y},\vec{z}) &= \mathrm{vol}_3(a\vec{e}_1 + b\vec{e}_2 + c\vec{e}_3, d\vec{e}_1 + e\vec{e}_2 + f\vec{e}_3, g\vec{e}_1 + h\vec{e}_2 + i\vec{e}_3)\\ &= aei \times \mathrm{vol}_3(\vec{e}_1,\vec{e}_2,\vec{e}_3) + afh \times \mathrm{vol}_3(\vec{e}_1,\vec{e}_3,\vec{e}_2) + bdi \times \mathrm{vol}_3(\vec{e}_2,\vec{e}_1,\vec{e}_3)\\ &\quad + bfg \times \mathrm{vol}_3(\vec{e}_1,\vec{e}_2,\vec{e}_3) + cdh \times \mathrm{vol}_3(\vec{e}_3,\vec{e}_1,\vec{e}_2) + ceg \times \mathrm{vol}_3(\vec{e}_3,\vec{e}_2,\vec{e}_1).\end{aligned}$$

Since vol_3 is alternate, we have:

$$\begin{aligned}\mathrm{vol}_3(\vec{e}_1,\vec{e}_3,\vec{e}_2) &= -\mathrm{vol}_3(\vec{e}_1,\vec{e}_2,\vec{e}_3) = -1\\ \mathrm{vol}_3(\vec{e}_2,\vec{e}_1,\vec{e}_3) &= -\mathrm{vol}_3(\vec{e}_1,\vec{e}_2,\vec{e}_3) = -1\\ \mathrm{vol}_3(\vec{e}_2,\vec{e}_3,\vec{e}_1) &= -\mathrm{vol}_3(\vec{e}_1,\vec{e}_3,\vec{e}_2) = \mathrm{vol}_3(\vec{e}_1,\vec{e}_2,\vec{e}_3) = 1\\ \mathrm{vol}_3(\vec{e}_3,\vec{e}_1,\vec{e}_2) &= -\mathrm{vol}_3(\vec{e}_1,\vec{e}_3,\vec{e}_2) = \mathrm{vol}_3(\vec{e}_1,\vec{e}_3,\vec{e}_2) = 1\\ \mathrm{vol}_3(\vec{e}_3,\vec{e}_2,\vec{e}_1) &= -\mathrm{vol}_3(\vec{e}_1,\vec{e}_2,\vec{e}_3) = -1\end{aligned}$$

Putting all this together, we get

$$\mathrm{vol}_3(\vec{x},\vec{y},\vec{z}) = aei + bfg + cdh - ceg - afh - bdi.$$

Dimension n. We could keep computing in the same way in dimension n for vectors

$$\vec{x}_1 = \sum_{1\le i_1\le n} a_{i1}\vec{e}_i,\ \ldots,\ \vec{x}_n = \sum_{1\le i\le n} a_{in}\vec{e}_i.$$

Here again the volume would simplify as

$$\begin{aligned}\mathrm{vol}_n(\vec{x}_1,\ldots,\vec{x}_n) &= \mathrm{vol}_n\Big(\sum_{1\le i_1\le n} a_{i_1 1}\vec{e}_{i_1},\ldots,\sum_{1\le i_n\le n} a_{i_n n}\vec{e}_{i_n}\Big)\\ &= \sum_{1\le i_1\le n}\cdots\sum_{1\le i_n\le n} a_{i_1 1}\cdots a_{i_n n}\mathrm{vol}_n(\vec{e}_{i_1},\ldots,\vec{e}_{i_n})\\ &\quad \sum_{\sigma \text{ bijection of } \{1,\ldots,n\}} a_{\sigma(1)1}\cdots a_{\sigma(n)n}\mathrm{vol}_n(\vec{e}_{\sigma(1)},\ldots,\vec{e}_{\sigma(n)})\\ &= \sum_{\sigma \text{ bijection of } \{1,\ldots,n\}} \varepsilon(\sigma)a_{\sigma(1)1}\cdots a_{\sigma(n)n}\mathrm{vol}_n(\vec{e}_1,\ldots,\vec{e}_n),\\ &= \sum_{\sigma \text{ bijection of } \{1,\ldots,n\}} \varepsilon(\sigma)a_{\sigma(1)1}\cdots a_{\sigma(n)n},\end{aligned}$$

where $\varepsilon(\sigma)\in\{-1,1\}$ is called the *signature* of the bijection σ, and is determined by the number of swaps necessary to go from $\mathrm{vol}_n(\vec{e}_{\sigma(1)},\ldots,\vec{e}_{\sigma(n)})$ to $\mathrm{vol}_n(\vec{e}_1,\ldots,\vec{e}_n)$. This notion will be shown to be well-defined and investigated further in **Abstract Algebra (F19PL1)**. However, this formula is too complicated to be useful for large values of n, as it is a sum of $n!$ terms. We will see in the next paragraphs faster ways to compute determinants.

Theorem 2.26. There exists a unique function $\mathrm{vol}_n : \underbrace{\mathbb{R}^n \times \ldots \times \mathbb{R}^n}_{n \text{ times}} \to \mathbb{R}$, called the ($n$-dimensional) *volume*, satisfying property (V1)-(V3).
Moreover, we have for every family $\vec{v}_1, \ldots, \vec{v}_n$ of n vectors of $\mathbb{R}^n$:

$$\vec{v}_1, \ldots, \vec{v}_n \text{ is a basis of } \mathbb{R}^n \Leftrightarrow \mathrm{vol}_n(\vec{v}_1, \ldots, \vec{v}_n) \neq 0.$$

Volume and determinant of a matrix.

Definition 2.27. Let

$$A = (\vec{c}_1, \ldots, \vec{c}_n)$$

be an $n \times n$ matrix with columns denoted $\vec{c}_1, \ldots, \vec{c}_n$. We define the *determinant* of A, denoted $\det(A)$, as follows:

$$\det(A) := \mathrm{vol}_n(\, \vec{c}_1, \ldots, \vec{c}_n \,).$$

We already mentioned that an $n \times n$ matrix can be used to 'transform' a vector $\vec{v} \in \mathbb{R}^n$ into another vector $A\vec{v} \in \mathbb{R}^n$. From that perspective, the matrix A 'transform' the unit cube spanned by $\vec{e}_1, \ldots, \vec{e}_n$ (which has volume 1) into the parallelepiped spanned by $A\vec{e}_1 = \vec{c}_1, \ldots, A\vec{e}_n = \vec{c}_n$ (which has volume $\det(A)$).

Theorem 2.28. (i) For a 2×2 matrix $A = \begin{pmatrix} a & c \\ b & d \end{pmatrix}$, we have:

$$\det(A) = \mathrm{vol}_2(\begin{pmatrix} a \\ b \end{pmatrix}, \begin{pmatrix} c \\ d \end{pmatrix}) = ad - bc.$$

(ii) For a 3×3 matrix $A = \begin{pmatrix} a & d & g \\ b & e & h \\ c & f & i \end{pmatrix}$, we have (Sarrus' rule):

$$\det(A) = \mathrm{vol}_3(\begin{pmatrix} a \\ b \\ c \end{pmatrix}, \begin{pmatrix} d \\ e \\ f \end{pmatrix}, \begin{pmatrix} g \\ h \\ i \end{pmatrix}) = aei + bfg + cdh - ceg - afh - bdi.$$

We have the following criterion to check that a given family of n vectors forms a basis of $\mathbb{R}^n$.

Theorem 2.29. Let $\vec{v}_1, \ldots, \vec{v}_n$ be a family of n vectors of $\mathbb{R}^n$, and let

$$A := \left(\begin{array}{c|c|c|c} \vec{v}_1 & \vec{v}_2 & \cdots & \vec{v}_n \end{array} \right)$$

be the associated matrix. Then:

$$\vec{v}_1, \ldots, \vec{v}_n \text{ is a basis of } \mathbb{R}^n \iff \det(A) \neq 0.$$

The determinant of a matrix as a stretching factor. The determinant of a matrix A was defined as the volume of the parallelepiped spanned by the columns $A\vec{e}_1, \ldots, A\vec{e}_n$ of A (where $\vec{e}_1, \ldots, \vec{e}_n$ is

the standard basis of $\mathbb{R}^n$). Another way to think of the determinant is as a 'stretching factor' when applying the matrix A to the unit cube $\vec{e}_1,\dots,\vec{e}_n$ of $\mathbb{R}^n$. This geometric intuition of the determinant as a stretching factor holds in general:

Proposition 2.30. For every family $\vec{x}_1,\dots,\vec{x}_n$ of vectors of $\mathbb{R}^n$ and every $m\times n$ matrix A, we have

$$\mathrm{vol}_n\big(A\vec{x}_1,\dots,A\vec{x}_n\big)=\det(A)\times\mathrm{vol}_n\big(\vec{x}_1,\dots,\vec{x}_n\big).$$

Before proving this result, we will need an additional result about n-linear alternante maps. Notice that for an n-linear alternate function ψ, the same computations as in the previous paragraphs would lead to:

$$\begin{aligned}\psi(\vec{x}_1,\dots,\vec{x}_n)&=\sum_{\sigma\text{ bij.of }\{1,\dots,n\}}\varepsilon(\sigma)a_{\sigma(1)1}\cdots a_{\sigma(n)n}\psi(\vec{e}_1,\dots,\vec{e}_n).\\&=\psi(\vec{e}_1,\dots,\vec{e}_n)\times\mathrm{vol}_n(\vec{x}_1,\dots,\vec{x}_n).\end{aligned}$$

In particular, we see that such functions are completely determined by their image on the basis $\vec{e}_1,\dots,\vec{e}_n$. As a consequence, we have:

Proposition 2.31 (Uniqueness Principle). If ψ is a n-linear alternate map from $\mathbb{R}^n\times\ldots\times\mathbb{R}^n$ to $\mathbb{R}$, then ψ is proportional to the volume, that is, there exists a scalar $\lambda\in\mathbb{R}$ such that

$$\psi=\lambda\times\mathrm{vol}_n.$$

More precisely, we have $\lambda=\psi(\vec{e}_1,\dots,\vec{e}_n)$.

Proof of Proposition 2.30. We will apply the Uniqueness Principle. Define a map

$$\psi:\mathbb{R}^n\times\cdots\times\mathbb{R}^n\to\mathbb{R},(\vec{x}_1,\dots,\vec{x}_n)\mapsto\mathrm{vol}_n(A\vec{x}_1,\dots,A\vec{x}_n).$$

Since $\vec{x}\mapsto A\vec{x}$ is linear and vol_n is n-linear, one easily checks that ψ is also n-linear. Moreover, since vol_n is alternate, so is ψ. By the Uniqueness Principle, there exists a constant $\lambda\in\mathbb{R}$ such that $\psi=\lambda\times\mathrm{vol}_n$. We find the constant λ by evaluating the previous expression at the standard basis: gives

$$\lambda=\mathrm{vol}_n(A\vec{e}_1,\dots,A\vec{e}_n)=\det(A).$$

We thus have $\mathrm{vol}_n(A\vec{x}_1,\dots,A\vec{x}_n)=\det(A)\times\mathrm{vol}_n(\vec{x}_1,\dots,\vec{x}_n)$ for every $(\vec{x}_1,\dots,\vec{x}_n)$. □

Computing determinants via row/column expansion. For higher dimensions, we present here a faster way to compute determinants.

Definition 2.32. Let A be an $n\times n$ matrix. The (i,j)-**minor** is the determinant of the $(n-1)\times(n-1)$ matrix obtained by removing row i and column j from A, and is denoted $M_{i,j}$ (or $M_{i,j}(A)$ when there is a risk of confusion).

Theorem 2.33. Let A be a $n\times n$ matrix. Then we have

- **Expansion along column j:**

$$\det(A)=(-1)^{1+j}a_{1j}M_{1j}+\ldots+(-1)^{n+j}a_{nj}M_{nj}.$$

- **Expansion along row i:**

$$\det(A) = (-1)^{i+1} a_{i1} M_{i1} + \ldots + (-1)^{i+n} a_{in} M_{in}.$$

This formula is particularly useful when there are many zeros in a row or in a column.

Example 2.34. Let us compute the previous determinant using expansion along the first column. We have:

$$\begin{aligned}\det\begin{pmatrix} 0 & 1 & 2 & 3 \\ 1 & 2 & 3 & 0 \\ 2 & 3 & 0 & 1 \\ 3 & 0 & 1 & 2 \end{pmatrix} &= 0 - \det\begin{pmatrix} 1 & 2 & 3 \\ 3 & 0 & 1 \\ 0 & 1 & 2 \end{pmatrix} + 2\det\begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 0 \\ 0 & 1 & 2 \end{pmatrix} - 3\det\begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 0 \\ 3 & 0 & 1 \end{pmatrix} \\ &= -(0+0+9-0-1-12) + 2(6+0+6-0-0-8) - 3(3+0+0-27-0-4) \\ &= 4 + 2 \times 4 - 3 \times (-28) = 96.\end{aligned}$$

An important consequence is the following result:

Theorem 2.35. Let

$$A = \begin{pmatrix} d_1 & * & \cdots & * \\ 0 & d_2 & \ddots & \vdots \\ \vdots & \ddots & \ddots & * \\ 0 & \cdots & 0 & d_n \end{pmatrix}$$

be an upper-triangular $n \times n$ matrix. Then

$$\det(A) = d_1 \times \cdots \times d_n.$$

A similar result holds for lower-triangular matrices.

Chapter 3

Vector spaces

In the previous chapter, we introduced the space $\mathbb{R}^n$ of *vectors*. Recall that the key operations for vectors in $\mathbb{R}^n$ are adding two vectors and multiplying a vector by a scalar. You are already familiar with other examples of mathematical objects where one is allowed to add them and multiply them by a scalar. In this chapter, we will develop the general abstract framework allowing us to treat all the examples in a uniform way.

3.1 Examples and definition

We introduce the main examples that will be used throughout the course:

Definition 3.1.(i) **Spaces of functions.** We denote by $\mathscr{F}(\mathbb{R})$ the set of all functions from $\mathbb{R}$ to $\mathbb{R}$. One can define the sum of two functions pointwise, that is, for functions $f, g \in \mathscr{F}(\mathbb{R})$, the sum $f+g$ is the function defined by:

$$(f+g)(x) := f(x) + g(x) \quad \text{for every} \quad x \in \mathbb{R}.$$

We can also define the multiplication of a function by a scalar pointwise. That is, for a function $f \in \mathscr{F}(\mathbb{R})$ and a scalar $\lambda \in \mathbb{R}$, we define the function λf by:

$$(\lambda f)(x) := \lambda f(x) \quad \text{for every} \quad x \in \mathbb{R}.$$

It may happen that we need functions that are continuous or differentiable, so we also introduce the set $\mathscr{C}(\mathbb{R})$ of continuous functions from $\mathbb{R}$ to $\mathbb{R}$, and the set $\mathscr{C}^\infty(\mathbb{R})$ of smooth functions from $\mathbb{R}$ to $\mathbb{R}$.

(ii) **Spaces of polynomial functions.** An important subset of the set of all functions is the set of *polynomial functions*. We denote by $\mathscr{P}(\mathbb{R})$ the set of all polynomial functions from $\mathbb{R}$ to $\mathbb{R}$, and by $\mathscr{P}_n(\mathbb{R})$ the set of all polynomial functions from $\mathbb{R}$ to $\mathbb{R}$ of degree at most n. Addition of polynomial functions and multiplication of a polynomial function by a scalar are defined as before.

(iii) **Spaces of sequences.** We denote by $\mathbb{R}^{\mathbb{N}}$ the set of all sequences $(u_n)_{n\geq 0}$ of real-valued sequences. In analogy with $\mathbb{R}^n$, we can denote a sequence as an 'infinite vector' $(u_0, u_1, \ldots)^T$. Here again, one can define the sum of two sequences pointwise, that is, we define the sum of

two sequences as follows:

$$\begin{pmatrix} u_0 \\ u_1 \\ \vdots \end{pmatrix} + \begin{pmatrix} v_0 \\ v_1 \\ \vdots \end{pmatrix} = \begin{pmatrix} u_0 + v_0 \\ u_1 + v_1 \\ \vdots \end{pmatrix}.$$

Similarly, one defines the multiplication of a sequence by a scalar $\lambda \in \mathbb{R}$ as follows:

$$\lambda \times \begin{pmatrix} u_0 \\ u_1 \\ \vdots \end{pmatrix} := \begin{pmatrix} \lambda u_0 \\ \lambda u_1 \\ \vdots \end{pmatrix}.$$

We thus have several examples of sets of mathematical objects that display a similar type of structure. To be able to treat these various cases with a single mathematical framework, we now introduce the notion of an *abstract vector space*. This is the notion that makes rigorous this idea of 'spaces of objects where one can add vectors and multiply them by a scalar'.

Definition 3.2. A *real vector space* is a set V (whose elements are called *vectors*) endowed with two operations:

(i) an *addition* denoted + that associates to two vectors $\vec{u}, \vec{v} \in V$ a vector $\vec{u} + \vec{v} \in V$,

(ii) a *multiplication by a scalar* denoted $\cdot$ that associates to a vector $\vec{v} \in V$ and a scalar $\lambda \in \mathbb{R}$ a vector $\lambda \cdot \vec{v} \in V$ (or simply $\lambda \vec{v}$).

We further require that the following *vector space axioms* are satisfied:

(1) *null vector:* There is an element $\vec{0} = \vec{0}_V \in V$, the *zero* or *null vector*, such that $\vec{v} + \vec{0} = \vec{v}$ for all $\vec{v} \in V$.

(2) *opposite:* For all $\vec{v} \in V$, there is an element $-\vec{v} \in V$ such that $\vec{v} + (-\vec{v}) = \vec{0}$.

(3) *commutativity of the addition:* $\vec{v} + \vec{w} = \vec{w} + \vec{v}$ for all $\vec{v}, \vec{w} \in V$,

(4) *associativity of the addition:* $(\vec{v} + \vec{w}) + \vec{u} = \vec{v} + (\vec{w} + \vec{u})$ for all $\vec{v}, \vec{w}, \vec{u} \in V$.

(5) *distributivity of the scalar multiplication:* $a(\vec{v} + \vec{w}) = a\vec{v} + a\vec{w}$ and $(a + b)\vec{v} = a\vec{v} + b\vec{v}$, $a, b \in \mathbb{R}$, $\vec{v}, \vec{w} \in V$,

(6) *associativity of the scalar multiplication:* $a(b\vec{v}) = (ab)\vec{v}$, $a, b \in \mathbb{R}$, $\vec{v} \in V$,

(7) *compatibility:* $1 \cdot \vec{v} = \vec{v}$.

This list of axioms (1)-(7) may seem long and technical. However, you should convince yourself that they encode the usual properties that one expects from addition and multiplication by a number for vectors, functions, etc. Moreover, the strength of this abstract framework is that, once we know that something is a vector space, we can treat objects of that space (functions, sequences, or more exotic objects) as if they were vectors, and use our geometric intuition to solve problems.

Example 3.3.(i) The space $\mathbb{R}^n$ is a vector space for the addition and multiplication by a scalar introduced in the previous chapter. The null vector $(0, \dots, 0)^T$ satisfies axiom (1). The opposite of a vector $(v_1, \dots, v_n)^T$ is defined as $(-v_1, \dots, -v_n)^T$ to satisfy axiom (2). One then checks that axioms (3)-(7) hold. Indeed, one has to check the equations coordinate by coordinate, and each

of these equations then boils down to a standard rule about addition and multiplication of real numbers (commutativity of addition, distributivity of the multiplication, etc.)

(ii) The space $\mathscr{F}(\mathbb{R})$ is a vector space for the addition and multiplication by a scalar defined above. Here, a vector is a function, the zero vector is the *zero function*, that is, the function z such that $z(x) = 0$ for every $x \in \mathbb{R}$, which satisfies axiom (1). For a function $f \in \mathscr{F}(\mathbb{R})$, the opposite $-f$ is defined pointwise by $(-f)(x) := -f(x)$ for every $x \in \mathbb{R}$, and satisfies axiom (2). One then checks that axioms (3)-(7) hold. Indeed, by definition of the addition and multiplication by a scalar, each of the equations can be checked pointwise.

(iii) The space $\mathbb{R}^{\mathbb{N}}$ is a vector space for the addition and multiplication by a scalar defined above. Here, a vector is a sequence

$$\begin{pmatrix} u_0 \\ u_1 \\ \vdots \end{pmatrix},$$

the zero vector is the *zero sequence*

$$\begin{pmatrix} 0 \\ 0 \\ \vdots \end{pmatrix},$$

and the opposite of a vector is given by

$$-\begin{pmatrix} u_0 \\ u_1 \\ \vdots \end{pmatrix} := \begin{pmatrix} -u_0 \\ -u_1 \\ \vdots \end{pmatrix}.$$

Here again, one then checks that axioms (3)-(7) hold, by checking each equation pointwise.

Complex vector spaces. We have just defined *real* vector spaces, that is, vector spaces where the set of scalars is $\mathbb{R}$. These will be the almost sole focus of this course. However, one can analogously define *complex vectors spaces* by setting the set of scalars to be $\mathbb{C}$ and having the same list of axioms. Examples of complex vectors spaces are:

- The space $\mathbb{C}^n$ of vectors of the form $(z_1, \ldots, z_n)^T$ with $z_i \in \mathbb{C}$ for every i.
- The space $\mathscr{F}(\mathbb{C})$ of functions from $\mathbb{C}$ to $\mathbb{C}$.
- The space $\mathscr{P}(\mathbb{C})$ of polynomial functions with complex coefficients.
- The space $\mathbb{C}^{\mathbb{N}}$ of complex sequences $(z_0, z_1, \ldots)$ with $z_i \in \mathbb{C}$ for every $i \geq 0$.

More generally, one can define a notion of vector space for more general set of scalars called **fields**. Fields can be thought as sets of generalised numbers, where one can perform the usual four operations of addition, subtraction, multiplication, and division by a non-zero element. Examples of fields include of course $\mathbb{R}$ and $\mathbb{C}$, but also the set $\mathbb{Z}/p\mathbb{Z}$ of integers modulo a prime number p. Vector spaces over $\mathbb{Z}/p\mathbb{Z}$ (or more general finite fields) have many concrete applications, including the construction of error correcting codes. You will learn more about fields in **Abstract Algebra**.

3.2 Linear combinations in abstract vector spaces

We now generalise to abstract vector spaces the notions we introduced in $\mathbb{R}^n$.

Definition 3.4. A vector $\vec{v} \in V$ is called a *linear combination* of the vectors $\vec{u}_1, \ldots, \vec{u}_k \in V$, if it can be written as

$$\vec{v} = a_1\vec{u}_1 + a_2\vec{u}_2 + \ldots + a_k\vec{u}_k\,, \quad a_1, \ldots, a_k \in \mathbb{R}\,.$$

This is an extension of our definition of linear combination of vectors in $\mathbb{R}^n$.

Example 3.5. Consider the vector space $\mathscr{P}_2(\mathbb{R})$ of polynomials of degree at most 2. Linear combinations of the two functions (=vectors) defined $P_1(x) = x$ and $P_2(x) = x^2$ are polynomial functions of the form

$$a_1 x + a_2 x^2, \quad \text{for } a_1, a_2 \in \mathbb{R}.$$

For instance, the polynomial function defined $P(x) = 5x^2 - 2x$ is a linear combination of x and x^2, since we have $P = 5P_2 - 2P_1$.

Definition 3.6. The *span* of a family of vectors $\vec{u}_1, \ldots, \vec{u}_k$, usually denoted by $\text{span}(\vec{u}_1, \ldots, \vec{u}_k)$, is the set of all possible linear combinations of $\vec{u}_1, \ldots, \vec{u}_k$:

$$\text{span}(\vec{u}_1, \ldots, \vec{u}_k) := \{a_1\vec{u}_1 + \ldots + a_k\vec{u}_k | a_1, \ldots, a_k \in \mathbb{R}\}\,.$$

We also say that $\text{span}(\vec{u}_1, \ldots, \vec{u}_k)$ is spanned by $\vec{u}_1, \ldots, \vec{u}_k$ or that these vectors span $\text{span}(\vec{u}_1, \ldots, \vec{u}_k)$.

Example 3.7.(i) The vectors $(1,0,0)^T$, $(0,1,0)^T$, $(0,0,1)^T$ span $\mathbb{R}^3$.

(ii) The vector space of polynomials up to degree n, $\mathscr{P}_n(\mathbb{R})$, is spanned by the polynomials $1, x, x^2, \ldots, x^n$, as every polynomial function in $\mathscr{P}_n(\mathbb{R})$ can be written as $a_0 \times 1 + \ldots + a_n x^n$.

(iii) The polynomial function $2x^2 + x - 1$ belongs to $\text{span}(x^2 + x, x + 1)$ since $2x^2 + x - 1 = 2(x^2 + x) - (x + 1)$.

Definition 3.8. We say that vectors $\vec{v}_1, \ldots, \vec{v}_k$ are *linearly independent* if

$$c_1\vec{v}_1 + \ldots + c_k\vec{v}_k = \vec{0} \implies c_1 = \ldots = c_k = 0.$$

Otherwise, we say that the vectors are *linearly dependent*. This is an extension of our definition of linear independence for $\mathbb{R}^n$.

As in the case of $\mathbb{R}^n$, linear dependence as a simple interpretation: a family of vectors $\vec{u}_1, \ldots, \vec{u}_k$ is linearly dependent, if and only if one (i.e. at least one) of the $\vec{u}_i$s can be expressed as a linear combination of the others.

Example 3.9. Let us show that the functions $f_1(x) := \cos(x)$, $f_2(x) := \cos(2x)$, $f_3(x) := \cos(3x)$ are linearly independent vectors of $\mathscr{F}(\mathbb{R})$.
Let $a, b, c \in \mathbb{R}$ be scalars such that $af_1 + bf_2 + cf_3$ is the zero vector of $\mathscr{F}(\mathbb{R})$, that is, the zero function. This is equivalent to:

$$a\cos(x) + b\cos(2x) + c\cos(3x) = 0 \quad \text{for every } \ x \in \mathbb{R}.$$

By evaluating at $x = \pi/2$, we get $b = 0$, hence $a\cos(x) + c\cos(3x) = 0$ for every $x \in \mathbb{R}$. By evaluating this equation at $x = \pi/6$, we get $a \times \sqrt{3}/2 = 0$, hence $a = 0$. Finally, we have $c\cos(3x) = 0$ for every

$x \in \mathbb{R}$, and evaluating at $x = 0$ yields $c = 0$.
We thus have $a = b = c = 0$, and it follows that f_1, f_2, f_3 are linearly independent.

3.3 Basis and dimension

Basis of a vector space.

Definition 3.10. A family of vectors $\vec{u}_1, \ldots, \vec{u}_n$ is called a *basis* of V, if it is a linearly independent set of vectors that span V.

Example 3.11. We have already seen the standard basis $\vec{e}_1 = (1,0,\ldots,0)^T, \ldots, \vec{e}_n = (0,\ldots,0,1)^T$ of $\mathbb{R}^n$. Note that a real vector space has infinitely many bases. For instance, $(1,1)^T$ and $(1,-1)^T$ is also a basis of $\mathbb{R}^2$ (Why?).

Here is an important example:

Theorem 3.12. A basis for the vector space of polynomials of degree at most n is $1, x, x^2, \ldots, x^n$.

Proof. The family clearly spans $\mathscr{P}_n(\mathbb{R})$, as every polynomial function of $\mathscr{P}_n(\mathbb{R})$ can be written as a linear combination $a_0 + a_1 x + \ldots + a_n x^n$ for some scalars $a_0, \ldots, a_n \in \mathbb{R}$. Let us show that this family is free. Suppose that we have a linear combination $a_0 + a_1 x + \ldots + a_n x^n$ that is the zero vector, that is, $a_0 + a_1 x + \ldots + a_n x^n = 0$ for every $x \in \mathbb{R}$. By evaluating at $x = 0$, we get $a_0 = 0$. Now since the polynomial function $a_0 + a_1 x + \ldots + a_n x^n$ is the zero function, so its derivative, so we get $a_1 + 2a_2 x + \ldots + n a_n x^{n-1} = 0$ for every $x \in \mathbb{R}$. Evaluating again at $x = 0$, we get $a_1 = 0$. By repeating the same procedure (differentiating and evaluating at $x = 0$), we prove successively that $a_0 = a_1 = \ldots = a_n = 0$. Thus, the family of polynomial functions $1, x, \ldots, x^n$ is linearly independent, so it is a basis of $\mathscr{P}_n(\mathbb{R})$. □

In tutorials, you will see other examples of bases of polynomials and how to use them.

Dimension of a vector space. In order to define the dimension of a vector space, we need the following important result:

Proposition 3.13. Any two bases for a vector space V contain the same number of vectors.

Before proving it, we need the following result that generalises to general vector spaces Proposition 2.21:

Proposition 3.14. If a family of vectors $\vec{v}_1, \vec{v}_2, \ldots, \vec{v}_n$ is a basis of a vector space V, then every family of vectors of V containing more than n vectors is linearly dependent.

Proof. Let $\vec{w}_1, \vec{w}_2, \ldots, \vec{w}_m$ be a family of $m > n$ vectors of V. We will show that there exist $c_1, c_2, \ldots, c_m$ not all zero such that

$$c_1 \vec{w}_1 + c_2 \vec{w}_2 + \ldots + c_m \vec{w}_m = \vec{0}\,. \tag{3.1}$$

Since $\vec{v}_1, \vec{v}_2, \ldots, \vec{v}_n$ spans V, each $\vec{w}_i$ can be expressed as a linear combination of the $\vec{v}_i$'s:

$$\begin{aligned}
\vec{w}_1 &= a_{11}\vec{v}_1 + a_{12}\vec{v}_2 + \ldots + a_{1n}\vec{v}_n\,, \\
\vec{w}_2 &= a_{21}\vec{v}_1 + a_{22}\vec{v}_2 + \ldots + a_{2n}\vec{v}_n\,, \\
&\vdots \qquad\qquad \vdots \\
\vec{w}_m &= a_{m1}\vec{v}_1 + a_{m2}\vec{v}_2 + \ldots + a_{mn}\vec{v}_n\,.
\end{aligned}$$

Plugging this into (3.1), we have

$$c_1(a_{11}\vec{v}_1 + \ldots + a_{1n}\vec{v}_n) + \ldots + c_m(a_{m1}\vec{v}_1 + \ldots + a_{mn}\vec{v}_n) = \vec{0}\,.$$

To have all the coefficients of the $\vec{v}_1, \ldots, \vec{v}_n$ vanish, note that it is sufficient to find $c_1, \ldots, c_m$ such that

$$\begin{aligned}
a_{11}c_1 + a_{21}c_2 + \ldots + a_{m1}c_m &= 0\,, \\
a_{12}c_1 + a_{22}c_2 + \ldots + a_{m2}c_m &= 0\,, \\
&\vdots \qquad\qquad \vdots \\
a_{1n}c_1 + a_{2n}c_2 + \ldots + a_{mn}c_m &= 0\,.
\end{aligned}$$

This is a homogeneous system of linear equations with more unknowns (m) than equations (n) and thus there is a solution with not all of the c_i being zero (Theorem 1.14). It follows that (3.1) has a solution besides the trivial solution and so S' is linearly dependent. □

Proof of Proposition 3.13. Let $B = (\vec{v}_1, \vec{v}_2, \ldots, \vec{v}_n)$ and $B' = (\vec{v}'_1, \vec{v}'_2, \ldots, \vec{v}'_m)$ be bases for V. From the above lemma, we conclude that since B is a basis and B' is linearly independent, $m \leq n$. Equally, since B' is a basis and B is linearly independent, $n \leq m$. Altogether, we have $m = n$. □

We are now able to define properly the dimension of a vector space:

Definition 3.15. Let V be a vector space. We say that V is *finite-dimensional* if it has a finite basis, and *infinite-dimensional* otherwise. We define the *dimension* of V to be the number of vectors in any basis of V.

Example 3.16. We have constructed bases of several vector spaces, in Example 3.11 and Theorem 3.12. We thus have the following dimensions:

Space	$\mathbb{R}^2$	$\mathbb{R}^3$	$\mathbb{R}^n$	$\mathscr{P}_n(\mathbb{R})$
Dimension	2	3	n	$n+1$

The following result is useful in finding a lower bound for the dimension of a vector space:

Proposition 3.17. Let V be a vector space, and let $\vec{v}_1, \ldots, \vec{v}_k$ be a linearly independent family. Then we have:

$$\dim V \geq k.$$

As a consequence, we have the following:

Proposition 3.18. The vector space $\mathscr{P}(\mathbb{R})$ of all real polynomial functions has infinite dimension.

Proof. We know from Theorem 3.12 that $1, x, \ldots, x^n$ is linearly independent for all $n \geq 0$. The previous proposition thus implies that $\dim \mathscr{P}(\mathbb{R}) \geq n$ for all $n \geq 0$, hence $\dim \mathscr{P}(\mathbb{R}) = \infty$. □

3.4 'Coordinatisation': Thinking of vector spaces geometrically.

Once we choose a basis $\mathscr{B}$ of a n-dimensional vector space V, we can identify elements of V with vectors of $\mathbb{R}^n$ as follows:

Definition 3.19. Let V be an n-dimensional vector space with basis $\mathscr{B} = (\vec{e}_1, \ldots, \vec{e}_n)$. Let $\vec{x} \in V$ be a vector, which we write as a linear combination $\vec{x} = x_1\vec{e}_1 + \ldots + x_n\vec{e}_n$.
The *coordinate matrix of* $\vec{x}$ *in the basis* $\mathscr{B}$, denoted $[\vec{x}]_{\mathscr{B}}$ is the following $n \times 1$ matrix (in other words, the following vector of $\mathbb{R}^n$):

$$[\vec{x}]_{\mathscr{B}} := \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}.$$

Example 3.20. Let us consider the basis $\mathscr{B} := (1, x, x^2)$ of $\mathscr{P}_2(\mathbb{R})$, and let us consider the polynomial functions $P(x) := 2x^2 - 3x + 5$ and $Q(x) := 3x + 7$. We have

$$[P]_{\mathscr{B}} = \begin{pmatrix} 5 \\ -3 \\ 2 \end{pmatrix}, \quad [Q]_{\mathscr{B}} = \begin{pmatrix} 7 \\ 3 \\ 0 \end{pmatrix}.$$

Be careful with the order of the coefficients, which is determined by the order of the vectors in $\mathscr{B}$: Here, the first vector in our basis is the polynomial 1, so the first coordinate in the coordinate vector must be the constant coefficient of our polynomial.

Proposition 3.21. The following 'coordinatisation' map

$$T : V \to \mathbb{R}^n, \quad \vec{x} \mapsto [\vec{x}]_{\mathscr{B}}$$

is a bijection. Moreover it satisfies for every $\vec{x}, \vec{y} \in V$ and $\lambda \in \mathbb{R}$:

- $T(\vec{x} + \vec{y}) = T(\vec{x}) + T(\vec{y})$,
- $T(\lambda\vec{x}) = \lambda T(\vec{x})$.

Under the previous hypotheses, we say that the vector space V is *isomorphic* to $\mathbb{R}^n$. We will explain this concept further in Section 4.5. Intuitively, this result says that any real vector space of dimension n can be 'identified' with $\mathbb{R}^n$ in a way that is compatible with the vector space operations on V and $\mathbb{R}^n$. This allows us to use the geometric intuition we have from $\mathbb{R}^n$ to study more general vector spaces. This point of view will be extremely useful when dealing with general vector spaces. For instance, we have the following:

Theorem 3.22. Let V be an n-dimensional vector space with basis $\mathscr{B} = (\vec{e}_1, \dots, \vec{e}_n)$, and consider the coordinatisation map

$$T : V \to \mathbb{R}^n, \quad \vec{x} \mapsto [\vec{x}]_{\mathscr{B}}.$$

Then for every family $\vec{v}_1, \dots, \vec{v}_k$ of vectors of V, we have

- $\vec{v}_1, \dots, \vec{v}_k$ is linearly independent in $V \Leftrightarrow [\vec{v}_1]_{\mathscr{B}}, \dots, [\vec{v}_k]_{\mathscr{B}}$ is linearly independent in $\mathbb{R}^n$.
- $\vec{v}_1, \dots, \vec{v}_k$ spans $V \Leftrightarrow [\vec{v}_1]_{\mathscr{B}}, \dots, [\vec{v}_k]_{\mathscr{B}}$ spans $\mathbb{R}^n$.
- $\vec{v}_1, \dots, \vec{v}_k$ is a basis of $V \Leftrightarrow [\vec{v}_1]_{\mathscr{B}}, \dots, [\vec{v}_k]_{\mathscr{B}}$ is a basis of $\mathbb{R}^n$.

Since we saw in the previous chapter many tools to determine whether a given family of vectors of $\mathbb{R}^n$ is linearly independent / spanning / a basis, this gives us new tools to study family of vectors in more general vector spaces.
Here is an example of application, generalising Theorem 3.12:

Theorem 3.23. Let $P_0, \dots, P_n$ be polynomial functions such that for every i, the degree of P_i is exactly i. Then $P_0, \dots, P_n$ is a basis of $\mathscr{P}_n(\mathbb{R})$.

Proof. Consider the standard basis $\mathscr{B} = (1, x, \dots, x^n)$ of $\mathscr{P}_n(\mathbb{R})$. By the previous theorem, $P_0, \dots, P_n$ is a basis of $\mathscr{P}_n(\mathbb{R})$ if and only if $[P_0]_{\mathscr{B}}, \dots, [P_n]_{\mathscr{B}}$ is a basis of $\mathbb{R}^{n+1}$. But we know from Theorem 2.29 that $[P_0]_{\mathscr{B}}, \dots, [P_n]_{\mathscr{B}}$ is a basis of $\mathbb{R}^{n+1}$ if and only if the matrix

$$A = \left([P_0]_{\mathscr{B}} | \cdots | [P_n]_{\mathscr{B}}\right)$$

satisfies $\det(A) \neq 0$. But since by assumption $\deg(P_i) = i$ for every i, A is an upper-triangular matrix with non-zero diagonal entries. Thus, it follows from Theorem 2.35 that $\det(A) \neq 0$, which concludes the proof. □

Example 3.24. The polynomial functions 1, $1+x$, $2+x^2$, and $3-x^3$ form a basis of $\mathscr{P}_3(\mathbb{R})$, since it is a family of four vectors of degree $0, 1, 2, 3$ respectively.

3.5 Vector subspaces

The previous examples are the main sources of vectors spaces for this course. However, we are often not interested in the space of all vectors, or of all functions. Instead, we are often interested in particular subsets of elements that satisfy some equation: system of linear equations, differential equations, etc.
In this section, we introduce the notion of *vector subspace* as the natural notion of subset of a vector space that is compatible with the operations of addition and multiplication by a scalar. We will see that the set of solutions of various equations naturally form a vector subspace of the associated vector space.

Definition 3.25. Let V be a vector space and let W be a subset of V. We say that W is a *vector subspace* (or simply a subspace) of V if the following holds:

(i) $\vec{0} \in W$,

(ii) for all $\vec{w}_1, \vec{w}_2 \in W$, we have $\vec{w}_1 + \vec{w}_2 \in W$ and

(iii) for all $\lambda \in \mathbb{R}$ and $\vec{w} \in W$, we have $\lambda \vec{w} \in W$.

Proposition 3.26. A vector subspace W of V is itself a vector space, when endowed with the addition and scalar multiplication coming from V. (This is actually an equivalence: a subset W is a subspace if and only if it W is a vector space when endowed with the addition and scalar multiplication from V.)
As a consequence, we can talk of the dimension or of bases of a given subspace.

Example 3.27.(i) In a vector space V, the whole space V and the trivial subset $\{\vec{0}\}$ are always vector subspaces.

(ii) Consider the subset $W = \{(x,y)^T \in \mathbb{R}^2 : x+y=0\}$ of $\mathbb{R}^2$. That is, W consists of all the vectors of the form $(x,-x)^T$ for $x \in \mathbb{R}$. The null vector $(0,0)^T$ is clearly in W. If we add two vectors or multiply a vector by a scalar in W, we end up back in W:

$$\begin{pmatrix} x_1 \\ -x_1 \end{pmatrix} + \begin{pmatrix} x_2 \\ -x_2 \end{pmatrix} = \begin{pmatrix} x_1+x_2 \\ -(x_1+x_2) \end{pmatrix} \quad \text{and} \quad \lambda \begin{pmatrix} x_1 \\ -x_1 \end{pmatrix} = \begin{pmatrix} \lambda x_1 \\ -\lambda x_1 \end{pmatrix}.$$

Thus, W us a subspace of $\mathbb{R}^2$. Note that every vector of W is of the form

$$\begin{pmatrix} x \\ -x \end{pmatrix} = x \begin{pmatrix} 1 \\ -1 \end{pmatrix} \quad \text{for some } x \in \mathbb{R},$$

so W has dimension 1 and a basis of W is given by the vector $(1,-1)^T$.

Remark. If a subspace W contains a vector $\vec{w}$, it also contains its opposite $-\vec{w}$, as W is stable under multiplication by a scalar and $-\vec{w} = (-1).\vec{w}$.

Example 3.28. The subset $W' = \{(x,y)^T | x+y=1\}$ is not a vector subspace of $\mathbb{R}^2$, as $\vec{0} = (0,0)^T$ is not an element of W'. Another reason is that the sum of two elements $w_1, w_2 \in W'$ is not always in W'. For instance, $(1,0)^T$ and $(0,1)^T$ belong to W', but their sum $(1,1)^T$ does not.

Example 3.29.(i) Let $V = \mathbb{R}^2$. Then $W = \{(x,y)^T | x = 3y\}$ is a vector subspace, since vectors of W are of the form

$$\begin{pmatrix} 3y \\ y \end{pmatrix} = y \begin{pmatrix} 3 \\ 1 \end{pmatrix} \quad \text{for some } y \in \mathbb{R},$$

and we have:

$$\begin{pmatrix} 3y_1 \\ y_1 \end{pmatrix} + \begin{pmatrix} 3y_2 \\ y_2 \end{pmatrix} = \begin{pmatrix} 3(y_1+y_2) \\ y_1+y_2 \end{pmatrix} \quad \text{and} \quad \lambda \begin{pmatrix} 3y \\ y \end{pmatrix} = \begin{pmatrix} 3\lambda y \\ \lambda y \end{pmatrix}, \quad \lambda \in \mathbb{R}.$$

Thus, W is a subspace of $\mathbb{R}^2$ of dimension 1, and a basis of it is given by the vector $(3,1)^T$.
In general, lines through the origin of $\mathbb{R}^2$ form vector subspaces of $\mathbb{R}^2$.

(ii) Lines in $\mathbb{R}^2$ that do not pass through the origin do not contain $\vec{0}$ and thus are not vector subspaces of $\mathbb{R}^2$, cf. example in 3.28.

Example 3.30.(i) The set $\mathscr{P}(\mathbb{R})$ is a subspace of $\mathscr{F}(\mathbb{R},\mathbb{R})$ (the set of all functions f such that $f:\mathbb{R}\to\mathbb{R}$). Indeed, it contains the zero function, the sum of two polynomial functions is again a polynomial function, and multiplying a polynomial by a scalar yields another polynomial function.

(ii) Consider the following subset of $\mathscr{F}(\mathbb{R},\mathbb{R})$: $W=\{f\in\mathscr{F}(\mathbb{R},\mathbb{R}) : f(1)=0\}$. In contains the zero function. Moreover, we have:

$$\text{For } f_1,f_2\in W\,,\quad (f_1+f_2)(1)=f_1(1)+f_2(1)=0+0=0,\quad \text{hence } f_1+f_2\in W\,,$$
$$\text{For } \lambda\in\mathbb{R}, f\in W\,,\quad (\lambda f)(1)=\lambda f(1)=\lambda 0=0\quad \text{hence } \lambda f\in W\,.$$

Thus, W is a vector subspace of $\mathscr{F}(\mathbb{R},\mathbb{R})$.

Subspaces of $\mathbb{R}^2$. We can completely describe the subspaces of $\mathbb{R}^2$:

Proposition 3.31. The subspaces of $\mathbb{R}^2$ are the following:

(i) the trivial subspace $\{\vec{0}\}$,

(ii) lines through the origin,

(iii) the whole space $\mathbb{R}^2$.

Proof. Let V be a subspace of $\mathbb{R}^2$. We can assume that V contains a non-trivial vector $\vec{v}$, otherwise we are in case (i) and we are done. Since V is a subspace, it also contains all the multiples $\lambda\vec{v}$ for $\lambda\in\mathbb{R}$. If V contains no other vector, then V is the line generated by the vector $\vec{v}$ and we are in case (ii). Suppose now that V contains a vector $\vec{w}$ that is not of the form $\lambda\vec{v}$. Then $\vec{v}$ and $\vec{w}$ are linearly independent, and by Theorem 2.24, $\vec{v},\vec{w}$ form a basis of $\mathbb{R}^2$. Thus, every vector of $\mathbb{R}^2$ can be written a linear combination of $\vec{v}$ and $\vec{w}$. But since V is a subspace containing $\vec{v}$ and $\vec{w}$, it contains every linear combination of $\vec{v}$ and $\vec{w}$, hence it contains $\mathbb{R}^2$. It thus follows that $V=\mathbb{R}^2$, and we are done. □

There is a similar picture in $\mathbb{R}^3$, where subspaces can be lines through the origin, planes through the origin, etc. Geometrically, being stable under addition and scalar multiplication makes vector subspaces 'look flat'.

The solution space of a homogeneous system of linear equations.

Theorem 3.32. The set of solutions of a *homogeneous* system of linear equations in n unknowns is a vector subspace of $\mathbb{R}^n$ whose dimension is the number of free variables, after reducing the system to its echelon form.

Proof. Let W be the set of solutions. By definition, we have $W=\{\vec{x}\in\mathbb{R}^n \mid A\vec{x}=\vec{0}\}$ for some matrix A. In particular, the null vector belongs to W. Let $\vec{x}_1,\vec{x}_2\in W$ and $\lambda\in\mathbb{R}$. We have:

$$A(\vec{x}_1+\vec{x}_2)=A\vec{x}_1+A\vec{x}_2=\vec{0}+\vec{0}=\vec{0},\text{ hence } \vec{x}_1+\vec{x}_2\in W$$
$$A(\lambda\vec{x}_1)=\lambda A\vec{x}_1=\lambda\vec{0}=\vec{0},\text{ hence } \lambda\vec{x}_1\in W\,.$$

Thus, W is a vector subspace of $\mathbb{R}^n$. We solve the system $A\vec{x} = \vec{0}$ using Gaussian elimination. As usual, we can express solutions in terms of the $k \leq n$ free variables $\alpha_1, \ldots, \alpha_k$ of the system. In particular, we write the general solution as a linear combination of the form $\alpha_1 \vec{v}_1 + \ldots + \alpha_k \vec{v}_k$ for some vectors $\vec{v}_1, \ldots, \vec{v}_k$ of $\mathbb{R}^n$. In particular, we see that the family $\vec{v}_1, \ldots, \vec{v}_k$ spans the subspace of solutions. To show that this family is also free, notice that if a free variable α_i corresponds to the variable x_j, then the j-th component of the vector $\alpha_1 \vec{v}_1 + \ldots + \alpha_k \vec{v}_k$ is exactly α_i. In particular, if $\alpha_1 \vec{v}_1 + \ldots + \alpha_k \vec{v}_k = \vec{0}$, then all the components must be zero, and it follows that $\alpha_1 = \ldots = \alpha_k = 0$. □

Remark. Note that $\{x \in \mathbb{R}^n : A\vec{x} = \vec{b}\}$ with $\vec{b} \neq \vec{0}$ is *not* a vector subspace. Indeed, this subset does not contain the null vector as $A\vec{0} = \vec{0} \neq \vec{b}$. This is analogous to lines in $\mathbb{R}^2$ not running through the origin.

Method: basis of solutions of a homogeneous system of linear equations

To find a basis of the space of solutions of a homogeneous system of linear equations, we can apply the following algorithm:

- Solve the system using Gaussian elimination.
- Express solutions in terms of the free variables $\alpha_1, \ldots, \alpha_k$.
- Decompose the general solution as a linear combination of the form $\alpha_1 \vec{v}_1 + \ldots + \alpha_k \vec{v}_k$.
- A basis of the space of solutions is $\vec{v}_1, \ldots, \vec{v}_k$.

Example 3.33. Consider the following system of linear equations:

$$\begin{aligned} x_1 - x_2 + 2x_3 + x_4 &= 0\,, \\ 2x_1 + x_2 - x_3 + x_4 &= 0\,, \\ 4x_1 - x_2 + 3x_3 + 3x_4 &= 0\,, \\ x_1 + 2x_2 - 3x_3 &= 0\,. \end{aligned}$$

We would like to determine a basis of the subspace W of solutions. We first perform Gaussian elimination to reduce the system:

$$\left(\begin{array}{cccc|c} 1 & -1 & 2 & 1 & 0 \\ 2 & 1 & -1 & 1 & 0 \\ 4 & -1 & 3 & 3 & 0 \\ 1 & 2 & -3 & 0 & 0 \end{array}\right) \overset{\substack{R_2 \to R_2 - 2R_1 \\ R_3 \to R_3 - 4R_1 \\ R_4 \to R_4 - R_1}}{\rightsquigarrow} \left(\begin{array}{cccc|c} 1 & -1 & 2 & 1 & 0 \\ 0 & 3 & -5 & -1 & 0 \\ 0 & 3 & -5 & -1 & 0 \\ 0 & 3 & -5 & -1 & 0 \end{array}\right)$$

$$\overset{\substack{R_3 \to R_3 - R_2 \\ R_4 \to R_4 - R_2}}{\rightsquigarrow} \left(\begin{array}{cccc|c} 1 & -1 & 2 & 1 & 0 \\ 0 & 3 & -5 & -1 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right)$$

There are two free variables, x_3 and x_4, and the general solution is thus of the form

$$x_3 = \alpha\,, \quad x_4 = \beta\,, \quad x_2 = \frac{1}{3}(5\alpha + \beta)\,, \quad x_1 = x_2 - 2x_3 - x_4 = -\frac{1}{3}\alpha - \frac{2}{3}\beta\,.$$

Any solution can be rewritten as

$$\vec{x} = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} = \frac{1}{3}\begin{pmatrix} -\alpha - 2\beta \\ 5\alpha + \beta \\ 3\alpha \\ 3\beta \end{pmatrix} = \alpha\frac{1}{3}\begin{pmatrix} -1 \\ 5 \\ 3 \\ 0 \end{pmatrix} + \beta\frac{1}{3}\begin{pmatrix} -2 \\ 1 \\ 0 \\ 3 \end{pmatrix},$$

and we find that $\frac{1}{3}(-1,5,3,0)^T, \frac{1}{3}(-2,1,0,3)^T$ is a basis of the solution space of the system.

The span of a family of vectors . Another important type of subspaces, that will become important in Chapter 4, is given by spans of families of vectors. The following generalises a result we have seen in $\mathbb{R}^n$:

Theorem 3.34. Let V be a vector space and let $\vec{u}_1, \ldots, \vec{u}_k$ be vectors of V. Then $\operatorname{span}(\vec{u}_1, \ldots, \vec{u}_k)$ is a vector subspace of V.

Proof. The span contains the null vector since $\vec{0} = 0\vec{u}_1 + \ldots + 0\vec{u}_k$. Let $a_1\vec{u}_1 + \ldots + a_k\vec{u}_k, b_1\vec{u}_1 + \ldots + b_k\vec{u}_k$ be vectors in $\operatorname{span}(\vec{u}_1, \ldots, \vec{u}_k)$, and let $\lambda \in \mathbb{R}$. Then we have

$$(a_1\vec{u}_1 + \ldots + a_k\vec{u}_k) + (b_1\vec{u}_1 + \ldots + b_k\vec{u}_) = (a_1 + b_1)\vec{u}_1 + \ldots + (a_k + b_k)\vec{u}_k \in \operatorname{span}(\vec{u}_1, \ldots, \vec{u}_k),$$

$$\lambda(a_1\vec{u}_1 + \ldots + a_k\vec{u}_k) = (\lambda a_1)\vec{u}_1 + \ldots + (\lambda a_k)\vec{u}_k \in \operatorname{span}(\vec{u}_1, \ldots, \vec{u}_k).$$

Thus, $\operatorname{span}(\vec{u}_1, \ldots, \vec{u}_k)$ is a subspace of V. □

Example 3.35. You have probably seen in a previous course that a solution of the differential equation $y'' + \omega^2 y = 0$ is of the form $a\cos(\omega x) + b\sin(\omega x)$ with $a, b \in \mathbb{R}$. Thus, the set of solutions of the differential equation $y'' + \omega^2 y = 0$ is the span of the functions $x \mapsto \cos(\omega x)$ and $x \mapsto \sin(\omega x)$. In particular, we recover the fact that it is a vector subspace of $\mathscr{F}(\mathbb{R})$.

Remark. A vector subspace can sometimes be spanned by many different sets of vectors: for instance, both the pairs of vectors $\left((1,0)^T, (0,1)^T\right)$ and $\left((1,0)^T, (1,1)^T\right)$ span $\mathbb{R}^2$. Indeed, given a vector $(x,y)^T \in \mathbb{R}^2$, we have

$$\begin{pmatrix} x \\ y \end{pmatrix} = x\begin{pmatrix} 1 \\ 0 \end{pmatrix} + y\begin{pmatrix} 0 \\ 1 \end{pmatrix} \quad \text{and also} \quad \begin{pmatrix} x \\ y \end{pmatrix} = (x-y)\begin{pmatrix} 1 \\ 0 \end{pmatrix} + y\begin{pmatrix} 1 \\ 1 \end{pmatrix}.$$

Solutions of systems of linear differential equations as subspaces. We saw in a previous example how the set of solutions of certain differential equations may be described as a span, and hence is a subspace of $\mathscr{F}(\mathbb{R})$. Even without an explicit description of the solutions, it is possible to show that the set of solutions forms a subspace.

Proposition 3.36. Consider the following differential equation (linearised simple pendulum):

$$y'' + \omega^2 y = 0.$$

Then the set W of solutions of this equation is a subspace of $\mathscr{F}(\mathbb{R})$.

Proof. Let $y_1, y_2 \in W$ and $\lambda \in \mathbb{R}$. Then:

$$(y_1+y_2)''+\omega^2(y_1+y_2) = (y_1''+\omega^2 y_1)+(y_2''+\omega^2 y_2) = 0+0 = 0, \text{ hence } y_1+y_2 \in W$$
$$(\lambda y_1)''+\omega^2(\lambda y_1) = \lambda(y_1''+\omega^2 y_1) = \lambda 0 = 0, \text{ hence } \lambda y_1 \in W\,.$$

We conclude that W is a vector subspace of $\mathscr{F}(\mathbb{R})$. □

Dimension of subspaces. Since a subspace of vector space is itself a vector space, it also has a dimension. It is natural to wonder whether the dimension behaves well with respect to subspaces: For instance, is the dimension of a subspace at most the dimension of the original vector space? While this is intuitively obvious, it requires a proof. And indeed, things go extremely well:

Theorem 3.37. Let V be a finite-dimensional vector space, and let W be a subspace. Then

$$\dim W \leq \dim V.$$

Moreover, we have

$$\dim W = \dim V \iff W = V.$$

This result can provide a simple way to show the equality between two subspaces: Instead of showing both inclusions, it is only necessary to show one inclusion and the equality of dimensions, something that is generally easier to handle. We will see applications in the next chapter.
The previous theorem relies on the following results:

Proposition 3.38. Let V be an n-dimensional vector space and let $\vec{v}_1,\dots,\vec{v}_k$ be a linearly independent family of vectors. Then $\vec{v}_1,\dots,\vec{v}_k$ can be extended to a basis $\vec{v}_1,\dots,\vec{v}_n$ of V

Proposition 3.39. Suppose that a family $\vec{v}_1,\dots,\vec{v}_k$ of vectors spans a (finite-dimensional) vector space V. Then there exists a subfamily that is a basis for V.

which have been proven in one of tutorials.

Building on Proposition 3.39, there is a simple way to determine the dimension of the subspace spanned by $\vec{v}_1,\dots,\vec{v}_k$, and to find a basis for it. However, the proof of this result requires tools that will be introduced in the next chapter, so we postpone its proof for now.
We introduce the following definition which characterises the dimension of subspaces.

Definition 3.40. The *rank* of a family of vectors $\vec{v}_1,\dots,\vec{v}_k$ is the dimension of the subspace they span.

Method: Finding the dimension and a basis of the span of a family of vectors of $\mathbb{R}^n$.

We can find a basis for $\text{span}(\vec{v}_1,\dots,\vec{v}_k)$ as follows:

(1) Write down a matrix whose ith column is $\vec{v}_i$.

(2) Perform elementary row operations to bring the matrix into row echelon form.

(3) The rank of $(\vec{v}_1,\dots,\vec{v}_k)$ is the number of pivot variables in the row echelon form. If we denote by $j_1,\dots,j_k$ the columns of the row echelon form that contain a pivot, then a basis

for $\text{span}(\vec{v}_1,\ldots,\vec{v}_k)$ is given by $\vec{v}_{j_1},\ldots,\vec{v}_{j_k}$.

Example 3.41. We wish to find a basis for the subspace of $\mathbb{R}^4$ spanned by $(1,2,3,0)^T$, $(2,1,2,1)^T$, $(1,5,7,-1)^T$, $(0,0,1,2)^T$. We apply the previous algorithm:

$$\begin{pmatrix} 1 & 2 & 1 & 0 \\ 2 & 1 & 5 & 0 \\ 3 & 2 & 7 & 1 \\ 0 & 1 & -1 & 2 \end{pmatrix} \overset{\substack{R_2\to R_2-2R_1\\ R_3\to R_3-3R_1}}{\rightsquigarrow} \begin{pmatrix} 1 & 2 & 1 & 0 \\ 0 & -3 & 3 & -5 \\ 0 & -4 & 4 & 1 \\ 0 & 1 & -1 & 2 \end{pmatrix} \overset{\substack{R_3\to 3R_3-4R_2\\ R_4\to 3R_4+R_2}}{\rightsquigarrow}$$

$$\begin{pmatrix} 1 & 2 & -1 & 3 \\ 0 & -5 & 4 & -5 \\ 0 & 0 & 0 & 23 \\ 0 & 0 & 0 & 1 \end{pmatrix} \overset{R_4\to R_3-23R_4}{\rightsquigarrow} \begin{pmatrix} 1 & 2 & -1 & 3 \\ 0 & -5 & 4 & -5 \\ 0 & 0 & 0 & 23 \\ 0 & 0 & 0 & 0 \end{pmatrix}.$$

There are three pivot variables, corresponding to columns 1, 2, and 4, so the rank of this family of vectors (or the dimension of the subspace spanned by this family of vectors) is 3.

A basis for its span is given by $\{(1,2,3,0)^T,(2,1,2,1)^T,(0,0,1,2)^T\}$, which is a subfamily of the original family of vectors.

Chapter 4

Linear maps

We introduce a central concept in this chapter, that of *linear maps*. These are the maps from between vector spaces that 'behave well' with respect to the various operations available on vectors, namely addition and multiplication by a scalar. We have already seen one important example: Given an n-dimensional vector space, the coordinatisation map

$$T : V \to \mathbb{R}^n, \quad \vec{x} \mapsto [\vec{x}]_{\mathscr{B}}.$$

In this chapter, we study such maps in greater generality.

4.1 Linear maps from $\mathbb{R}^n$ to $\mathbb{R}^m$.

We start by looking at maps from $\mathbb{R}^n$ to $\mathbb{R}^m$.

Definition 4.1. A *linear map* (or *linear transformation*) from $\mathbb{R}^n$ to $\mathbb{R}^m$ is a map $T : \mathbb{R}^n \to \mathbb{R}^m$ such that:

(1) $T(\vec{u}_1 + \vec{u}_2) = T(\vec{u}_1) + T(\vec{u}_2)$ for all $\vec{u}_1, \vec{u}_2 \in \mathbb{R}^n$.

(2) $T(\lambda \vec{u}) = \lambda T(\vec{u})$ for all $\lambda \in \mathbb{R}$ and $\vec{u} \in \mathbb{R}^n$.

Example 4.2. (i) The map $T : \mathbb{R} \to \mathbb{R}, x \mapsto 2x$ is linear.

(ii) The map $T : \mathbb{R} \to \mathbb{R}, x \mapsto x^2$ is not linear. For instance, we have $T(1+1) = 2^2 = 4$ but $T(1) + T(1) = 2$.

(iii) The map $T : \mathbb{R} \to \mathbb{R}, x \mapsto x+1$ is not linear. For instance, we have $T(2 \times 1) = 3$ but $2 \times T(1) = 4$.

Elementary properties of linear maps. Here are a couple of facts about linear maps that can be useful in practice:

Proposition 4.3. Let $T : \mathbb{R}^n \to \mathbb{R}^m$ be a linear map. We have:

(1) *Linear maps preserve linear combinations:* If $\vec{u}_1, \ldots, \vec{u}_n \in \mathbb{R}^n$, $c_1, \ldots, c_n \in \mathbb{R}$, then

$$T(c_1\vec{u}_1 + c_2\vec{u}_2 + \ldots + c_n\vec{u}_n) = c_1 T(\vec{u}_1) + c_2 T(\vec{u}_2) + \ldots + c_n T(\vec{u}_n).$$

(2) *Linear maps preserve zero vectors:* $T(\vec{0}) = \vec{0}$.

Proof. (1) We iterate: $T(c_1\vec{u}_1 + c_2\vec{u}_2 + \ldots + c_n\vec{u}_n) = T(c_1\vec{u}_1) + T(c_2\vec{u}_2 + \ldots + c_n\vec{u}_n) = c_1T(\vec{u}_1) + T(c_2\vec{u}_2 + \ldots + c_n\vec{u}_n)$, and we reason by induction.
(2) $T(\vec{0}) = T(0\cdot\vec{0}) = 0\cdot T(\vec{0}) = \vec{0}$ by linearity of the map and the axioms of vector spaces. □

The matrix associated to a linear map.

Proposition 4.4. Let $\vec{e}_1, \ldots, \vec{e}_n$ be a basis of $\mathbb{R}^n$. A linear map $T : \mathbb{R}^n \to \mathbb{R}^m$ is completely determined by the images $T(\vec{e}_1), \ldots, T(\vec{e}_n)$.

Proof. Let $\vec{x} \in \mathbb{R}^n$ and let $\lambda_1, \ldots, \lambda_n \in \mathbb{R}$ such that

$$\vec{x} = \lambda_1\vec{e}_1 + \ldots + \lambda_n\vec{e}_n.$$

Then the linearity of T implies that

$$T(\vec{x}) = T(\lambda_1\vec{e}_1 + \ldots + \lambda_n\vec{e}_n) = \lambda_1T(\vec{e}_1) + \ldots + \lambda_nT(\vec{e}_n).$$

□

Note that the situation is completely different for non-linear maps from $\mathbb{R}$ to $\mathbb{R}$, where no finite collection of values is ever enough to completely characterise a function.

Example 4.5. Let $T : \mathbb{R}^2 \to \mathbb{R}^3$ be a linear map such that $T((1,1)^T) = (4,3,5)^T$ and $T((0,1)^T) = (1,2,0)^T$. Let us compute $T((5,3)^T)$. Note that $(1,1)^T$ and $(0,1)^T$ form a basis of $\mathbb{R}^2$ (Why?). Because $(5,3)^T = 5(1,1)^T - 2(0,1)^T$, we have by linearity: $T((5,3)^T) = 5T((1,1)^T) - 2T((0,1)^T) = (18,11,25)^T$.

We have already seen many examples of linear maps in previous chapters. Indeed, given an $n \times m$ matrix A, matrix multiplication defines a linear map

$$T_A : \mathbb{R}^n \to \mathbb{R}^m, \vec{x} \mapsto A\vec{x}.$$

As it turns out, *all* linear maps from $\mathbb{R}^n$ to $\mathbb{R}^m$ come from matrices.

Theorem 4.6. For every linear map $T : \mathbb{R}^n \to \mathbb{R}^m$, there is a unique $m \times n$ matrix A such that for every vector $\vec{v} \in \mathbb{R}^n$:

$$T(\vec{v}) = A\vec{v}.$$

The matrix A is given in column form by

$$A = \left(\begin{array}{c|c|c} T(\vec{e}_1) & \cdots & T(\vec{e}_n) \end{array} \right)$$

where $\vec{e}_1, \ldots, \vec{e}_n$ denotes the standard basis of $\mathbb{R}^n$.

Proof. For a vector $\vec{x} = (x_1, \ldots, x_n)^T$, we have

$$\vec{x} = x_1 \begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix} + \ldots + x_n \begin{pmatrix} 0 \\ \vdots \\ 0 \\ 1 \end{pmatrix},$$

and it follows from the linearity of T that

$$T(\vec{x}) = x_1 \times T\begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix} + \ldots + x_n \times T\begin{pmatrix} 0 \\ \vdots \\ 0 \\ 1 \end{pmatrix} = \underbrace{\left(T\begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix} \middle| T\begin{pmatrix} 0 \\ 1 \\ \vdots \\ 0 \end{pmatrix} \middle| \cdots \middle| T\begin{pmatrix} 0 \\ \vdots \\ 0 \\ 1 \end{pmatrix} \right)}_{=:A} \vec{x}.$$

□

Definition 4.7. With the notations from the previous theorem, the matrix A is the *matrix associated to* T and is also denoted $[T]$. Reciprocally, an $m \times n$ matrix A can be thought of as a linear map $\mathbb{R}^n \to \mathbb{R}^m$, and we will denote by T_A this associated linear map.

Example 4.8.(i) The *zero map* $0_n : \mathbb{R}^n \to \mathbb{R}^n, \vec{x} \mapsto \vec{0}$ is linear, and its associated matrix is the zero matrix

$$0_n := \begin{pmatrix} 0 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & 0 \end{pmatrix}.$$

(ii) The *identity map* $I_n : \mathbb{R}^n \to \mathbb{R}^n, \vec{x} \mapsto \vec{x}$ is linear, and its associated matrix is the identity matrix

$$I_n := \begin{pmatrix} 1 & & \\ & \ddots & \\ & & 1 \end{pmatrix}.$$

Here is a more interesting example:

Example 4.9. Let $r_\theta : \mathbb{R}^2 \to \mathbb{R}^2$ be the rotation around the origin by an oriented angle θ. For a vector $\vec{v} = (v_1, v_2)^T \in \mathbb{R}^2$, we have

$$r_\theta\left(\begin{pmatrix} v_1 \\ v_2 \end{pmatrix}\right) = \begin{pmatrix} v_1\cos(\theta) - v_2\sin(\theta) \\ v_1\sin(\theta) + v_2\cos(\theta) \end{pmatrix}.$$

(Check it!) In particular, we have $r_\theta(\vec{v}) = A_\theta \vec{v}$, where A_θ is the 2×2 matrix

$$A_\theta := \begin{pmatrix} \cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{pmatrix}.$$

It follows that r_θ is a linear map.

In other words, a linear map $\mathbb{R}^n \to \mathbb{R}^m$ is 'the same thing' as an $m \times n$ matrix. This can be made a bit more precise. Let $\mathscr{L}(\mathbb{R}^n, \mathbb{R}^m)$ denote the set of all linear maps from $\mathbb{R}^n$ to $\mathbb{R}^m$. The map

$$M_{m,n}(\mathbb{R}) \to \mathscr{L}(\mathbb{R}^n, \mathbb{R}^m), \quad A \mapsto T_A$$

is a bijection, and its inverse is the bijection

$$\mathscr{L}(\mathbb{R}^n, \mathbb{R}^m) \to M_{m,n}(\mathbb{R}), \quad T \mapsto [T].$$

4.2 Operations on linear maps and their associated matrices.

As we will see, there are various operations that one can define on linear maps, to construct new linear maps out of existing ones. Since a linear map is 'essentially the same as' a matrix, we will also see the corresponding operations on matrices.
One can obtain new linear maps by combining linear maps, as explained below.

Proposition 4.10. Let $T_1, T_2 : \mathbb{R}^n \to \mathbb{R}^m$ be two linear maps and let $\lambda \in \mathbb{R}$. Then the following maps are also linear maps:

(1) *addition:* $(T_1 + T_2)$, where $(T_1 + T_2)(\vec{u}) := T_1(\vec{u}) + T_2(\vec{u})$ for all $\vec{u} \in \mathbb{R}^n$.

(2) *scalar multiplication:* λT, where $(\lambda T)(\vec{u}) := \lambda T(\vec{u})$ for all $\vec{u} \in \mathbb{R}^n$.

Let $T : \mathbb{R}^n \to \mathbb{R}^m$ and $S : \mathbb{R}^m \to \mathbb{R}^p$ be two linear maps. Then the following is also a linear map:

(3) *composition:* $S \circ T : \mathbb{R}^n \to \mathbb{R}^p$, where $(S \circ T)(\vec{u}) := S(T(\vec{u}))$ for all $\vec{u} \in \mathbb{R}^n$.

Proof. (1) and (2): Exercise.
(3) Let $\vec{u}_1, \vec{u}_2 \in \mathbb{R}^n$ and $\lambda \in \mathbb{R}$. We have:
$(S \circ T)(\vec{u}_1 + \vec{u}_2) = S(T(\vec{u}_1 + \vec{u}_2)) = S(T(\vec{u}_1) + T(\vec{u}_2)) = S(T(\vec{u}_1)) + S(T(\vec{u}_2)) = (S \circ T)(\vec{u}_1) + (S \circ T)(\vec{u}_2)$.
$(S \circ T)(\lambda \vec{u}_1) = S(T(\lambda \vec{u}_1)) = S(\lambda T(\vec{u}_1)) = \lambda S(T(\vec{u}_1)) = \lambda (S \circ T)(\vec{u}_1)$. □

Warning! The notation $S \circ T(\vec{x})$ can be source of mistakes when it comes to doing concrete computations, as you are supposed to read *from right to left*: start from the vector $\vec{x}$, then apply T to get $T(\vec{x})$, then apply S to this new vector to get $S(T(\vec{x}))$.

Definition 4.11. Let $A = (a_{ij})$ and $B = (b_{ij})$ be two $m \times n$ matrices, and let $\lambda \in \mathbb{R}$. We define the following matrices:

(1) The matrix $A + B$ is defined as the $m \times n$ matrix whose (i, j)-entry is $a_{ij} + b_{ij}$.

(2) The matrix λA is defined as the $m \times n$ matrix whose (i, j)-entry is λa_{ij}.

Let $A = (a_{ij})$ be an $p \times m$ matrix, and let $B = (b_{ij})$ be an $m \times n$ matrix.

(3) The matrix $A \times B$ is defined as the $p \times n$ matrix whose (i, j)-entry is:

$$(AB)_{ij} = \sum_{k=1}^{m} a_{ik} b_{kj}.$$

The first two definitions are rather intuitive: The operations are performed entry by entry in the natural way. By contrast, the definition of the product of two matrices may seem perplexing at first. However, this definition is justified by the following:

Theorem 4.12. Let $T, S : \mathbb{R}^n \to \mathbb{R}^m$ be two linear maps, and let $\lambda \in \mathbb{R}$. The associated matrices satisfy the following:

(1) $[T + S] = [T] + [S]$.

(2) $[\lambda T] = \lambda [T]$.

Let $T : \mathbb{R}^n \to \mathbb{R}^m$ be a linear map and let $S : \mathbb{R}^m \to \mathbb{R}^p$ be a linear map. The associated matrices satisfy the following:

(3) $[S \circ T] = [S] \times [T]$.

Basic arithmetic of matrices. With the operations defined above, matrices enjoy many of the 'arithmetic rules' you have seen for numbers:

Proposition 4.13. Whenever the product is well-defined, we have

$$(AB)C = A(BC).$$

Moreover, we have the standard 'rules of matrix arithmetic'. Whenever the operations are well-defined, we have:

- $A + B = B + A$,
- $A + (B + C) = (A + B) + C$
- $A(B + C) = AB + AC$,
- $(A + B)C = AC + BC$,
- $\lambda(AB) = (\lambda A)B = A(\lambda B)$.

However, here is an important difference. As a consequence of the definition of the product of two matrices, we have that $AB \neq BA$ in general. For instance, for the matrices

$$A = \begin{pmatrix} 0 & 1 \\ 2 & 3 \end{pmatrix} \text{ and } B = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix},$$

we find

$$AB = \begin{pmatrix} 3 & 4 \\ 11 & 16 \end{pmatrix} \text{ and } BA = \begin{pmatrix} 4 & 6 \\ 8 & 15 \end{pmatrix}.$$

In particular, the standard formulas for computing $(A+B)^2$, $(A-B)(A+B)$, etc. do not hold in general for matrices (but do hold if we have $AB = BA$).

The vector space of matrices (and linear maps). We have just define a notion of addition and multiplication by a scalar for matrices. These operations were exactly what is needed when defining an abstract vector space, and indeed, we have:

Proposition 4.14. The set $M_{m,n}(\mathbb{R})$ of $m \times n$ real matrices is a vector space for the addition of matrices and multiplication by a scalar. It has dimension $m \times n$, and a basis of it is given by

$$\{E^{i,j}\}_{1 \leq i \leq m, 1 \leq j \leq n},$$

where $E^{i,j}$ is the $m \times n$ matrix with 1 in its (i, j)-entry and zeros everywhere else.

4.3 A second look at the determinant

We have introduced the determinant of a matrix $A \in M_n(\mathbb{R})$, which could be thought of as the way volumes are stretched when going from the unit cube $\vec{e}_1, \ldots, \vec{e}_n$ to the parallelepiped $A\vec{e}_1, \ldots, A\vec{e}_n$.

Product of matrices and determinant. The geometric intuition we have of the determinant should make the following result intuitively plausible:

Theorem 4.15. Let A, B be two $n \times n$ matrices. Then

$$\det(AB) = \det(A) \times \det(B).$$

Proof. Evaluating Proposition 2.30 at $\vec{x}_1 = B\vec{e}_1, \ldots, \vec{x}_n = B\vec{e}_n$, we get:

$$\det(AB) = \mathrm{vol}_n(AB\vec{e}_1, \ldots, AB\vec{e}_n) = \det(A) \times \mathrm{vol}_n(B\vec{e}_1, \ldots, B\vec{e}_n) = \det(A) \times \det(B) \times \underbrace{\mathrm{vol}_n(\vec{e}_1, \ldots, \vec{e}_n)}_{=1}.$$

□

Using row operations to simplify determinants. In order to simplify computations, one may first perform row operations on the matrix to make many zeros appear. In order to do this, we need to understand the effect of row operations on the determinant.

Proposition 4.16. Let A be an $n \times n$ matrix.

- Let B be the matrix obtained by switching two distinct rows $R_i \leftrightarrow R_j$. Then we have
$$\det B = -\det A.$$
- Let B be the matrix obtained by performing the row operation $R_i \rightsquigarrow \alpha R_i + \beta R_j$. Then we have
$$\det B = \alpha \det A.$$

In practice, when computing a $n \times n$ determinant for $n \geq 4$, it might be useful to first perform some row operations to make several zeros appear (as with usual Gaussian elimination), before developing along a column or a row.

4.4 Inverses of matrices.

Definition 4.17. A matrix $A \in M_n(\mathbb{R})$ is called *invertible* if for every $\vec{y} \in \mathbb{R}^n$, the system of linear equations

$$A\vec{x} = \vec{y}$$

admits a unique solution $\vec{x} \in \mathbb{R}^n$.
In other words, $A \in M_n(\mathbb{R})$ is invertible if and only if the associated linear map $T_A : \mathbb{R}^n \to \mathbb{R}^n$ is a bijection.

Since we understand exactly when a given system has a unique solution, we have the following characterisation:

Theorem 4.18. Let $A \in M_n(\mathbb{R})$. The following are equivalent:

- A is invertible.
- When A is reduced to its row echelon form, every variable is a pivot variable.

- When A is reduced to a row echelon form, the matrix we get is triangular with non-zero diagonal entries.

In particular, we get the following useful criterion to determine whether a given square matrix is invertible.

Theorem 4.19. We have

$$A \text{ is invertible } \Leftrightarrow \det(A) \neq 0.$$

Here is a 'matrix version' of the definition of an invertible matrix:

Definition 4.20. An $n \times n$ matrix A is invertible if and only if there exists an $n \times n$ matrix A^{-1}, called the *inverse* of A, such that

$$AA^{-1} = A^{-1}A = I_n.$$

Note that once we know the inverse of a given matrix A, we can easily solve any system of the form $A\vec{x} = y$:

Theorem 4.21. Let A be an invertible $n \times n$ matrix, and let $\vec{x}, \vec{y}$ be vectors of $\mathbb{R}^n$. Then:

$$A\vec{x} = \vec{y} \Longleftrightarrow \vec{x} = A^{-1}\vec{y}.$$

Proof. We have, by associativity of the the matrix products:

$$A\vec{x} = \vec{y} \Rightarrow A^{-1}(A\vec{x}) = A^{-1}\vec{y} \Rightarrow (A^{-1}A)\vec{x} = A^{-1}\vec{y} \Rightarrow \vec{x} = A^{-1}\vec{y}.$$

The other implication is completely analogous. □

We already know how to solve the system of linear equations $A\vec{x} = \vec{y}$ (with variable $\vec{x}$) by using Gaussian elimination. However, in the case of an invertible system (that is, when A is an invertible square matrix), computing A^{-1} can be more practical when one wants to solve similar systems of the form $A\vec{x} = \vec{y}_i$ for various vectors $\vec{y}_1, \ldots, \vec{y}_k$. Indeed, instead of solving the k systems by Gaussian elimination, one may just compute once the inverse A^{-1}, and then compute the various solutions using matrix products, since $\vec{x}_i = A^{-1}\vec{y}_i$. We thus present an algorithm to compute the inverse of a matrix, based on Gaussian elimination. Given an invertible $n \times n$ matrix A, we write the inverse of A in column form as

$$A^{-1} = (\vec{c}_1|\vec{c}_2|\cdots|\vec{c}_n)\,.$$

It follows that $A(\vec{c}_1|\vec{c}_2|\cdots|\vec{c}_n) = I_n = (\vec{e}_1|\vec{e}_2|\ldots|\vec{e}_n)$, and thus $\vec{c}_1, \ldots, \vec{c}_n$ satisfy $A\vec{c}_1 = \vec{e}_1$, $A\vec{c}_2 = \vec{e}_2$, …, $A\vec{c}_n = \vec{e}_n$. We thus have to solve the systems of linear equations $A\vec{x}_1 = \vec{e}_1$, $A\vec{x}_2 = \vec{e}_2$, …, $A\vec{x}_n = \vec{e}_n$. We solve all these systems at once, using the following method:

Method: Computing the inverse of a matrix

Let A be an $n \times n$ matrix A that we want to invert (if possible).

1) Consider the augmented matrix

$$\left(\begin{array}{c|ccc} & \vec{e}_1 & \cdots & \vec{e}_n \end{array}\right) = \left(\begin{array}{c|ccc} & 1 & 0 & 0 \\ A & 0 & \ddots & 0 \\ & 0 & 0 & 1 \end{array}\right).$$

2) Use elementary row operations so that the left-hand side is in echelon form. If the left

hand side contains free variable, for instance

$$\left(\begin{array}{ccc|ccc} 1 & \times & \times & \times & \times & \times \\ 0 & 0 & 3 & \times & \times & \times \\ 0 & 0 & 0 & \times & \times & \times \end{array}\right),$$

then A is not invertible.

3) Otherwise, the left hand side has a row echelon form with non-zero entries on its diagonal, and we can perform further row operations to reduce the augmented matrix to the following form:

$$\left(\begin{array}{ccc|c} \vec{e}_1 & \cdots & \vec{e}_n & B \end{array}\right) = \left(\begin{array}{ccc|c} 1 & 0 & 0 & \\ 0 & \ddots & 0 & B \\ 0 & 0 & 1 & \end{array}\right).$$

It follows that $A^{-1} = B$.

Example 4.22. We want to invert the following matrix:

$$A = \begin{pmatrix} 1 & 2 & 4 \\ -1 & -1 & 5 \\ 2 & 1 & -3 \end{pmatrix}.$$

We apply the previous algorithm:

$$\left(\begin{array}{ccc|ccc} 1 & 2 & 4 & 1 & 0 & 0 \\ -1 & -1 & 5 & 0 & 1 & 0 \\ 2 & 1 & -3 & 0 & 0 & 1 \end{array}\right) \overset{\substack{R_2\to R_2+R_1 \\ R_3\to R_3-2R_1}}{\rightsquigarrow} \left(\begin{array}{ccc|ccc} 1 & 2 & 4 & 1 & 0 & 0 \\ 0 & 1 & 9 & 1 & 1 & 0 \\ 0 & -3 & -11 & -2 & 0 & 1 \end{array}\right)$$

$$\overset{\substack{R_1\to R_1-2R_2 \\ R_3\to R_3+3R_2}}{\rightsquigarrow} \left(\begin{array}{ccc|ccc} 1 & 0 & -14 & -1 & -2 & 0 \\ 0 & 1 & 9 & 1 & 1 & 0 \\ 0 & 0 & 16 & 1 & 3 & 1 \end{array}\right) \overset{\substack{R_1\to R_1+\frac{14}{16}R_3 \\ R_2\to R_2-\frac{9}{16}R_3}}{\rightsquigarrow} \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & -\frac{2}{16} & \frac{10}{16} & \frac{14}{16} \\ 0 & 1 & 0 & \frac{7}{16} & -\frac{11}{16} & -\frac{9}{16} \\ 0 & 0 & 16 & 1 & 3 & 1 \end{array}\right)$$

$$\overset{R_3\to\frac{1}{16}R_3}{\rightsquigarrow} \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & -\frac{1}{8} & \frac{5}{8} & \frac{7}{8} \\ 0 & 1 & 0 & \frac{7}{16} & -\frac{11}{16} & -\frac{9}{16} \\ 0 & 0 & 1 & \frac{1}{16} & \frac{3}{16} & \frac{1}{16} \end{array}\right), \quad \text{thus} \quad A^{-1} = \frac{1}{16}\begin{pmatrix} -2 & 10 & 14 \\ 7 & -11 & -9 \\ 1 & 3 & 1 \end{pmatrix}$$

For instance, the unique solution of the equation $A\vec{x} = \begin{pmatrix} 16 \\ 0 \\ 0 \end{pmatrix}$ is $\vec{x} = A^{-1}\begin{pmatrix} 16 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} -2 \\ 7 \\ 1 \end{pmatrix}$, and the unique solution of the equation $A\vec{x} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$ is $\vec{x} = A^{-1}\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} = \frac{1}{16}\begin{pmatrix} 22 \\ -13 \\ 5 \end{pmatrix}$.

Note that we have the following:

Proposition 4.23. For every invertible matrix A, we have

$$\det(A^{-1}) = \frac{1}{\det(A)}.$$

Proof. By Theorem 4.15: $\det(A) \times \det(A^{-1}) = \det(A \times A^{-1}) = \det(I_n) = 1$. □

4.5 Linear maps between general vector spaces

Definition 4.24. Let U and V be vectors spaces. A *linear map* (or *linear transformation*) from U to V is a map $T : U \to V$ preserving vector space operations. That is:

(1) $T(\vec{u}_1 + \vec{u}_2) = T(\vec{u}_1) + T(\vec{u}_2)$ for all $\vec{u}_1, \vec{u}_2 \in U$.

(2) $T(\lambda \vec{u}) = \lambda T(\vec{u})$ for all $\lambda \in \mathbb{R}$ and $\vec{u} \in U$.

Note that by induction, we automatically get that for $\vec{u}_1, \ldots, \vec{u}_k \in U$ and $\lambda_1, \ldots, \lambda_k \in \mathbb{R}$:

$$T(\lambda_1 \vec{u}_1 + \cdots + \lambda_k \vec{u}_k) = \lambda_1 T(\vec{u}_1) + \cdots + \lambda_k T(\vec{u}_k).$$

We also get:

$$T(\vec{0}) = \vec{0}.$$

Example 4.25.(i) Let $\mathscr{C}^\infty(\mathbb{R})$ be the vector space of smooth functions on $\mathbb{R}$. (Note that this is a subspace of $\mathscr{F}(\mathbb{R})$.) Then the map $D : f \mapsto f'$ is a linear map from $\mathscr{C}^\infty(\mathbb{R})$ to itself. Indeed, we have that for functions $f, g \in \mathscr{C}^\infty(\mathbb{R})$ and scalars $\lambda \in \mathbb{R}$:

$$D(f+g) = (f+g)' = f' + g' = D(f) + D(g),$$

$$D(\lambda f) = (\lambda f)' = \lambda f' = \lambda D(f).$$

(ii) Let $\mathscr{C}(\mathbb{R})$ be the vector space of continuous functions on $\mathbb{R}$. (Note that this is a subspace of $\mathscr{F}(\mathbb{R})$.) Then the map $I : \mathscr{C}(\mathbb{R}) \to \mathbb{R}, f \mapsto \int_0^1 f(s)\mathrm{d}s$ is a linear map from $\mathscr{C}(\mathbb{R})$ to $\mathbb{R}$. Indeed, we have that for functions $f, g \in \mathscr{C}(\mathbb{R})$ and scalars $\lambda \in \mathbb{R}$:

$$I(f+g) = \int_0^1 (f+g)(s)ds = \int_0^1 f(s)ds + \int_0^1 g(s)ds = I(f) + I(g),$$

$$I(\lambda f) = \int_0^1 \lambda f(s)ds = \lambda \int_0^1 f(s)ds = \lambda I(f).$$

Isomorphisms.

Definition 4.26. A linear map $T : U \to V$ is called an *isomorphism* if it is bijective. If such an isomorphism exists, we say that U is *isomorphic* to V, or that U and V are *isomorphic*.

Intuitively, two isomorphic vector spaces ought to be considered as 'essentially the same', at least from the point of view of linear algebra.

Theorem 4.27. Let V be a real vector space of dimension n and let $\mathscr{B}$ be a basis of it. The coordinatisation map

$$T : V \to \mathbb{R}^n, \quad \vec{x} \mapsto [\vec{x}]_{\mathscr{B}}$$

is an isomorphism. In particular, a real vector space of dimension n is isomorphic to $\mathbb{R}^n$.

Example 4.28. The vector space $\mathscr{P}_2(\mathbb{R})$ is isomorphic to $\mathbb{R}^3$.

Here is another example of isomorphism, making precise what we meant earlier when saying that a linear map from $\mathbb{R}^n$ to $\mathbb{R}^m$ is 'essentially the same as an $m \times n$ matrix' (or vice versa).

Theorem 4.29. The map

$$\Phi : \mathscr{L}(\mathbb{R}^n, \mathbb{R}^m) \to M_{m,n}(\mathbb{R}), \quad T \mapsto [T],$$

that assigns to each linear map $\mathbb{R}^n \to \mathbb{R}^m$ its matrix, is an isomorphism.

Matrices associated to linear maps. As we have seen in the previous chapter, once we know a basis $\mathscr{B}$ of a finite-dimensional vector space, it is possible to 'identify V with $\mathbb{R}^n$', or to 'think in coordinates', by using the coordinatisation map

$$T : V \to \mathbb{R}^n, \quad \vec{x} \mapsto [\vec{x}]_{\mathscr{B}}.$$

Doing so can make computations simpler in certain cases. Similarly, when dealing with a linear map $T : U \to V$ between abstract vector spaces with bases $\mathscr{B}_U$ and $\mathscr{B}_V$ respectively, one may hope that thinking in coordinates may make things easier. Intuitively, if we identify U with $\mathbb{R}^n$ (via our choice of basis $\mathscr{B}_U$) and V with $\mathbb{R}^m$ (via our choice of basis $\mathscr{B}_V$), then T ought to be identified with a linear map from $\mathbb{R}^n$ to $\mathbb{R}^m$, which we already know is the same as an $m \times n$ matrix.
This is what we will do now. We will associate to each linear map a matrix. For simplicity, we will only consider linear maps $T : V \to V$ from a vector to itself in this course. The results however can be generalised to arbitrary linear maps.
This construction relies on the following crucial observation, which is the complete analogue of what happens for linear maps from $\mathbb{R}^n$ to $\mathbb{R}^m$.

Proposition 4.30. Let V be a vector space with basis $(\vec{e}_1, \ldots, \vec{e}_n)$. A linear map $T : V \to V$ is completely determined by the images of the basis vectors $T(\vec{e}_1), \ldots, T(\vec{e}_n)$.

Proof. Given a vector $\vec{v} \in U$, we can write it as a linear combination $\vec{v} = c_1\vec{e}_1 + \ldots + c_n\vec{e}_n$. The linear map $T(\vec{v})$ can be evaluated as $T(\vec{v}) = c_1 T(\vec{e}_1) + \ldots + c_n T(\vec{e}_n)$, and all we need are the images $T(\vec{e}_1), \ldots, T(\vec{e}_n)$. □

Definition 4.31. Let V be an n-dimensional vector space with basis $\mathscr{B} = (\vec{e}_1, \ldots, \vec{e}_n)$. Let $T : V \to V$ be a linear map, and let $\vec{x} \in V$ be a vector, which we write as a linear combination $\vec{x} = x_1\vec{e}_1 + \ldots + x_n\vec{e}_n$. Recall that we defined the matrix of $\vec{x}$ in the basis $\mathscr{B}$ to be the following vector of $\mathbb{R}^n$:

$$[\vec{x}]_{\mathscr{B}} := \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}.$$

The *matrix of T in the basis $\mathscr{B}$*, denoted $[T]_{\mathscr{B}}$, is the following $m \times n$ matrix:

$$[T]_{\mathscr{B}} := \left(\begin{array}{c|c|c|c} & & & \\ [T(\vec{e}_1)]_{\mathscr{B}} & [T(\vec{e}_2)]_{\mathscr{B}} & \cdots & [T(\vec{e}_n)]_{\mathscr{B}} \\ & & & \end{array} \right).$$

We have the following result. In a nutshell, it says that once we have chosen $\mathscr{B}$ of our vector space, we can perform computations in $\mathbb{R}^n$ instead:

Theorem 4.32. Let V be an n-dimensional vector space with basis $\mathscr{B}$. Let $T : V \to V$ be a linear map. Let $\vec{v}$ be a vector of V. Then:

$$[T(\vec{v})]_{\mathscr{B}} = [T]_{\mathscr{B}} [\vec{v}]_{\mathscr{B}}.$$

Example 4.33. Let us consider the map $T : \mathscr{P}_2(\mathbb{R}) \to \mathscr{P}_2(\mathbb{R})$ that sends a polynomial function $P(x)$ to the polynomial function $P(x+1)$. (You should check that this is indeed a linear map!) We choose basis $\mathscr{B} := (1, x, x^2)$ for the vector space $\mathscr{P}_2(\mathbb{R})$. We compute:

$$T(1) = 1 = 1 + 0.x + 0.x^2, \text{ hence } [T(1)]_{\mathscr{B}} = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix},$$

$$T(x) = x + 1 = 1 + 1.x + 0.x^2, \text{ hence } [T(x)]_{\mathscr{B}} = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix},$$

$$T(x^2) = (x+1)^2 = x^2 + 2x + 1 = 1 + 2.x + 1.x^2, \text{ hence } \left[T(x^2)\right]_{\mathscr{B}} = \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}.$$

We thus have

$$[T]_{\mathscr{B}} = \begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{pmatrix}.$$

Let us now use the previous result to compute $T(x^2 - x)$. We compute

$$\left[x^2 - x\right]_{\mathscr{B}} = \begin{pmatrix} 0 \\ -1 \\ 1 \end{pmatrix},$$

hence

$$\left[T(x^2 - x)\right]_{\mathscr{B}} = \begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 0 \\ -1 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}.$$

We thus find $T(x^2 - x) = x^2 + x$. We can check this computation directly:

$$T(x^2 - x) = (x+1)^2 - (x+1) = x^2 + x.$$

Operations on linear maps and matrices

Proposition 4.34. Let $T, S : U \to U$ be two linear maps and let $\lambda \in \mathbb{R}$. Then the following maps are also linear maps:

(1) *addition:* $(T+S)$, where $(T+S)(\vec{u}) := T(\vec{u}) + S(\vec{u})$ for all $\vec{u} \in U$.

(2) *scalar multiplication:* λT, where $(\lambda T)(\vec{u}) := \lambda T(\vec{u})$ for all $\vec{u} \in Un$.

(3) *composition:* $S \circ T : U \to U$, where $(S \circ T)(\vec{u}) := S(T(\vec{u}))$ for all $\vec{u} \in U$.

Proposition 4.35. Let $T, S : U \to U$ be two linear maps, let $\mathscr{B}$ be a basis of U, and let $\lambda \in \mathbb{R}$. The associated matrices satisfy the following:

(1) $[T+S]_{\mathscr{B}} = [T]_{\mathscr{B}} + [S]_{\mathscr{B}}$.

(2) $[\lambda T]_{\mathscr{B}} = \lambda [T]_{\mathscr{B}}$,

(3) $[S \circ T]_{\mathscr{B}} = [S]_{\mathscr{B}} \times [T]_{\mathscr{B}}$.

We can now make sense of the intuition we had at the beginning of this paragraph, namely that 'once we have chosen a basis of V, a linear map from V to V is essentially the same thing as a matrix':

Theorem 4.36. Let V be an n-dimensional vector space, and let $\mathscr{B}$ be a basis of V. Then the map

$$\Phi : \mathscr{L}(V,V) \to M_n(\mathbb{R}), \quad T \mapsto [T]_{\mathscr{B}},$$

that associated to a linear map from V to itself its matrix in the basis $\mathscr{B}$, is an isomorphism.

4.6 Change of basis

In $\mathbb{R}^2$, let us consider the orthogonal reflection s_θ across the line making an oriented angle θ with the horizontal axis. One can check that this a linear map, and the matrix associated to s_θ is the following:

$$[s_\theta] = \begin{pmatrix} \cos(2\theta) & \sin(2\theta) \\ \sin(2\theta) & -\cos(2\theta) \end{pmatrix}.$$

This matrix is the matrix of the linear map $s_\theta : \mathbb{R}^2 \to \mathbb{R}^2$ computed in the standard basis $\mathscr{B} = ((0,1)^T, (1,0)^T)$. However, in certain circumstances, it may be easier to work in another basis. For instance, let us consider the basis $\mathscr{B}_\theta = ((\cos(\theta), \sin(\theta)^T, (-\sin(\theta), \cos(\theta))$ obtained by rotating $\mathscr{B}$ of an angle θ. Then we have

$$[s_\theta]_{\mathscr{B}_\theta} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

We see that the matrix in this new basis is considerably simpler. In Chapter 6, we will investigate the existence of such 'nice' bases in greater generality. For now, we would like to understand how to go from the matrix in one basis to the matrix in another basis.

Matrices of change of basis.

Definition 4.37. Suppose that we have two bases, $\mathscr{B} = (\vec{e}_1, \ldots, \vec{e}_n)$ and $\mathscr{B}' = (\vec{e}'_1, \ldots, \vec{e}'_n)$, of a vector space V. $\mathscr{B}$ should be thought of as the original basis, and we would like to find the matrix of a given linear map in the new basis $\mathscr{B}'$. We define the *matrix of change of basis*, denoted $[\mathscr{B}']_{\mathscr{B}}$,

as the $n \times n$ matrix defined by

$$[\mathscr{B}']_{\mathscr{B}} = \left([\vec{e}'_1]_{\mathscr{B}} \middle| \cdots \middle| [\vec{e}'_n]_{\mathscr{B}} \right).$$

In other words, we are writing elements of the new basis in terms of the old basis.

Remark. We have a priori two possible matrices of change of basis, namely $[\mathscr{B}']_{\mathscr{B}}$ and $[\mathscr{B}]_{\mathscr{B}'}$. These two matrices are related by the following equation:

$$[\mathscr{B}']_{\mathscr{B}} = [\mathscr{B}]^{-1}_{\mathscr{B}'}.$$

Theorem 4.38 (Change of basis for a vector). Let V be a finite-dimensional vector space, let $\vec{v} \in V$ be a vector. Consider two bases $\mathscr{B}$ and $\mathscr{B}'$ of V. Let $P := [\mathscr{B}']_{\mathscr{B}}$ be the matrix of change of bases. Then we have:

$$[\vec{v}]_{\mathscr{B}'} = P^{-1} \times [\vec{v}]_{\mathscr{B}}.$$

Theorem 4.39 (Change of basis for a matrix). Let V be a finite-dimensional vector space, let $T : V \to V$ be a linear map. Consider two bases $\mathscr{B}$ and $\mathscr{B}'$ of V. Let $P := [\mathscr{B}']_{\mathscr{B}}$ be the matrix of change of bases. Then we have:

$$[T]_{\mathscr{B}'} = P^{-1} \times [T]_{\mathscr{B}} \times P.$$

Warning! One has to be careful about the choice of matrix of change of basis, and the place of the inverse.

Example 4.40. Let us go back to the previous example. To simplify computations, we will focus on

$$s := s_{\pi/4},$$

whose matrix in the canonical basis $\mathscr{B}$ of $\mathbb{R}^2$ is

$$[s]_{\mathscr{B}} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

We consider the 'new basis' $\mathscr{B}' = \mathscr{B}_{\pi/4} = (\frac{1}{\sqrt{2}}(1,1)^T, \frac{1}{\sqrt{2}}(-1,1)^T)$. We have the following matrix of change of basis:

$$[\mathscr{B}']_{\mathscr{B}} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}.$$

Applying Gaussian elimination, we find the inverse of this matrix:

$$[\mathscr{B}']^{-1}_{\mathscr{B}} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix}.$$

Finally, we compute:

$$[s]_{\mathscr{B}'} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix} \times \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \times \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix},$$

which does give us the same result as in the introduction of this section.

Remark 4.41. Given an n-dimensional vector space U, we can define the determinant of a linear map $T : U \to U$ as:

$$\det(T) = \det([T]_{\mathscr{B}})$$

for any choice of basis $\mathscr{B}$ of U. Indeed, it follows from the Theorem 4.15 and Proposition 4.23 that this quantity does not depend on the choice of basis. This is compatible with our intuition that the determinant measures the way volumes are stretched, and this should not depend on the basis we choose to measure this stretching effect.

4.7 The Rank-Nullity Theorem

Image and rank of a matrix.

Definition 4.42. Given an $m \times n$ matrix written in column form as

$$A = \left(\begin{array}{c|c|c} \vec{c}_1 & \cdots & \vec{c}_n \end{array} \right),$$

the *image* of A, denoted $\mathrm{Im}(A)$, is the subspace of $\mathbb{R}^m$ spanned by $\vec{c}_1, \ldots, \vec{c}_n$. Its dimension is called the *rank* of A and denoted $\mathrm{rk}(A)$.

Example 4.43.

$$\begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \text{ has rank } 0 \qquad \begin{pmatrix} 1 & 1 & 1 \\ 1 & -1 & 1 \\ 0 & 0 & 0 \end{pmatrix} \text{ has rank } 2$$

$$\begin{pmatrix} 1 & 1 & 1 \\ 2 & 2 & 2 \\ 3 & 3 & 3 \end{pmatrix} \text{ has rank } 1 \qquad \begin{pmatrix} 1 & 2 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \text{ has rank } 3$$

Rank and systems of linear equations. Let us reinterpret some old results in term of the rank. There is nothing new in this paragraph (and you should convince yourself of it):

Proposition 4.44. Let A be an $n \times n$ matrix, Then:

$$A \text{ is invertible} \iff \mathrm{rk}(A) = n.$$

Proposition 4.45. Let A be an $m \times n$ matrix. Then the system of linear equations $A\vec{x} = \vec{b}$ is consistent for all $\vec{b} \in \mathbb{R}^m$ if and only if $\mathrm{rk}(A) = m$.

Proof. The statement that $A\vec{x} = \vec{b}$ is consistent for all $\vec{b} \in \mathbb{R}^m$ is equivalent to the fact that the image of A is $\mathbb{R}^m$ and thus that the (column) rank of A is maximal: $\mathrm{rk}(A) = m$. □

Remark 4.46. Let A be an $m \times n$-matrix. We have:

(i) $\mathrm{rk}(A) < n \Rightarrow A\vec{x} = \vec{0}$ has infinitely many solutions.

(ii) $\mathrm{rk}(A) = n \Rightarrow A\vec{x} = \vec{0}$ has only one solution $\vec{x} = \vec{0}$.

(iii) $\mathrm{rk}(A) > n$ is not possible, as the row space of A is a subspace of $\mathbb{R}^n$.

The Rank-Nullity Theorem for matrices.

Definition 4.47. Let A be an $m \times n$ matrix. The *kernel* (also called the *nullspace*) of A is the subspace of $\mathbb{R}^n$ consisting of the solutions of the system of linear equations

$$A\vec{x} = \vec{0}.$$

Its dimension is called the *nullity* of A.

Theorem 4.48. Let A be an $m \times n$ matrix. Then we have:

$$\mathrm{rk}(A) + \dim \ker(A) = n.$$

Proof. Performing Gaussian elimination, we obtain a matrix with exactly $\mathrm{rk}(A)$ pivots. In particular, $\mathrm{rk}(A)$ of the n variables are pivot variables, while the other $n - \mathrm{rk}(A)$ are free variables. But we know from previous methods that the rank of a matrix is the number of pivot variables in any row echelon form, while the nullity of a matrix is the number of free variables in any row echelon form. □

The Rank-Nullity Theorem for linear maps. We have seen that, up to a choice of bases, a linear map between finite-dimensional vector spaces is essentially a matrix. It should not be surprising then to expect an analogue of the rank-nullity theorem for general linear maps.

Definition 4.49. Let $T : U \to V$ be a linear map between two vector spaces. The *image* (or *range*) of T is the following set:

$$\mathrm{Im}(T) := \{T(\vec{u}) \mid \vec{u} \in U\}.$$

The *kernel* (or *nullspace*) of T is the following set:

$$\ker(T) := \{\vec{u} \in U \mid T(\vec{u}) = \vec{0}\}.$$

Proposition 4.50. Let $T : U \to V$ be a linear map between two vector spaces. The kernel of T is a subspace of U, and the image of T is a subspace of V.

Theorem 4.51. *(Rank-Nullity Theorem)* If $T : U \to V$ is a linear map between finite-dimensional vector spaces, then we have:

$$\dim \mathrm{Im}(T) + \dim \ker(T) = \dim U.$$

Here is a direct consequence, which provides a simple criterion to check that a given linear map is an isomorphism:

Theorem 4.52. Let $T : U \to V$ be a linear map between finite-dimensional vector spaces **of the same dimension**. Then

$$T \text{ is an isomorphism} \iff \ker(T) = \{\vec{0}\}.$$

This result is very powerful, and quite surprising at first. Indeed, if we unpack that statement, we are essentially saying that, under certain circumstances, showing that

$$\text{The equation } T(\vec{x}) = \vec{y} \text{ has } \textit{at least} \text{ one solution } \vec{x}$$

is equivalent to showing that

$$\text{The equation } T(\vec{x}) = \vec{y} \text{ has } \textit{at most} \text{ one solution } \vec{x},$$

and the second problem may be simpler to solve. In other words, our problem will have at least one solution if we can show that it cannot have too many solutions...!

Application: Polynomial Interpolation, revisited again. In tutorials, you saw that for every choice of pairwise distinct $x_0, \dots, x_n \in \mathbb{R}$ and every $y_0, \dots, y_n \in \mathbb{R}$, there exists a unique polynomial $P \in \mathscr{P}_n(\mathbb{R})$ such that

$$P(x_0) = y_0, \dots, P(x_n) = y_n.$$

We did this by constructing an appropriate basis of $\mathscr{P}_n(\mathbb{R})$ and constructing explicitly an interpolating polynomial. Here is a short (non-constructive) proof that only uses the Rank-Nullity Theorem.

Proposition 4.53. Let $x_0, \dots, x_n \in \mathbb{R}$ be a family of pairwise distinct real numbers. For every $y_0, \dots, y_n \in \mathbb{R}$, there exists a unique polynomial $P \in \mathscr{P}_n(\mathbb{R})$ such that

$$P(x_0) = y_0, \dots, P(x_n) = y_n.$$

Proof. Consider the map

$$T : \mathscr{P}_n(\mathbb{R}) \to \mathbb{R}^{n+1}, \quad P \mapsto \begin{pmatrix} P(x_0) \\ \vdots \\ P(x_n) \end{pmatrix}.$$

Then it is straightforward to check that T is linear (Do it!). The statement we want is equivalent to showing that T is an isomorphism. Moreover, both $\mathscr{P}_n(\mathbb{R})$ and $\mathbb{R}^{n+1}$ have dimension $n+1$. By Theorem 4.52, it is enough to show that $\ker(T) = \{\vec{0}\}$.
Let $P \in \ker(T)$. We thus have $P(x_0) = \cdots = P(x_n) = 0$. But the only real polynomial of degree $\leq n$ with $n+1$ roots is the zero polynomial. Thus, P is the zero polynomial, which implies that $\ker(T) = \{\vec{0}\}$, hence T is an isomorphism. □

Isn't that proof neat? Of course, one could just have turned the equations $P(x_0) = y_0, \dots, P(x_n) = y_n$ into a system of equations to solve (where the variables are the coefficients of P), but this would be cumbersome. By contrast, the previous proof only uses the geometric picture we have developed about vector spaces and linear maps between them to give a short proof that does not involve any computation.

Chapter 5

Inner product spaces

It happens very often in real life situations, when trying to solve a given problem by means of a system of linear equations, to have an *overdetermined* system, that is, a system with more equations than unknowns. In such a case, the system may be inconsistent. Still, one may wonder if there exists good 'approximations' of a solution. We will see how to make sense of such a problem in this chapter, using the geometric framework we have developed.

5.1 A motivating problem: Approximating statistical data

When trying to make sense of a given statistical data, a natural strategy is to approximate it by a simple function. For instance, consider the following data set (image taken from the Wikipedia article 'Polynomial regression'):

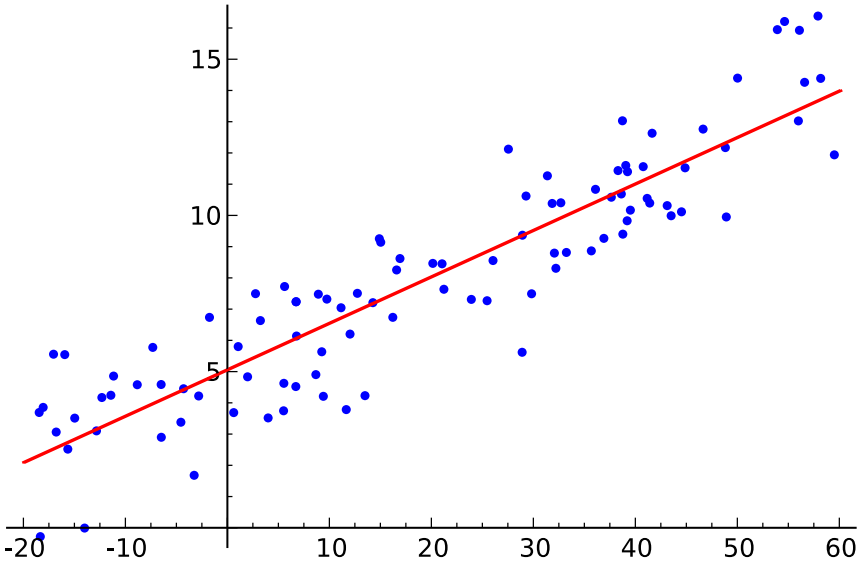


Figure 5.1: Some statistical data and a line approximating it.

We would like to approximate this data set by a polynomial function of the form $P(x) = a + bx$, to capture the 'linear trend' of this data. It should be clear from the picture that no such function perfectly matches our problem, i.e. no straight line goes through all our points at once.
Let us reinterpret this observation mathematically. Each of our N data point has coordinate (x_i, y_i), $1 \leq i \leq N$, for well-chosen values $x_i, y_i \in \mathbb{R}$, and the x_i are pairwise distinct. We assimilate the data

set with a *data vector* $\vec{v} := (y_1, \ldots, y_N)^T \in \mathbb{R}^N$, encoding the various values at $x_1, \ldots, x_N$. At the same values $x_1, \ldots, x_N$, a polynomial function of the form $P(x) = a + bx$ gives the vector

$$\begin{pmatrix} P(x_1) \\ \vdots \\ P(x_N) \end{pmatrix} = a \begin{pmatrix} 1 \\ \vdots \\ 1 \end{pmatrix} + b \begin{pmatrix} x_1 \\ \vdots \\ x_N \end{pmatrix}.$$

In particular, we see that the set of all the vectors $(P(x_1), \ldots, P(x_N))^T$, for P a polynomial of degree at most 1, forms a vector subspace of $\mathbb{R}^N$ of dimension 2 spanned by $(1, \ldots, 1)^T$ and $(x_1, \ldots, x_N)^T$. From the picture, we see that no line contains all points from our data set. Said in algebraic terms, our data vector $\vec{v}$ does not belong to the linear subspace spanned by $(1, \ldots, 1)^T$ and $(x_1, \ldots, x_N)^T$. We are interested in finding a good approximation of the data vector $\vec{v}$ by a data vector associated to a polynomial. This can be reinterpreted as finding a vector of the subspace W of 'polynomial data vectors' that minimises the distance to $\vec{v}$.

In this chapter, we will see that such vectors are given by taking the orthogonal projection of $\vec{v}$ on the subspace W, and we will study an algorithm to compute such projections.

5.2 Inner product in $\mathbb{R}^n$ and distance

Definition 5.1. The *inner product* (or *dot product* or *scalar product*) of two vectors $\vec{u} = (u_1, \ldots, u_n)^T$ and $\vec{v} = (v_1, \ldots, v_n)^T$ is defined as follows:

$$\langle \vec{u}, \vec{v} \rangle = \langle \begin{pmatrix} u_1 \\ \vdots \\ u_n \end{pmatrix}, \begin{pmatrix} v_1 \\ \vdots \\ v_n \end{pmatrix} \rangle := u_1 v_1 + \ldots + u_n v_n.$$

The *norm* of a vector is defined as

$$||\vec{v}|| := \sqrt{\langle \vec{v}, \vec{v} \rangle}\,.$$

The *distance* between two vectors $\vec{u}, \vec{v} \in \mathbb{R}^n$ is defined as

$$d(\vec{u}, \vec{v}) := ||\vec{u} - \vec{v}||.$$

Elementary properties.

Proposition 5.2. The inner product satisfies the following properties. For every $\vec{u}, \vec{v}, \vec{w} \in \mathbb{R}^n$ and every scalar $\lambda \in \mathbb{R}$, we have:

(i) $\langle \vec{u}, \vec{v} \rangle = \langle \vec{v}, \vec{u} \rangle$ (*symmetry*),

(ii$_1$) $\langle \vec{u} + \vec{v}, \vec{w} \rangle = \langle \vec{u}, \vec{w} \rangle + \langle \vec{v}, \vec{w} \rangle$ (*linearity 1*),

(ii$_2$) $\langle \lambda \vec{u}, \vec{v} \rangle = \lambda \langle \vec{u}, \vec{v} \rangle$ (*linearity 2*).

(iii) $\langle \vec{v}, \vec{v} \rangle \geq 0$ and $\langle \vec{v}, \vec{v} \rangle = 0$ if and only if $\vec{v} = \vec{0}$ (*positive definiteness*),

It follows from (ii$_1$) and (ii$_2$) that the Euclidean inner product behaves well with linear combinations. Indeed, for vectors $\vec{v}, \vec{v}_1, \ldots, \vec{v}_k \in \mathbb{R}^n$ and scalars $\lambda_1, \ldots, \lambda_k \in \mathbb{R}$, we have

$$\langle \sum_i \lambda_i \vec{v}_i, \vec{v} \rangle = \sum_i \lambda_i \langle \vec{v}_i, \vec{v} \rangle.$$

Orthogonality

Definition 5.3. Let $\vec{u}, \vec{v} \in \mathbb{R}^n$.

(i) The vectors $\vec{u}$, $\vec{v}$ are said to be *orthogonal*, if $\langle \vec{u}, \vec{v} \rangle = 0$.

(ii) If the vector $\vec{u}$ is orthogonal to each vector $\vec{x}$ in a set $S \subset \mathbb{R}^n$, then we say that $\vec{u}$ is orthogonal to S.

Example 5.4. In $\mathbb{R}^4$, $\vec{u} = (1,2,2,-1)^T$ and $\vec{v} = (2,-1,1,2)^T$ are orthogonal with respect to the Euclidean inner product.

Here is a useful remark that is a direct consequence of the linearity property of inner products:

Proposition 5.5. Let W be a subspace of $\mathbb{R}^n$ and let $\vec{e}_1, \ldots, \vec{e}_k$ be a family spanning W. Then, for $\vec{v} \in \mathbb{R}^n$, we have

$$\vec{v} \text{ is orthogonal to } W \iff \vec{v} \text{ is orthogonal to each of } \vec{e}_1, \ldots, \vec{e}_k.$$

Theorem 5.6. *(Pythagoras)* Two vectors $\vec{u}$ and $\vec{v}$ are orthogonal if and only if

$$||\vec{u} + \vec{v}||^2 = ||\vec{u}||^2 + ||\vec{v}||^2.$$

Proof. This follows immediately from:

$$||\vec{u} + \vec{v}||^2 = \langle \vec{u} + \vec{v}, \vec{u} + \vec{v} \rangle = \langle \vec{u}, \vec{u} \rangle + 2\langle \vec{u}, \vec{v} \rangle + \langle \vec{v}, \vec{v} \rangle = ||\vec{u}||^2 + ||\vec{v}||^2 + 2\langle \vec{u}, \vec{v} \rangle .$$

□

Transpose of a matrix and inner product.

Definition 5.7. The *transpose* of an $m \times n$ matrix A is an $n \times m$ matrix denoted A^T such that

$$(A^T)_{ij} = A_{ji}.$$

That is, the k-th row of A is 'turned into' the k-th column of A^T.

Example 5.8.

$$A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{pmatrix} \quad \rightsquigarrow \quad A^T = \begin{pmatrix} 1 & 3 & 5 \\ 2 & 4 & 6 \end{pmatrix}.$$

This justifies the notation we used for vectors of $\mathbb{R}^n$ in previous chapters: when writing for instance $(1,2,3)^T$ instead of the vector

$$\begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix},$$

this was a way to write this vector as the transpose of the 1×3 matrix $(1 \ \ 2 \ \ 3)$.

Using matrix multiplication can be used to reinterpret the Euclidean inner product in $\mathbb{R}^n$:

Proposition 5.9. Let $\vec{x}, \vec{y}$ be two vectors of $\mathbb{R}^n$, seen as $n \times 1$ matrices. Then:

$$\langle \vec{x}, \vec{y} \rangle = \vec{x}^T \vec{y}.$$

The following property of transposes can be useful:

Proposition 5.10. For an $m \times n$ matrix A and an $n \times p$ matrix B, we have:

$$(AB)^T = B^T A^T.$$

Proof. We have $((AB)^T)_{ij} = (AB)_{ji} = \sum_{k=1}^n a_{jk} b_{ki}$, and $(B^T A^T)_{ij} = \sum_{k=1}^n (B^T)_{ik} (A^T)_{kj} = \sum_{k=1}^n b_{ki} a_{jk}$. □

5.3 Abstract inner product spaces

Optimisation problems are very common in mathematics and their applications. For instance, one may want to approximate a given function (that represents for instance a signal) by a simpler function: polynomials, trigonometric polynomials, etc. Such approximations allow to encode and work with complicated with a small amount of information while minimising loss of information. This is particularly important in signal transmission or audio compression, for instance.
Thus, it is natural to ask whether one can develop a similar framework to work with approximations in spaces of functions (and more general vector spaces). Such a framework does exist, and relies on a generalisation of the idea of inner product, which was crucial to our study of optimisation problems in $\mathbb{R}^n$.

Definition 5.11. An *inner product* or *scalar product* on a vector space V over $\mathbb{R}$ is a map that associates to each pair of vectors $\vec{u}, \vec{v} \in V$ a real number $\langle \vec{u}, \vec{v} \rangle$ such that the following rules are satisfied for all vectors $\vec{u}, \vec{v}, \vec{w} \in V$ and $\lambda \in \mathbb{R}$:

(i) $\langle \vec{u}, \vec{v} \rangle = \langle \vec{v}, \vec{u} \rangle$ (*symmetry*),

(ii) $\langle \vec{v}, \vec{v} \rangle \geq 0$ and $\langle \vec{v}, \vec{v} \rangle = 0$ if and only if $\vec{v} = \vec{0}$ (*positive definiteness*),

(iii) $\langle \vec{u} + \vec{v}, \vec{w} \rangle = \langle \vec{u}, \vec{w} \rangle + \langle \vec{v}, \vec{w} \rangle$ (*linearity 1*),

(iv) $\langle \lambda \vec{u}, \vec{v} \rangle = \lambda \langle \vec{u}, \vec{v} \rangle$ (*linearity 2*).

A vector space endowed with an inner product is called an *inner product space*.

$\mathbb{R}^n$ with the Euclidean inner product is an inner product space. Another important example of inner product space is the following:

Proposition 5.12. Let $\mathscr{C}([0,1])$ be the set of continuous functions defined on the interval $[0,1]$. We can endow $\mathscr{C}([0,1])$ with the following inner product:

$$\langle f, g\rangle = \int_0^1 f(x)g(x)\mathrm{d}x\,.$$

Proof. Indeed, we check:

(i) $\langle f, g\rangle = \int_0^1 f(x)g(x)\mathrm{d}x = \int_0^1 g(x)f(x)\mathrm{d}x = \langle g, f\rangle$.

(ii) $\langle f, f\rangle = \int_0^1 f(x)^2\mathrm{d}x \geq 0$, as the integral of a non-negative function. Moreover, if f is not the zero-function, then by continuity there exists an interval $[a,b] \subset [0,1]$ (with $a < b$)and a constant $\varepsilon > 0$ such that $|f(x)| > \varepsilon$ for every $x \in [a,b]$. Thus, $\langle f, f\rangle = \int_0^1 f(x)^2\mathrm{d}x \geq \int_a^b f(x)^2\mathrm{d}x \geq \int_a^b \varepsilon^2\mathrm{d}x = (b-a)\varepsilon^2 > 0$, hence $\langle f, f\rangle > 0$.

(iii) $\langle f+g, h\rangle = \int_0^1 (f(x)+g(x))h(x)\mathrm{d}x = \int_0^1 f(x)h(x)\mathrm{d}x + \int_0^1 g(x)h(x)\mathrm{d}x = \langle f, h\rangle + \langle g, h\rangle$.

(iv) $\langle \lambda f, g\rangle = \int_0^1 (\lambda f(x))g(x)\mathrm{d}x = \lambda \int_0^1 f(x)g(x)\mathrm{d}x = \lambda\langle f, g\rangle$.

□

Definition 5.13. The *norm* of a vector $\vec{u} \in V$ in an inner product space is given by $||\vec{u}|| := \sqrt{\langle \vec{u}, \vec{u}\rangle}$.
The *distance* between two vectors $\vec{u}, \vec{v} \in V$ is defined as $d(\vec{u}, \vec{v}) := ||\vec{u} - \vec{v}||$.

Example 5.14. In the case of the vector space $\mathscr{C}(\mathbb{R})$ of continuous functions defined on the interval $[0,1]$, we have $||f|| = \left(\int_0^1 f(x)^2\mathrm{d}x\right)^{1/2}$ and $d(f,g) = \left(\int_0^1 |f(x)-g(x)|^2\mathrm{d}x\right)^{1/2}$ for all $f, g \in \mathscr{C}(\mathbb{R})$.

5.4 Optimisation and orthogonal projections

In this chapter, we are interested in the following general optimisation problem: Let V be an inner product space and W a subspace, and let $\vec{v} \in V$. We wish to find a vector $\vec{w} \in W$ minimising the distance to $\vec{v}$, that is such that

$$d(\vec{v}, \vec{w}) = \min_{\vec{w}' \in W} d(\vec{v}, \vec{w}').$$

We will see in this chapter that such minimising vectors are linked to the notion of orthogonal projection that you have already studied in $\mathbb{R}^2$ and $\mathbb{R}^3$ in past courses.

Definition 5.15. Let V be a vector space, and let W be a finite-dimensional subspace. For a vector $\vec{v} \in V$, an *orthogonal projection* of $\vec{v}$ on W is a vector $\vec{p} \in W$ such that $\vec{v} - \vec{p}$ is orthogonal to W, or in other words:

$$\langle \vec{v} - \vec{p}, \vec{w}\rangle = 0 \quad \text{for every } \vec{w} \in W.$$

Theorem 5.16. Let V be a inner product space, $\vec{v} \in V$, and let W be a subspace. A vector $\vec{w} \in W$ minimises the distance to $\vec{v}$, that is,

$$d(\vec{v}, \vec{w}) = \min_{\vec{w}' \in W} d(\vec{v}, \vec{w}'),$$

if and only if $\vec{w}$ is an orthogonal projection of $\vec{v}$ on W. Moreover, this orthogonal projection, if it exists, is unique.

Proof. Let us show one implication. Let $\vec{w} \in W$ be such that $\langle \vec{v} - \vec{w}, \vec{w}' \rangle = 0$ for every other vector $\vec{w}' \in W$. Since $\vec{v} - \vec{w}$ is orthogonal to $\vec{w} - \vec{w}' \in W$ by construction, Pythagoras' Theorem implies that

$$d(\vec{v}, \vec{w}')^2 = ||\vec{v} - \vec{w}'||^2 = ||(\vec{v} - \vec{w}) + (\vec{w} - \vec{w}')||^2 = ||\vec{v} - \vec{w}||^2 + ||\vec{w} - \vec{w}'||^2 > ||\vec{v} - \vec{w}||^2 = d(\vec{v}, \vec{w})^2.$$

In particular, such a vector $\vec{w}$ minimises the distance to $\vec{v}$, and the previous strict equality shows that it must be unique. We will admit the existence of a solution to the previous system of equations. □

We now show that there always exist an orthogonal projection on a finite-dimensional subspace of V.

Theorem 5.17. Let V be a inner product space, $\vec{v} \in V$, and let W be a **finite-dimensional** subspace. If W has a basis $\vec{e}_1, \ldots, \vec{e}_k$, then a vector $\vec{x} \in W$ is an orthogonal projection of $\vec{v}$ on W if and only if it is a solution of the following system of equations:

$$\begin{cases} \langle \vec{e}_1, \vec{x} \rangle &= \langle \vec{e}_1, \vec{v} \rangle, \\ \quad \vdots & \\ \langle \vec{e}_k, \vec{x} \rangle &= \langle \vec{e}_k, \vec{v} \rangle. \end{cases}$$

Let

$$A := \left(\vec{e}_1 | \cdots | \vec{e}_k \right)$$

be the $n \times k$ matrix whose columns are $\vec{e}_1, \ldots, \vec{e}_k$. If we write a vector $\vec{x} \in W$ as a linear combination of the $\vec{e}_i$:

$$\vec{x} = \lambda_1 \vec{e}_1 + \ldots + \lambda_k \vec{e}_k, \quad \text{with} \quad \lambda_1, \ldots, \lambda_k \in \mathbb{R},$$

then the previous system can be written in matrix form as

$$A^T A \begin{pmatrix} \lambda_1 \\ \vdots \\ \lambda_k \end{pmatrix} = A^T \vec{v},$$

where the unknown is the vector $(\lambda_1, \ldots, \lambda_k)^T$. Moreover, the $k \times k$ matrix $A^T A$ associated to that system is invertible. In particular, there is always exist a unique orthogonal projection of $\vec{v}$ on W.

Remark. The previous result requires that the subspace W be finite-dimensional. And indeed, things can become much more complicated in infinite dimension. For instance, there does not always exist an orthogonal projection on an infinite-dimensional subspace. Worse still, it is possible to construct a vector space V and a subspace W such that there exists a vector $\vec{v} \notin W$, but $\vec{v}$ is at distance 0 from W...!

Definition 5.18. Let V be a inner product space, $\vec{v} \in V$, and let W be a finite-dimensional subspace. The orthogonal projection of $\vec{v}$ on W is denoted $\pi_W(\vec{v})$. It is characterised by the fact that this is the unique vector of W minimising the distance to $\vec{v}$.
If W is the line generated by a vector $\vec{u}$, we simply denote by $\pi_{\vec{u}}(\vec{v})$ the orthogonal projection of $\vec{v}$ on W.

5.5 Application: Data fitting and polynomial regression

Let us go back to our initial problem. Given a data set $\{(x_1, y_1), \ldots, (x_N, y_N)\}$ for N very large, we would like to approximate it by a polynomial of low degree m (generally $m \leq 3$).
We assimilate the data set with a *data vector* $\vec{v} := (y_1, \ldots, y_N)^T \in \mathbb{R}^N$. A polynomial $P(x) = a_0 + a_1 x + \ldots + a_m x^m$ gives the vector $(P(x_1), \ldots, P(x_N))^T$. Note that we have

$$\begin{pmatrix} P(x_1) \\ \vdots \\ P(x_m) \end{pmatrix} = a_0 \begin{pmatrix} 1 \\ \vdots \\ 1 \end{pmatrix} + a_1 \begin{pmatrix} x_1 \\ \vdots \\ x_N \end{pmatrix} + \ldots + a_m \begin{pmatrix} x_1^m \\ \vdots \\ x_N^m \end{pmatrix}.$$

In particular, we see that the set of all the vectors $(P(x_1), \ldots, P(x_N))^T$ for a polynomial function of a degree at most m forms a vector subspace of $\mathbb{R}^N$ of dimension at most m (actually, exactly m). We are interested in finding a good approximation of the data vector $\vec{v}$ by a data vector associated to a polynomial. This can be reinterpreted as finding the orthogonal projection of the vector $\vec{v}$ on the subspace W of 'polynomial data vectors'.
Let

$$A : \mathbb{R}^m \to \mathbb{R}^N, \begin{pmatrix} a_0 \\ \vdots \\ a_m \end{pmatrix} \mapsto a_0 \begin{pmatrix} 1 \\ \vdots \\ 1 \end{pmatrix} + a_1 \begin{pmatrix} b_1 \\ \vdots \\ b_N \end{pmatrix} + \ldots + a_m \begin{pmatrix} b_1^m \\ \vdots \\ b_N^m \end{pmatrix}.$$

The equation $A\vec{x} = \vec{v}$ generally has no solution. Thus, instead of solving the problem

'$A\vec{x}$ is equal to $\vec{v}$'

which admits no solution, we will solve instead the approximation problem

'$A\vec{x}$ is as close as possible to $\vec{v}$.'

When $\vec{x}$ runs over $\mathbb{R}^m$, the vector $A\vec{x}$ runs over the image $\mathrm{Im}(A)$. From the previous section, the vector of $\mathrm{Im}(A)$ closest to $\vec{v}$ is the orthogonal projection of $\vec{v}$ on $W := \mathrm{Im}(A)$. We are thus back to a different problem:

Definition 5.19. For an $N \times m$ matrix A, a *least square solution* to the equation $A\vec{x} = \vec{v}$ is a vector $\vec{x} \in \mathbb{R}^m$ such that

$$A\vec{x} = \pi_{\mathrm{Im}(A)}(\vec{v}).$$

Intuitively, a least square solution is a 'best possible approximation' of a solution to the system equation $A\vec{x} = \vec{v}$, when such a system is inconsistent. The term 'least square' is a reference to the fact that we are minimising a norm, which when written in coordinates involves a sum of squares.

Theorem 5.20. A vector $\vec{x}$ is a least square solution to the equation $A\vec{x} = \vec{v}$ if and only if it satisfies

$$A^T A\vec{x} = A^T \vec{v}.$$

This last system of equations is consistent, and least square solutions are computed as usual using Gaussian elimination (or by inverting $A^T A$ when such a matrix is invertible).

The previous theorem reduces the problem of finding a best approximation to a system of m equations with m unknowns (a much smaller and more manageable system).

Example 5.21. Let us find the best polynomial of degree 2 approximating the data the system of equations $P(0)=0, P(1)=1, P(2)=4, P(3)=7$. One can first show wihout any computation that this system of equation has no solution (Can you see why?), so let us find the least square solution to this problem. For a polynomial of degree 2 of the form $P(x)=a+bx+cx^2$, the system of equations $P(0)=0, P(1)=1, P(2)=4, P(3)=7$ can be rewritten as

$$\underbrace{\begin{pmatrix} 1 & 0 & 0 \\ 1 & 1 & 1 \\ 1 & 2 & 4 \\ 1 & 3 & 9 \end{pmatrix}}_{=:A} \begin{pmatrix} a \\ b \\ c \end{pmatrix} = \underbrace{\begin{pmatrix} 0 \\ 1 \\ 4 \\ 7 \end{pmatrix}}_{=:\vec{v}}$$

We thus get

$$A^T A = \begin{pmatrix} 4 & 6 & 14 \\ 6 & 14 & 36 \\ 14 & 36 & 98 \end{pmatrix}$$

and

$$A^T \vec{v} = \begin{pmatrix} 12 \\ 30 \\ 80 \end{pmatrix}.$$

We thus have to solve the system

$$\begin{pmatrix} 4 & 6 & 14 \\ 6 & 14 & 36 \\ 14 & 36 & 98 \end{pmatrix} \begin{pmatrix} a \\ b \\ c \end{pmatrix} = \begin{pmatrix} 12 \\ 30 \\ 80 \end{pmatrix}.$$

Gaussian elimination yields:

$$\begin{pmatrix} a \\ b \\ c \end{pmatrix} = \begin{pmatrix} -0.1 \\ 0.9 \\ 0.5 \end{pmatrix}.$$

Thus, the best approximating polynomial of degree 2 is $0.5x^2+0.9x-0.1$.

5.6 Orthogonal families and orthogonal bases

In this section, we introduce particular families of vectors that make computations of orthogonal projections particularly simple.

Definition 5.22. An *orthogonal family of vectors* is a family of vectors such that each pair of distinct vectors in this family are orthogonal. An orthogonal family in which each vector has norm 1 is called *orthonormal*.

Example 5.23. In $\mathbb{R}^3$ with Euclidean inner product, $(1,-2,1)^T, (1,1,1)^T, (1,0,-1)^T$ form an orthogonal family, and $(1,0,0)^T$, $(0,1,0)^T, (0,0,1)^T$ form an orthonormal family.

Theorem 5.24. An orthogonal family of vectors that does not contain $\vec{0}$ is linearly independent.

Proof. Let $\vec{u}_1, \ldots, \vec{u}_n$ be an orthogonal family not containing $\vec{0}$. Consider the equation $c_1\vec{u}_1 + c_2\vec{u}_2 + \ldots + c_n\vec{u}_n = \vec{0}$. We need to show that $c_1 = c_2 = \ldots = c_n = 0$ is the only solution. Taking the inner product of both sides with $\vec{u}_k$ for some $1 \leq k \leq n$ yields

$$0 = \langle \vec{0}, \vec{u}_k \rangle = \langle c_1\vec{u}_1 + \ldots + c_n\vec{u}_n, \vec{u}_k \rangle = c_1\langle \vec{u}_1, \vec{u}_k \rangle + \ldots + c_n\langle \vec{u}_n, \vec{u}_k \rangle = c_k \underbrace{\langle \vec{u}_k, \vec{u}_k \rangle}_{>0},$$

hence $c_k = 0$. Since k was arbitrary, we conclude that $c_1, c_2, \ldots, c_n = 0$. □

Decomposition in an orthogonal basis. If $(\vec{e}_1, \ldots, \vec{e}_n)$ is a basis for $\mathbb{R}^n$ and we want to write a vector $\vec{v} \in \mathbb{R}^n$ as a linear combination $\vec{v} = c_1\vec{e}_1 + \ldots + c_n\vec{e}_n$ of the basis vectors, we have to solve a system of n equations in n unknowns in general. If $(\vec{e}_1, \ldots, \vec{e}_n)$ is an orthogonal basis, i.e. an orthogonal family of vectors that is also a basis of $\mathbb{R}^n$, then this problem becomes much simpler. If $\vec{v} = c_1\vec{e}_1 + \ldots + c_n\vec{e}_n$, then we have

$$\langle \vec{v}, \vec{e}_k \rangle = \langle c_1\vec{e}_1 + \ldots + c_n\vec{e}_n, \vec{u}_k \rangle = c_1\langle \vec{e}_1, \vec{e}_k \rangle + \ldots + c_n\langle \vec{e}_n, \vec{e}_k \rangle = c_k\langle \vec{e}_k, \vec{e}_k \rangle,$$

hence

$$c_k = \frac{\langle \vec{v}, \vec{e}_k \rangle}{\langle \vec{e}_k, \vec{e}_k \rangle}.$$

If $(\vec{e}_1, \ldots, \vec{e}_n)$ is even an *orthonormal basis*, i.e. $\langle \vec{e}_k, \vec{e}_k \rangle = 1$ for all $k = 1, \ldots, n$, then the formula for the coefficients simplifies further: $c_k = \langle \vec{v}, \vec{e}_k \rangle$.

Theorem 5.25 (Decomposition along an orthogonal basis). Let V be a finite dimensional inner product space, and let $\vec{e}_1, \ldots, \vec{e}_n$ be an orthogonal basis. For any $\vec{v} \in V$, we have

$$\vec{v} = c_1\vec{e}_1 + \ldots + c_n\vec{e}_n$$

with for every $1 \leq k \leq n$:

$$c_k = \frac{\langle \vec{v}, \vec{e}_k \rangle}{\langle \vec{e}_k, \vec{e}_k \rangle}.$$

Example 5.26. Consider the orthogonal basis $\left((1,-1,1)^T, (1,0,-1)^T, (1,2,1)^T\right)$ of $\mathbb{R}^3$. To write the vector $(3,4,5)^T$ as a linear combination of the basis vectors, we compute

$$c_1 = \frac{\left\langle \begin{pmatrix} 3 \\ 4 \\ 5 \end{pmatrix}, \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix} \right\rangle}{\left\langle \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix} \right\rangle} = \frac{4}{3}, \quad c_2 = \frac{\left\langle \begin{pmatrix} 3 \\ 4 \\ 5 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix} \right\rangle}{\left\langle \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix} \right\rangle} = -1, \quad c_3 = \frac{\left\langle \begin{pmatrix} 3 \\ 4 \\ 5 \end{pmatrix}, \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix} \right\rangle}{\left\langle \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix} \right\rangle} = \frac{8}{3},$$

hence

$$\begin{pmatrix} 3 \\ 4 \\ 5 \end{pmatrix} = \frac{4}{3}\begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix} - \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix} + \frac{8}{3}\begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}.$$

Orthogonal projections when an orthogonal basis is known.

Theorem 5.27 (Orthogonal projection on a line). The orthogonal projection of a vector $\vec{v}$ on the line generated by the vector $\vec{e}$ is given by

$$\pi_{\vec{e}}(\vec{v}) := \frac{\langle \vec{v}, \vec{e} \rangle}{\langle \vec{e}, \vec{e} \rangle} \vec{e}.$$

Proof. Indeed, it is enough to check that $\vec{v} - \pi_{\vec{e}}(\vec{v})$ is orthogonal to $\vec{e}$:

$$\langle \vec{e}, \vec{v} - \pi_{\vec{e}}(\vec{v}) \rangle = \left\langle \vec{e}, \vec{v} - \frac{\langle \vec{v}, \vec{e} \rangle}{\langle \vec{e}, \vec{e} \rangle} \vec{e} \right\rangle = \langle \vec{e}, \vec{v} \rangle - \frac{\langle \vec{v}, \vec{e} \rangle}{\langle \vec{e}, \vec{e} \rangle} \langle \vec{e}, \vec{e} \rangle = 0 \, .$$

□

Suppose now that we want to project a vector $\vec{v}$ on a subspace W of V, such that W is finite dimensional and has an orthogonal basis $\vec{e}_1, \ldots, \vec{e}_n$.

Theorem 5.28 (Orthogonal projection on a subspace, given an orthogonal basis). Let V be an inner product space, let W be a finite-dimensional subspace with an orthogonal basis $\vec{e}_1, \ldots, \vec{e}_n$, and let $\vec{v} \in V$. The orthogonal projection of $\vec{v}$ on W is given by

$$\pi_W(\vec{v}) := \pi_{\vec{e}_1}(\vec{v}) + \ldots + \pi_{\vec{e}_n}(\vec{v}) = \frac{\langle \vec{v}, \vec{e}_1 \rangle}{\langle \vec{e}_1, \vec{e}_1 \rangle} \vec{e}_1 + \ldots + \frac{\langle \vec{v}, \vec{e}_n \rangle}{\langle \vec{e}_n, \vec{e}_n \rangle} \vec{e}_n.$$

Proof. One checks this again by verifying that $\vec{v} - \pi_W(\vec{v})$ is orthogonal to $\vec{e}_1, \ldots, \vec{e}_n$, hence is orthogonal to $W = \text{span}(\vec{e}_1, \ldots, \vec{e}_n)$. Note that if $W = V$, we recover the formula for writing a vector as a linear combination of vectors from an orthogonal basis. □

5.7 Gram-Schmidt algorithm

The next natural question is: How to find an orthonormal basis of a space / of a subspace? This is done with the algorithm presented in this section.

Method: Gram-Schmidt algorithm

Let V be an inner product space with a basis $(\vec{u}_1, \ldots, \vec{u}_n)$. The following algorithm turns this basis into an orthogonal basis $(\vec{e}_1, \ldots, \vec{e}_n)$:

(1) Let $\vec{e}_1 = \vec{u}_1$.

(2) Take the next vector and make it perpendicular to all the previously obtained ones:

$$\vec{e}_i = \vec{u}_i - \pi_{\text{span}(\vec{e}_1, \ldots, \vec{e}_{i-1})}(\vec{u}_i) = \vec{u}_i - \sum_{j=1}^{i-1} \pi_{\vec{e}_j}(\vec{u}_i)$$

Explicitly, we have for example

$$\vec{e}_2 = \vec{u}_2 - \frac{\langle \vec{u}_2, \vec{e}_1 \rangle}{\langle \vec{e}_1, \vec{e}_1 \rangle} \vec{e}_1 \, , \quad \vec{e}_3 = \vec{u}_3 - \frac{\langle \vec{u}_3, \vec{e}_1 \rangle}{\langle \vec{e}_1, \vec{e}_1 \rangle} \vec{e}_1 - \frac{\langle \vec{u}_3, \vec{e}_2 \rangle}{\langle \vec{e}_2, \vec{e}_2 \rangle} \vec{e}_2 \, .$$

In the above procedure, $\vec{e}_i$ can always be replaced by $\lambda \vec{e}_i$ with $\lambda \in \mathbb{R} \setminus \{0\}$ to remove fractions. This can sometimes simplify computations.

The resulting vectors are all orthogonal and still span V. They are also linearly independent due to Proposition 5.24. Therefore, they form an orthogonal basis of V.

Remark. If $(\vec{e}_1,\dots,\vec{e}_n)$ is an orthogonal basis, then $\left(\frac{\vec{e}_1}{||\vec{e}_1||},\dots,\frac{\vec{e}_n}{||\vec{e}_n||}\right)$ is an orthonormal basis.

Example 5.29. Let us find an orthonormal basis for the subspace of $\mathbb{R}^4$ spanned by the vectors $(1,1,1,1)^T,(0,1,0,1)^T,(-1,1,3,2)^T$. We start with $\vec{e}_1=(1,1,1,1)$ and use the above formulas:

$$\begin{aligned}\vec{e}_2 &= (0,1,0,1)^T - \frac{\langle (0,1,0,1)^T,(1,1,1,1)^T\rangle}{\langle (1,1,1,1)^T,(1,1,1,1)^T\rangle}(1,1,1,1)^T = (0,1,0,1)^T - \frac{2}{4}(1,1,1,1)^T\\ &= \tfrac{1}{2}(-1,1,-1,1)^T,\\ \vec{e}_3 &= (-1,1,3,2)^T - \frac{\langle (-1,1,3,2)^T,(1,1,1,1)^T\rangle}{\langle (1,1,1,1)^T,(1,1,1,1)^T\rangle}(1,1,1,1)^T\\ &\qquad - \frac{\left\langle (-1,1,3,2)^T,\tfrac{1}{2}(-1,1,-1,1)^T\right\rangle}{\left\langle \tfrac{1}{2}(-1,1,-1,1)^T,\tfrac{1}{2}(-1,1,-1,1)^T\right\rangle}\tfrac{1}{2}(-1,1,-1,1)^T\\ &= \ldots = \tfrac{1}{2}(-4,-1,4,1)^T.\end{aligned}$$

An orthogonal basis is $\left((1,1,1,1)^T,\frac{1}{2}(-1,1,-1,1)^T,(-4,-1,4,1)^T\right)$ and therefore an orthonormal basis is $\left(\frac{1}{2}(1,1,1,1)^T,\frac{1}{2}(-1,1,-1,1)^T,\frac{1}{\sqrt{34}}(-4,-1,4,1)^T\right)$.

Remark.

(i) As mentioned above, $\vec{e}_i$ can always be replaced by $\lambda\vec{e}_i$ with $\lambda\in\mathbb{R}^*$ to remove fractions. Therefore, we could take $\vec{e}_2=(-1,1,-1,1)^T$ instead of $\vec{e}_2=\frac{1}{2}(-1,1,-1,1)^T$.

(ii) Normalising the vectors after each step produces an orthonormal basis and can be faster.

Method: Computing the orthogonal projection of a vector on a subspace

To compute the orthogonal projection of a vector $\vec{v}\in V$ on a finite-dimensional subspace $W\subset V$, we have the following algorithm:

(1) Find a basis of W. We have ways to construct a basis for several types of subspaces in this course.

(2) Turn this basis of W into an orthogonal basis $\vec{e}_1,\dots,\vec{e}_k$ using Gram-Schmidt.

(3) Finally, the orthogonal projection is given by

$$\pi_W(\vec{v}) := \pi_{\vec{e}_1}(\vec{v})+\ldots+\pi_{\vec{e}_n}(\vec{v}) = \frac{\langle\vec{v},\vec{e}_1\rangle}{\langle\vec{e}_1,\vec{e}_1\rangle}\vec{e}_1+\ldots+\frac{\langle\vec{v},\vec{e}_n\rangle}{\langle\vec{e}_n,\vec{e}_n\rangle}\vec{e}_n.$$

5.8 Application: Towards audio compression and Fourier Series

A sound corresponds to the variation in air pressure as perceived by our ears over time. This can be encoded by a function $f:\mathbb{R}\to\mathbb{R}$, called the *waveform* of that sound, that associates to a given time

the measured variation in air pressure. The loudness of the sound is associated to the amplitude in variation of air pressure, while the frequency of that waveform is associated to the pitch of the sound (high frequencies corresponding to high-pitched notes).
Simple periodic functions are *sine and cosine waves* of the form $\sin(2\pi\omega x)$ and $\cos(2\pi\omega x)$ (where the frequency here is ω). They correspond to 'pure tones' and have a single pitch, meaning that what you here is a single note. While very simple, these waveforms don't occur in nature.
A same note with frequency ω (for instance $\omega = 440Hz$ for the standard pitch) played by different instruments will not sound the same even though they are playing the same note. This comes from the fact that waveforms vary from one instrument to the other. This variation in waveforms is responsible for the *timbre*, or *tone color* of the instrument, i.e. its specific character. While they are still periodic functions with a frequency of ω, they are not a simple sine or cosine wave, but tend to be much more complicated periodic functions:

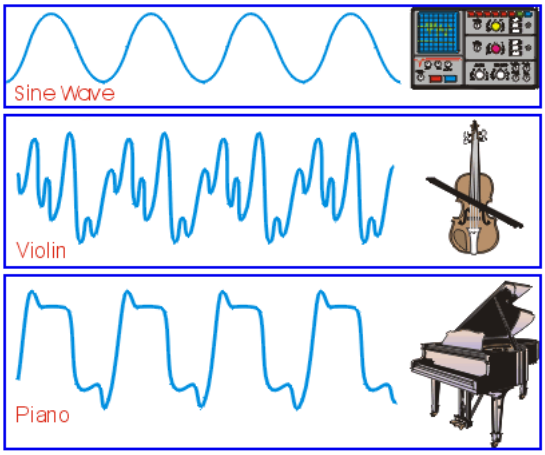


Figure 5.2: Simplified sketches of waveforms associated to different instruments (Imgur image)

Let us focus on a single note with frequency ω played by an instrument, which we would like to store on a computer. The waveform of that sound is generally a function that is too complicated to be encoded and stored as is. When trying to store this sound, one would like to approximate the corresponding waveform by a 'simpler' function, such as a linear combination of sine and cosine waves of the form

$$\sum_{1\leq k\leq n} a_k \cos(2\pi\omega kx) + b_k \sin(2\pi\omega kx).$$

Indeed, such functions can be stored simply by remembering the $2n$ coefficients $a_1, b_1, \ldots, a_n, b_n$.

A basic idea behind audio compression is thus the following. Starting from the waveform $f(t)$ of your instrument playing a note with frequency ω:

- approximate your waveform f by a simpler $2\pi\omega$-periodic function of the form

$$\sum_{1\leq k\leq n} a_k \cos(2\pi\omega kx) + b_k \sin(2\pi\omega kx)$$

  in the best possible way,

- store this as the 'approximated sound'.

There are two natural questions:

- How do we find these approximations?
- Are these approximations good enough to recognise the original sound?

Let us look at the first question. The pitch we are interested is encoded as a $2\pi\omega$-periodic function f, and thus represents an element of the vector space $\mathscr{C}_{1/\omega}(\mathbb{R})$ of $1/\omega$-periodic continuous functions. Finding the function of the form $\sum_{1\leq k\leq n} a_k \cos(2\pi\omega kx) + b_k \sin(2\pi\omega kx)$ that best approximates f amounts to finding the vector of

$$W_n := \operatorname{span}\big(\cos(2\pi\omega x), \sin(2\pi\omega x), \ldots, \cos(2\pi\omega nx), \sin(2\pi\omega nx)\big)$$

that best approximates f. There is a natural inner product on $\mathscr{C}_{1/\omega}(\mathbb{R})$, namely

$$\langle u, v\rangle := \int_0^{1/\omega} u(x)\,v(x)\,dx.$$

Having this inner product, and thus the corresponding notion of distance, allows us to rephrase this problem as finding the vector of W_n that minimises the distance to f. As we have learned in this chapter, this is exactly the orthogonal projection of f on W_n.

Computing this orthogonal projection turns out to be quite simple. Indeed, the family of functions

$$c_1 := \cos(2\pi\omega x), s_1 := \sin(2\pi\omega x), c_2 := \cos(2\pi\omega \times 2x), s_2 := \sin(2\pi\omega \times 2x), \ldots$$

is orthogonal for that inner product. Thus, the orthogonal projection of f on the subspace W_n is simply given by the standard formula:

$$S_n f(x) := \frac{\langle f, c_1\rangle}{\langle c_1, c_1\rangle}\cos(2\pi\omega x) + \frac{\langle f, s_1\rangle}{\langle s_1, s_1\rangle}\sin(2\pi\omega x) + \cdots + \frac{\langle f, c_n\rangle}{\langle c_n, c_n\rangle}\cos(2\pi\omega nx) + \frac{\langle f, s_n\rangle}{\langle s_n, s_n\rangle}\sin(2\pi\omega nx).$$

This orthogonal projection is called the *(truncated) Fourier series* of f, and it is this function that plays the role of 'good approximation' stored in computers (for a good value of n).

Now, how good are these approximations? As we take more and more *harmonics*, that is, as n grows larger, this approximation of f become more and more faithful to the original sound. Actually, it was a breakthrough result of Fourier (in his study of the heat equation) that, in a certain sense that I won't make more explicit here, the sequence of approximating functions $S_n f$ converges to our original waveform f. Put simply, every periodic function can be decomposed as an infinite sum of sine and cosine waves in an appropriate sense. In our case, we get:

$$f(x) = \sum_{k=0}^{\infty} a_k \cos(2k\pi\omega x) + b_k \sin(2k\pi\omega x).$$

Moreover, the *Fourier coefficients* a_k, b_k converge to zero as k grows larger. Thus, taking a truncated sum by removing the harmonics corresponding to very small Fourier coefficients, or to frequencies that are too high for the human ear (the human ear can only hear sounds whose frequency is between 20 and 20.000Hz), we obtain an approximation of the sound that is extremely faithful to the original.

Chapter 6

Eigenvalues and diagonalisability

6.1 A motivating problem: predicting the dynamics of a population

A simple model to study a population of animals is the following: the population is divided into two age groups: adults and juveniles. At the start of the observation, the population consists of a_0 adults and j_0 juveniles. From one year to the next, the adults will each produce on average γ juveniles. Adults will survive from one year to the next with probability α, while juveniles will survive into the next year and become adults with the probability β.
We wish to understand the evolution of the population over time: does the population collapse, does it converge to a stable state? And how does this behaviour depend on the fertility rate γ? Notice that we have the following equations between the population in two consecutive years:

$$\begin{aligned} a_{n+1} &= \alpha a_n + \beta j_n \\ j_{n+1} &= \gamma a_n \end{aligned}$$

This can be rewritten as

$$\begin{pmatrix} a_{n+1} \\ j_{n+1} \end{pmatrix} = A \begin{pmatrix} a_n \\ j_n \end{pmatrix} \quad \text{with } A := \begin{pmatrix} \alpha & \beta \\ \gamma & 0 \end{pmatrix}.$$

In particular, we see by induction that

$$\begin{pmatrix} a_n \\ j_n \end{pmatrix} = A^n \begin{pmatrix} a_0 \\ j_0 \end{pmatrix}.$$

To understand the population at a given time, we thus need to compute powers of the matrix A. This is a priori a non-trivial problem. There is however one case where computing powers poses no problem: the case of diagonal matrices.
Let us for a moment think of our matrix A as a linear map $A : \mathbb{R}^2 \to \mathbb{R}^2$. In previous chapters, we have seen examples of linear maps whose associated matrices become diagonal after an appropriate change of basis (see Section 4.6). Thus, a natural strategy would be to try to find a 'simplest possible basis' where the matrix associated to the linear map A becomes diagonal, compute the powers of the matrix in that basis, and go back to the original basis. If we restate this problem in terms of matrices, we want to find an invertible matrix P and a diagonal matrix D such that $A = PDP^{-1}$. We have:

$$A^n = (PDP^{-1})^n = PD\underbrace{P^{-1}P}_{=I_2}DP^{-1}\cdots\underbrace{P^{-1}P}_{=I_2}DP^{-1} = PD^nP^{-1}.$$

Since D^n is very easy to compute, it follows that computing A^n itself becomes much easier to compute, provided we know how to compute P. This leads to the following questions:

- Given a square matrix A, does there always exist an invertible matrix P such that $P^{-1}AP$ is diagonal?
- If so, how to compute such a matrix P?

The goal of this chapter is to answer these questions and see applications to various problems.

6.2 Eigenvalues and diagonalisability

Eigenvalues.

Definition 6.1. Let V be an n-dimensional vector space and let $T : V \to V$ be a linear map. Then $\lambda \in \mathbb{R}$ is called a (real) *eigenvalue* of T, if there is a vector $\vec{x} \in V$, $\vec{x} \neq \vec{0}$, such that

$$T(\vec{x}) = \lambda \vec{x}.$$

The vector $\vec{x}$ is called an *eigenvector* of T for the eigenvalue λ.

Recall that an $n \times n$ matrix A can be seen as a linear map $T_A : \mathbb{R}^n \to \mathbb{R}^n, \vec{x} \mapsto A\vec{x}$. In particular, the notions of eigenvalues and eigenvectors are well defined for matrices: $\lambda \in \mathbb{R}$ is an *eigenvalue* of A, if there is a vector $\vec{x} \in \mathbb{R}^n$, $\vec{x} \neq \vec{0}$, such that $A\vec{x} = \lambda\vec{x}$, and the vector $\vec{x}$ is called an *eigenvector* of A for the eigenvalue λ.

Example 6.2.(i) For $A = I_n$ the identity matrix, every vector $\vec{x} \neq \vec{0}$ is an eigenvector with eigenvalue 1.

(ii) The rotation of angle $\pi/2$, $r_{\pi/2} : \mathbb{R}^2 \to \mathbb{R}^2$ does not have any eigenvalue, since $r_{\pi/2}(\vec{x})$ is never collinear to $\vec{x}$ (but is orthogonal to it and of the same norm).

Example 6.3. Consider the following diagonal matrix:

$$D = \begin{pmatrix} d_1 & 0 & \dots & 0 \\ 0 & d_2 & \dots & 0 \\ \vdots & & \ddots & \vdots \\ 0 & 0 & \dots & d_n \end{pmatrix}, \quad d_1, \dots, d_n \in \mathbb{R}.$$

Then the eigenvalues of D are $d_1, \dots, d_n$. Indeed, for a vector $\vec{x} = (x_1, \dots, x_n)^T$, we have $D\vec{x} = (d_1 x_1, \dots, d_n x_n)^T$. If $\vec{x}$ is an eigenvector of D for an eigenvalue λ, then we also have $D\vec{x} = (\lambda x_1, \dots, \lambda x_n)$. As one of the x_i is non-zero (since $\vec{x} \neq \vec{0}$), it follows that $\lambda = d_i$ for some i.
Note that if the $d_1, \dots, d_n$ are pairwise distinct, then the eigenvectors associated to $d_1, \dots, d_n$ are of the form $(c_1, 0, \dots, 0)^T, \dots, (0, \dots, 0, c_n)^T$ respectively, where $c_1, \dots, c_n \in \mathbb{R}$.

Daigonalisability.

Definition 6.4. A linear map $T : V \to V$ is called *diagonalisable* if there exists a basis of eigenvectors.
Equivalently, T is diagonalisable if there exists a basis $\mathscr{B}$ of U such that the associated matrix

$[T]_{\mathscr{B}}$ is of the form:

$$[T]_{\mathscr{B}} = \begin{pmatrix} \lambda_1 & 0 & \dots & 0 \\ 0 & \lambda_2 & \dots & 0 \\ \vdots & & \ddots & \vdots \\ 0 & 0 & \dots & \lambda_n \end{pmatrix}.$$

Example 6.5.(i) The rotation $r_{\pi/4} : \mathbb{R}^2 \to \mathbb{R}^2$ is not diagonalisable as it has no (real) eigenvalue.

(ii) The orthogonal projection $s_{\pi/4} : \mathbb{R}^2 \to \mathbb{R}^2$ (see Example 4.40 for the notations) is diagonalisable as the matrix of $s_{\pi/4}$ in the basis $\mathscr{B}_{\pi/4}$ is

$$[s_{\pi/4}]_{\mathscr{B}_{\pi/4}} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}.$$

(iii) The map

$$T : \mathscr{P}_n(\mathbb{R}) \to \mathscr{P}_n(\mathbb{R}), \quad P(x) \mapsto xP'(x)$$

is diagonalisable. Indeed, in the basis $\mathscr{B} = (1, x, \dots, x^n)$, we have

$$[T]_{\mathscr{B}} = \begin{pmatrix} 0 & & & & \\ & 1 & & & \\ & & 2 & & \\ & & & \ddots & \\ & & & & n \end{pmatrix}.$$

By seeing an $n \times n$ matrix as a linear map $T_A : \mathbb{R}^n \to \mathbb{R}^n$, we are lead to the following definition:

Definition 6.6. An $n \times n$ matrix A is *diagonalisable* (over $\mathbb{R}$) if there exists a basis of $\mathbb{R}^n$ made of eigenvectors of A.
Equivalently, A is diagonalisable if there exists an invertible $n \times n$ matrix P such that $P^{-1}AP$ is of the form:

$$P^{-1}AP = \begin{pmatrix} \lambda_1 & 0 & \dots & 0 \\ 0 & \lambda_2 & \dots & 0 \\ \vdots & & \ddots & \vdots \\ 0 & 0 & \dots & \lambda_n \end{pmatrix}.$$

Remark. The two definitions of diagonalisability for maps and matrices are compatible: a linear map T is diagonalisable if and only if the associated matrix $[T]_{\mathscr{B}}$ is diagonalisable for some (hence every) basis $\mathscr{B}$.

Distinct eigenvalues and linear independence. Here is an useful criterion:

Theorem 6.7. An $n \times n$ matrix has at most n eigenvalues.

This theorem is a direct consequence of the following result, together with the fact that a linearly independent family in an n-dimensional vector space contains at most n vectors.

Proposition 6.8. Let $\vec{x}_1, \ldots, \vec{x}_k$ be eigenvectors of a matrix A corresponding to distinct eigenvalues $\lambda_1, \ldots, \lambda_k$. Then $\vec{x}_1, \ldots, \vec{x}_k$ are linearly independent.

Proof. Assume that we have constants $c_1, \ldots c_k$ such that $c_1\vec{u}_1 + c_2\vec{u}_2 + \ldots + c_k\vec{u}_k = \vec{0}$. Let us denote this equation (E_1). By applying A on both sides of the equation, we get $c_1\lambda_1\vec{u}_1 + c_2\lambda_2\vec{u}_2 + \ldots + c_k\lambda_k\vec{u}_k = \vec{0}$. Let us denote this equation (E_2). Now by considering $(E_2) - \lambda_1(E_1)$, we get the equation $c_2(\lambda_2 - \lambda_1)\vec{u}_2 + \ldots + c_k(\lambda_k - \lambda_1)\vec{u}_k = \vec{0}$. We are thus back to an other linear combination of fewer eigenvectors, and we can now prove the result by induction. □

Here is a useful corollary:

Theorem 6.9. Let V be an n-dimensional vector space, and let $T : V \to V$ be a linear map that has exactly n distinct real eigenvalues $\lambda_1, \ldots, \lambda_n$, then T is diagonalisable.

Remark. The previous result is not an equivalence, there exist diagonalisable maps that have fewer than n eigenvalues. For instance, the identity map is diagonalisable (Why?), but its only eigenvalue is 1.

Eigenspaces and diagonalisability.

Definition 6.10. Given an eigenvalue $\lambda \in \mathbb{R}$, the associated *eigenspace* is the following subset:

$$E_\lambda(T) := \{\vec{x} \in U \mid T(\vec{x}) = \lambda\vec{x}\} = \{\vec{0}\} \cup \{\vec{x} \in U \mid \vec{x} \text{ is an eigenvector for } \lambda\}.$$

Theorem 6.11. An eigenspace is a subspace of V.

Proof. By definition, we have that $E_\lambda(T) = \ker(T - \lambda Id)$, and we know that kernels of linear maps are vector subspaces. □

The proof of the following result is a slight variation on the proof of Proposition 6.8:

Proposition 6.12. Let $\lambda_1, \ldots, \lambda_k$ be distinct eigenvalues of a linear map T. Let $\mathscr{B}_1, \ldots, \mathscr{B}_k$ be bases of the eigenspaces $E_{\lambda_1}(T), \ldots, E_{\lambda_k}(T)$ respectively. Then $\mathscr{B}_1 \cup \ldots \cup \mathscr{B}_k$ is a linearly independent family of vectors.

Theorem 6.13. Let $T : V \to V$ be a linear map and let $\lambda_1, \ldots, \lambda_k$ be the distinct eigenvalues of T. Then T is diagonalisable if and only if:

$$\dim E_{\lambda_1}(T) + \ldots + \dim E_{\lambda_k}(T) = n.$$

The next natural question to answer is: how do we find the eigenvalues of a given linear map or matrix?

6.3 Characteristic polynomial

Finding an eigenvalue λ and an associated eigenvector $\vec{x}$ amounts to finding solutions to the equation $T(\vec{x}) = \lambda\vec{x}$, or equivalently $(T - \lambda I_n)\vec{x} = \vec{0}$. Up to choosing coordinates, we can assume that T is represented by an $n \times n$ matrix A. The problem thus amounts to solving a linear system of equation $A\vec{x} = \lambda\vec{x}$, or equivalently $(A - \lambda I_n)\vec{x} = \vec{0}$. This is a homogeneous system of n equations in n unknowns. It has a non-trivial solution if and only if $(A - \lambda I_n)$ is not invertible, and by Proposition 4.19 this is equivalent to $\det(A - \lambda I_n) = 0$. This motivates the following definition:

Definition 6.14. Given an $n \times n$-matrix A, the expression

$$\chi_A(\lambda) := \det(A - \lambda I_n) = \det \begin{pmatrix} a_{11} - \lambda & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} - \lambda & \dots & a_{2n} \\ \vdots & & \ddots & \vdots \\ a_{n1} & a_{n2} & \dots & a_{nn} - \lambda \end{pmatrix}$$

is a polynomial of degree n in λ, called the *characteristic polynomial* of the matrix A.

Remark. In this course, we will mostly compute characteristic polynomials for 2×2 and 3×3 matrices. When computing characteristic polynomials in general, one compute this determinant using the expansion technique.

The definition of the characteristic polynomial is motivated by the following result:

Theorem 6.15. Given an $n \times n$-matrix A, we have:

$$\lambda \text{ is an eigenvalue of } A \Leftrightarrow \chi_A(\lambda) = 0.$$

Proof. We have:

$$\begin{aligned} \lambda \text{ is an eigenvalue of } A &\Leftrightarrow \exists \vec{x} \neq \vec{0} \text{ such that } A\vec{x} = \lambda\vec{x}. \\ &\Leftrightarrow \exists \vec{x} \neq \vec{0} \text{ such that } (A - \lambda I_n)\vec{x} = \vec{0}. \\ &\Leftrightarrow \ker(A - \lambda I_n) \neq \{\vec{0}\}. \\ &\Leftrightarrow A - \lambda I_n \text{ is not invertible.} \\ &\Leftrightarrow \det(A - \lambda I_n) = 0 \quad \square \end{aligned}$$

Example 6.16. Let us find the eigenvalues of the matrix

$$A = \begin{pmatrix} 3 & 2 & 4 \\ 2 & 0 & 2 \\ 4 & 2 & 3 \end{pmatrix}.$$

The characteristic polynomial of A reads as (rule of Sarrus)

$$\det \begin{pmatrix} 3 - \lambda & 2 & 4 \\ 2 & -\lambda & 2 \\ 4 & 2 & 3 - \lambda \end{pmatrix} = -(3 - \lambda)^2\lambda + 16 + 16 + 16\lambda - 4(3 - \lambda) - 4(3 - \lambda),$$

or $\chi_A(\lambda) = -\lambda^3 + 6\lambda^2 + 15\lambda + 8$. We see that $\lambda = -1$ is a root of this polynomial and factorise it as $-(\lambda+1)(\lambda+1)(\lambda-8)$. That is, the eigenvalues of A are -1 and 8.

The characteristic polynomial seen as a polynomial over $\mathbb{C}$. By the fundamental theorem of algebra, the characteristic polynomial of an $n \times n$ matrix A can be factorised into n linear factors:

$$\chi_A(\lambda) = \pm(\lambda - \lambda_1)(\lambda - \lambda_2)\dots(\lambda - \lambda_n)\,,$$

where λ_i are (not necessarily distinct) *complex*. These λ_i are called the *complex eigenvalues* of A, and the eigenvalues we defined so far are the complex eigenvalues that are real. Since a polynomial of degree $n \geq 1$ has at most n roots, we recover the fact that A has at most n complex eigenvalues, hence at most n real eigenvalues.

Remark. A matrix may have no real eigenvalues, e.g.

$$A = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \quad \det\begin{pmatrix} -\lambda & 1 \\ -1 & -\lambda \end{pmatrix} = \lambda^2 + 1 = (\lambda + \mathrm{i})(\lambda - \mathrm{i}) = \chi_A(\lambda)$$

has a characteristic polynomial $\chi_A(\lambda)$ without real roots. However, since a polynomial of degre ≥ 1 always has at least one root by the fundamental theorem of algebra, A always has at least one complex eigenvalue.

There is one simple case where the eigenvalues can be read directly from a matrix:

Theorem 6.17. Let

$$A = \begin{pmatrix} d_1 & * & \cdots & * \\ 0 & d_2 & \ddots & \vdots \\ \vdots & \ddots & \ddots & * \\ 0 & \cdots & 0 & d_n \end{pmatrix} \in M_n(\mathbb{R})$$

be an upper-triangular $n \times n$ matrix. Then the eigenvalues of A are $d_1, \dots, d_n$.
The same result holds for lower-triangular matrices.

Proof. By Theorem 2.35, $\chi_A(x) = (d_1 - x)\cdots(d_n - x)$, and the result follows from Theorem 6.15. □

Warning! When computing the row echelon form of a matrix via Gaussian elimination, the diagonal entries in the echelon form are **not** the eigenvalues of A in general.

Trace, determinants, and consistency check.

Definition 6.18. The *trace* of a square matrix A, denoted $\mathrm{tr}(A)$, is the sum of its diagonal coefficients.

Proposition 6.19. We have the following properties:

- The trace defines a linear map $tr : M_n(\mathbb{R}) \to \mathbb{R}, A \mapsto \mathrm{tr}(A)$.
- We have $\mathrm{tr}(AB) = \mathrm{tr}(BA)$ for every pair of matrices $A, B \in M_n(\mathbb{R})$.

In particular, we get that for every invertible matrix P, $\mathrm{tr}(PAP^{-1}) = \mathrm{tr}(AP^{-1}P) = \mathrm{tr}(A)$. Similarly, we have $\det(PAP^{-1}) = \det(P)\det(A)\det(P)^{-1} = \det(A)$. In other words, the trace and the determinant remain the same when changing basis. If A is diagonalisable, then these quantities can be computed in terms of the eigenvalues of A in an appropriate basis. This leads to the following useful result.

Theorem 6.20. Let A be a diagonalisable $n \times n$ matrix, and let $\lambda_1, \ldots, \lambda_n$ be its eigenvalues counted with multiplicity, i.e. the roots of χ_A counted with multiplicity. Then we have

$$\det(A) = \lambda_1 \times \ldots \times \lambda_n.$$

$$\mathrm{tr}(A) = \lambda_1 + \ldots + \lambda_n.$$

This result is useful to check your computations, as the trace in particular is very easy to compute. Try to always use this consistency check!

Example 6.21. Let us go back to the matrix

$$A = \begin{pmatrix} 3 & 2 & 4 \\ 2 & 0 & 2 \\ 4 & 2 & 3 \end{pmatrix}$$

whose eigenvalues we computed previously: $-1, -1, 8$. In particular, we get that the sum of the eigenvalues is 6, which does indeed coincide with $\mathrm{tr}(A) = 3 + 0 + 3$. Similarly, the product of the eigenvalues is 8, which we recover from Sarrus' rule. We can thus be reasonably confident that we did not make mistakes in computing the eigenvalues of A.

6.4 Diagonalisability of a matrix

Definition 6.22. Let $A \in M_n(\mathbb{R})$ and let λ be an eigenvalue of A.

- the *geometric multiplicity* of λ is the dimension of the corresponding eigenspace $E_\lambda(A)$.
- the *algebraic multiplicity* of λ is number of factors $(x - \lambda)$ that appear in χ_A.

Example 6.23. Let us consider again the matrix

$$A = \begin{pmatrix} 3 & 2 & 4 \\ 2 & 0 & 2 \\ 4 & 2 & 3 \end{pmatrix}.$$

We have $\chi_A(x) = (8 - x)(x + 1)^2$, so the eigenvalues of A are -1 and 8, and their algebraic multiplicity are 2 and 1 respectively.
Let us compute the dimension of the corresponding eigenspaces using Gaussian elimination. For the eigenvalue 8, we obtain the system:

$$\begin{cases} 3x + 2y + 4z &= 8x \\ 2x + 2z &= 8y \\ 4x + 2y + 3z &= 8z \end{cases} \quad \rightsquigarrow \quad \begin{cases} 2x - 8y + 2z &= 0 \\ -5x + 2y + 4z &= 0 \\ 4x + 2y - 5z &= 0 \end{cases}$$

We use Gaussian elimination:

$$\left(\begin{array}{ccc|c} 1 & -4 & 1 & 0 \\ -5 & 2 & 4 & 0 \\ 4 & 2 & -5 & 0 \end{array}\right) \rightsquigarrow \left(\begin{array}{ccc|c} 1 & -4 & 1 & 0 \\ 0 & -18 & 9 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right).$$

The row echelon form has exactly one free variable, so the dimension of the corresponding eigenspace is 1, so the geometric dimension of the eigenvalue $\lambda = 8$ is 1.
For the eigenvalue $\lambda = -1$, we get the system of equations

$$\left\{\begin{array}{rcl} 3x+2y+4z & = & -x \\ 2x+2z & = & -y \\ 4x+2y+3z & = & -z \end{array}\right. \rightsquigarrow \left\{\begin{array}{rcl} 4x+2y+4z & = & 0 \\ 2x+y+2z & = & 0 \\ 4x+2y+4z & = & 0 \end{array}\right.$$

All of these equations are multiples of $2x+y+2z=0$, so the system boils down to the equation $2x+y+2z=0$. This system has exactly two free variables (y and z), and so the dimension of the corresponding eigenspace is 2. In other words, the geometric multiplicity of the eigenvalue $\lambda = -1$ is 2.
In particular, we see in this particular example that geometric and algebraic multiplicity coincide.

Example 6.24. Consider the following matrix:

$$A = \left(\begin{array}{ccc} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{array}\right).$$

Since A is upper-triangular, we have $\chi_A(x) = (1-x)^3$. Thus, $\lambda = 1$ is the unique eigenvalue of A and its algebraic multiplicity is 3.
We find the geometric multiplicity by solving the system $(A - 1 \times I_3)\vec{x} = \vec{0}$, which yields the system:

$$\left(\begin{array}{ccc|c} 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right).$$

This system is already in row echelon form and has exactly one free variable. So the geometric multiplicity of $\lambda = 1$ is 1.
In this particular example, the algebraic and geometric multiplicity differ.

Proposition 6.25. For every eigenvalue of A, the geometric multiplicity is less than or equal to the algebraic multiplicity.

Theorem 6.26. An $n \times n$ matrix A is diagonalisable (over $\mathbb{R}$) if and only if:

- all the complex roots of χ_A are real numbers.
- for every eigenvalue λ of A, the algebraic multiplicity of λ equals the geometric multiplicity of λ.

Example 6.27. Here are examples of matrices which are not diagonalisable:

$$A = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \quad B = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}.$$

We have $\chi_A(x) = x^2 + 1$, so the matrix A has no real eigenvalue (but two complex eigenvalues: i and $-i$), and therefore cannot be diagonalised as a real matrix.
We have $\chi_B(x) = (x-1)^2$, so the only eigenvalue of the matrix B is 1 (note that B is upper triangular), which has algebraic multiplicity 2, but the associated eigenspace has dimension 1, so in particular the algebraic and geometric multiplicity do not coincide. Thus, B is not diagonalisable over $\mathbb{R}$.

Example 6.28. Let us continue the study of the matrix $A = \begin{pmatrix} 3 & 2 & 4 \\ 2 & 0 & 2 \\ 4 & 2 & 3 \end{pmatrix}$. We already know that $\chi_A(x) = (8-x)(x+1)^2$, so the eigenvalues of A are -1 and 8, and we showed in previous examples that the algebraic and geometric multiplicity coincide for each eigenvalue. By the previous theorem, A is diagonalisable.

Method: Diagonalisability of a matrix (over $\mathbb{R}$)

We have the following method to determine whether a given matrix is diagonalisable (over $\mathbb{R}$):

(1) Compute the characteristic polynomial χ_A of A and compute its roots. For $n > 3$, you will generally be given some eigenvalues to help you with the computations.

(2) If χ_A possesses a non-real complex root, then A is not diagonalisable over $\mathbb{R}$.

(3) Otherwise, let $\lambda_1, \ldots, \lambda_k$ be the (real) roots of χ_A. Compute the algebraic multiplicity of each eigenvalue by factorising χ_A.

(4) For each eigenvalue λ_i, compute the geometric multiplicity of λ_i by computing the dimension of the solution space of the system $A\vec{x} - \lambda_i \vec{x} = \vec{0}$ using Gaussian elimination.

(5) If for some eigenvalue λ, the geometric and algebraic multiplicities of λ differ, then A is not diagonalisable.

(6) To find a basis of eigenvectors, compute for each eigenvalue λ_i a basis $\mathscr{B}_i$ of the corresponding eigenspace, using Gaussian elimination. A basis of eigenvectors is then given by $\mathscr{B}_1 \cup \ldots \cup \mathscr{B}_k$.

Remark 6.29. In the previous theorem, it is enough to check that algebraic and geometric multiplicity coincide for eigenvalues of algebraic multiplicity at least 2, as the algebraic and geometric multiplicity necessarily coincide for eigenvalues of algebraic multiplicity 1 (Why?).

Example 6.30. Let us finish the study of the matrix $A = \begin{pmatrix} 3 & 2 & 4 \\ 2 & 0 & 2 \\ 4 & 2 & 3 \end{pmatrix}$. We already know that $\chi_A(x) = (8-x)(x+1)^2$, so the eigenvalues of A are -1 and 8, and we showed in previous examples that the algebraic and geometric multiplicity coincide for each eigenvalue. By the previous

theorem, A is diagonalisable.
Let us compute a basis of eigenvectors. For the eigenvalue 8, we solve the system:

$$\begin{cases} 3x+2y+4z &= 8x \\ 2x+2z &= 8y \\ 4x+2y+3z &= 8z \end{cases} \rightsquigarrow \begin{cases} 2x-8y+2z &= 0 \\ -5x+2y+4z &= 0 \\ 4x+2y-5z &= 0 \end{cases}$$

We use Gaussian elimination:

$$\left(\begin{array}{ccc|c} 1 & -4 & 1 & 0 \\ -5 & 2 & 4 & 0 \\ 4 & 2 & -5 & 0 \end{array}\right) \overset{\substack{R_2 \to R_2+5R_1 \\ R_3 \to R_3-4R_1}}{\rightsquigarrow} \left(\begin{array}{ccc|c} 1 & -4 & 1 & 0 \\ 0 & -18 & 9 & 0 \\ 0 & 18 & -9 & 0 \end{array}\right).$$

By substitution, the solutions of this homogeneous system are $\vec{x} = (\alpha, \frac{1}{2}\alpha, \alpha)^T$, so a basis for $E_8(A)$ is $\mathscr{B}_8 = \{(1, \frac{1}{2}, 1)^T\}$.
For the eigenvalue $\lambda = -1$, we get the system of equations

$$\begin{cases} 3x+2y+4z &= -x \\ 2x+2z &= -y \\ 4x+2y+3z &= -z \end{cases} \rightsquigarrow \{\ 2x+y+2z \ = \ 0$$

By substitution, we find that the solution to this system of linear equations is of the form $(-\frac{1}{2}(2\alpha+\beta), \beta, \alpha)^T$ for $\alpha, \beta \in \mathbb{R}$. Thus, a basis of $E_{-1}(A)$ is given by $\mathscr{B}_{-1} = \{(-1,0,1)^T, (-\frac{1}{2}, 1, 0)^T\}$. Finally, a basis of eigenvectors is $\mathscr{B} = \{(1, \frac{1}{2}, 1)^T, (-1,0,1)^T, (-\frac{1}{2}, 1, 0)^T\}$.
Seeing A as a linear map $T_A : \mathbb{R}^3 \to \mathbb{R}^3$, let us perform a change of basis. Let

$$P := \begin{pmatrix} 1 & -1 & -\frac{1}{2} \\ -\frac{1}{2} & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix}.$$

be the matrix of change of basis from the standard basis to our basis of eigenvectors. In particular, the matrix of T_A in the basis of eigenvectors $\mathscr{B}$ is:

$$P^{-1}AP = \begin{pmatrix} 8 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{pmatrix}.$$

6.5 Application: Dynamical systems

Definition 6.31. The *dynamical system* associated to an $n \times n$ matrix A and a choice of initial condition $\vec{x}_0 \in \mathbb{R}^n$ is the sequence $(\vec{x}_k)_{k \geq 0}$ of vectors of $\mathbb{R}^n$ defined by:

$$\vec{x}_{k+1} = A\vec{x}_k.$$

In particular, we get by induction that:

$$\vec{x}_k = A^k \vec{x}_0.$$

Theorem 6.32. Let A be a diagonalisable $n \times n$ matrix with a chosen basis of eigenvectors $\vec{e}_1, \ldots, \vec{e}_n$ and corresponding eigenvalues $\lambda_1, \ldots, \lambda_n$. Let $(\vec{x}_k)$ be the dynamical system associated to a given initial condition $\vec{x}_0 \in \mathbb{R}^n$. Let $\alpha_1, \ldots, \alpha_k \in \mathbb{R}$ such that

$$\vec{x}_0 = \alpha_1 \vec{e}_1 + \ldots + \alpha_n \vec{e}_n.$$

Then we have for every $k \geq 0$,

$$\vec{x}_k = A^k \vec{x}_0 = \alpha_1 \lambda_1^k \vec{e}_1 + \ldots + \alpha_k \lambda_n^k \vec{e}_n.$$

Method: Asymptotic behaviour of dynamical systems

To determine the asymptotic behaviour (convergence, divergence, possible limit) of a dynamical system associated to a $n \times n$ diagonalisable matrix A:

(1) Compute a basis of eigenvectors $\vec{e}_1, \ldots, \vec{e}_n$ of A, with corresponding eigenvalues $\lambda_1, \ldots, \lambda_n$, as in the previous method.

(2) Using Gaussian elimination, find constants $\alpha_1, \ldots, \alpha_n$ such that

$$\vec{x}_0 = \alpha_1 \vec{e}_1 + \ldots + \alpha_n \vec{e}_n.$$

(3) We then have

$$\vec{x}_k = A^k \vec{x}_0 = \alpha_1 \lambda_1^k \vec{e}_1 + \ldots + \alpha_k \lambda_n^k \vec{e}_n.$$

This formula can be used to determine the asymptotic behaviour of the dynamical system.

Application: Dynamics of populations. Let us go back to the problem of understanding the dynamics of some animal population. We have:

$$\begin{pmatrix} a_{n+1} \\ j_{n+1} \end{pmatrix} = A \begin{pmatrix} a_n \\ j_n \end{pmatrix} \quad \text{with } A := \begin{pmatrix} \alpha & \beta \\ \gamma & 0 \end{pmatrix}.$$

For simplicity and concreteness, assume that $\alpha = \frac{1}{2}$, $\beta = \frac{3}{10}$. The characteristic polynomial of A is:

$$\chi_A(x) = (\frac{1}{2} - x)(-x) - \frac{3}{10}\gamma = x^2 - \frac{1}{2}x - \frac{3}{10}\gamma.$$

The eigenvalues of A are $\lambda_1 = \frac{1}{20}(5 - \sqrt{5}\sqrt{5+24\gamma})$ and $\lambda_2 = \frac{1}{20}(5 + \sqrt{5}\sqrt{5+24\gamma})$. In particular, A has two distinct eigenvalues $\lambda_1 < \lambda_2$, hence it is diagonalisable by Theorem 6.9. We can thus find a basis $\vec{e}_1, \vec{e}_2$ of eigenvectors for $\mathbb{R}^2$. If we write the initial vector in this basis as:

$$\begin{pmatrix} a_0 \\ j_0 \end{pmatrix} = \alpha_1 \vec{e}_1 + \alpha_2 \vec{e}_2,$$

we get

$$\begin{pmatrix} a_n \\ j_n \end{pmatrix} = A^n \begin{pmatrix} a_0 \\ j_0 \end{pmatrix} = \alpha_1 \lambda_1^n \vec{e}_1 + \alpha_2 \lambda_2^n \vec{e}_2.$$

We thus see that the dynamics of this population depend on the eigenvalues, and hence on the fecundity rate γ:

(i) If $\lambda_1, \lambda_2 < 1$, that is, if $\gamma < \frac{5}{3}$, then $a_n, j_n \to 0$ and the population collapses.

(ii) If $\lambda_2 > 1$, that is, if $\gamma > \frac{5}{3}$, then $a_n, j_n \to \infty$ and the population explodes.

(iii) Finally, if $\lambda_2 = 1$ and $\lambda_1 < 1$, that is, if $\gamma = \frac{5}{3}$, then

$$\begin{pmatrix} a_n \\ j_n \end{pmatrix} \to \alpha_2 \vec{e}_2 = \alpha_2 \begin{pmatrix} \frac{3}{5} \\ 1 \end{pmatrix}.$$

In particular, the population converges to a stable population where the number a, j of adults and juveniles satisfy:

$$a = \frac{3}{5} j.$$

This mathematical model to understand the evolution of a population by dividing it into several age groups is known as the *Leslie model*.

Application: The Fibonacci sequence. Consider the sequence satisfying the following recurrence relation:

$$a_{n+1} = a_n + a_{n-1},$$

and with the initial values $a_0 = 0$ and $a_1 = 1$. The numbers a_i are known as the *Fibonacci numbers*: $0, 1, 1, 2, 3, 5, 8, 13, \ldots$. Let us introduce the following vector of $\mathbb{R}^2$:

$$\vec{x}_n := \begin{pmatrix} a_{n+1} \\ a_n \end{pmatrix} \quad \text{for } n \geq 0.$$

The recurrence relation gets encoded into the following matrix equation (Check it!):

$$\vec{x}_{n+1} = A\vec{x}_n$$

with

$$A := \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}.$$

To compute a_{n+1} from given a_1 and a_0, we just have to apply A^n to the initial vector. The characteristic polynomial χ_A is

$$\chi_A(x) = (1-x)(-x) - 1 = x^2 - x - 1,$$

and its eigenvalues are $\varphi := \frac{1+\sqrt{5}}{2}$ and $\varphi' := \frac{1-\sqrt{5}}{2}$. In particular, the 2×2 matrix A has two distinct eigenvalues, hence it is diagonalisable by Theorem 6.9. We compute the corresponding eigenvectors $\vec{e}_\varphi = (\frac{1}{2}(1+\sqrt{5}), 1)^T$ and $\vec{e}_{\varphi'} = (\frac{1}{2}(1-\sqrt{5}), 1)^T$ (Check it!). One then writes the initial vector $\vec{x}_0$ in this basis of eigenvectors by solving the system of linear equations

$$\vec{x}_0 = a\vec{e}_\varphi + b\vec{e}_{\varphi'},$$

for some $a, b \in \mathbb{R}$, using Gaussian elimination. One finds (Check it!):

$$\vec{x}_0 = \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \frac{1}{\sqrt{5}}\vec{e}_\varphi - \frac{1}{\sqrt{5}}\vec{e}_{\varphi'}.$$

If we apply A^n to our initial vector, we obtain

$$\vec{x}_n = A^n \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \frac{1}{\sqrt{5}}\varphi^n \vec{e}_\varphi - \frac{1}{\sqrt{5}}(\varphi')^n \vec{e}_{\varphi'}.$$

and by taking the second component of $\vec{x}_n$, one gets

$$a_n = \frac{\varphi^n - (\varphi')^n}{\sqrt{5}}.$$

Sequences satisfying a linear recurrence relation. More generally, for a sequence (a_n) satisfying a linear recurrence relation of the form

$$a_{n+m} = c_{m-1}a_{n+m-1} + \ldots + c_0 a_n$$

for some constants $c_0, \ldots, c_m \in \mathbb{R}$, we can compute a_n using the same procedure. First introduce the following vector of $\mathbb{R}^m$:

$$\vec{x}_n = \begin{pmatrix} a_n \\ \vdots \\ a_{n+m-1} \end{pmatrix} \text{ for } n \geq 0,$$

And notice that we have

$$\vec{x}_{n+1} = A\vec{x}_n$$

with

$$A = \begin{pmatrix} c_{m-1} & c_{m-2} & \ldots & c_1 & c_0 \\ 1 & 0 & \ldots & 0 & 0 \\ 0 & 1 & 0 & \ldots & 0 \\ \vdots & & \ddots & \ddots & \vdots \\ 0 & \ldots & 0 & 1 & 0 \end{pmatrix}.$$

One then compute the characteristic polynomial χ_A. One can show that

$$\chi_A(x) = x^m - c_{m-1}x^{m-1} - \ldots - c_1 x - c_0.$$

One can then compute the eigenvalues and eigenvectors. If A turns out to be diagonalisable, with a basis $\mathscr{B}$ of eigenvectors $\vec{e}_1, \ldots, \vec{e}_m$ and associated eigenvalues $\lambda_1, \ldots, \lambda_m$ respectively, one then express the initial vector $\vec{x}_0$ in the basis $\mathscr{B}$ by solving the associated system of linear equations. If

$$\vec{x}_0 = \alpha_1\vec{e}_1 + \ldots + \alpha_n\vec{e}_n,$$

then it follows that

$$\vec{x}_n = A^n\vec{x}_0 = \alpha_1\lambda_1^n\vec{e}_1 + \ldots + \alpha_n\lambda_m^n\vec{e}_m,$$

and one finally obtains a_n by taking the first component of $\vec{x}_n$.

Diagonalisability over $\mathbb{C}$ and dynamical systems. In this course, we have mostly been dealing with real vector spaces and diagonalisability over $\mathbb{R}$. When it comes to diagonalisability however, using complex numbers instead can be very useful: A matrix may be diagonalisable over $\mathbb{C}$ (in which case the previous methods apply) even though the same matrix is not diagonalisable over $\mathbb{R}$.
Here is a concrete example. Consider a sequence satisfying the following linear recurrence relation:

$$u_{n+2} = -u_{n+1} - u_n.$$

By applying the previous method, we end up studying a dynamical system associated to a 2×2 matrix whose characteristic polynomial is $x^2 + x + 1$. This polynomial does not admit any real root, so the associated matrix is not diagonalisable *over* $\mathbb{R}$. However, over $\mathbb{C}$, the characteristic polynomial has two roots, $\lambda_1 = e^{2i\pi/3}$ and $\lambda_2 = e^{-2i\pi/3}$. One can show that the associated matrix is diagonalisable over $\mathbb{C}$, and as a result, there exists constants $\alpha_1, \alpha_2 \in \mathbb{C}$ such that for every $n \geq 0$, we have

$$u_n = \alpha_1\lambda_1^n + \alpha_2\lambda_2^n.$$

We can then compute the values of α_1, α_2 using the initial conditions u_0, u_1. For instance, for $u_0 = 2$ and $u_1 = 1$, we find $\alpha_1 = \alpha_2 = 1$, and finally

$$u_n = e^{2in\pi/3} + e^{-2in\pi/3} = 2\cos(2n\pi/3).$$

6.6 Powers of a matrix.

Diagonalising a matrix helps to compute its powers. Assume that we have diagonalised an $n \times n$ matrix A: $A = PDP^{-1}$, where D is a diagonal matrix with entries $d_1, \ldots, d_n$. We then have

$$A^k = (PDP^{-1})^k = PDP^{-1}PDP^{-1}\ldots PDP^{-1} = PD^kP^{-1}\,.$$

Here, D^k is simply the diagonal matrix with entries $d_1^k, \ldots, d_n^k$.

Method: Computing the powers of a diagonalisable matrix

We have the following algorithm to compute the powers of a diagonalisable matrix A:

1) Using the previous method, compute a diagonal matrix D and an invertible matrix P such that $A = PDP^{-1}$.
2) We then have $A^k = PD^kP^{-1}$ for every $k \in \mathbb{Z}$.

Example 6.33. Let

$$A = \begin{pmatrix} 2 & 2 \\ 1 & 3 \end{pmatrix}.$$

Let us compute A^{20}.
We compute the characteristic polynomial $\chi_A(\lambda) = \lambda^2 - 5\lambda + 4$, and so the eigenvalues of A are $\lambda = 1$ and $\lambda = 4$. In particular, A is a 2×2 matrix with two distinct eigenvalues so it is diagonalisable. Using Gaussian elimination, we find that a basis of $E_1(A)$ is the vector $(2,-1)^T$ and a basis of $E_4(A)$ is the vector $(1,1)^T$ (Check it!). We thus define the matrix of change of basis

$$P = \begin{pmatrix} 2 & 1 \\ -1 & 1 \end{pmatrix}$$

whose inverse we compute by Gaussian elimination (Do it!):

$$P^{-1} = \frac{1}{3}\begin{pmatrix} 1 & -1 \\ 1 & 2 \end{pmatrix}.$$

We thus have

$$A = \begin{pmatrix} 2 & 2 \\ 1 & 3 \end{pmatrix} = PDP^{-1} = \begin{pmatrix} 2 & 1 \\ -1 & 1 \end{pmatrix}\begin{pmatrix} 1 & 0 \\ 0 & 4 \end{pmatrix}\frac{1}{3}\begin{pmatrix} 1 & -1 \\ 1 & 2 \end{pmatrix}.$$

and it follows that

$$\begin{aligned} A^{20} = PD^{20}P^{-1} &= \tfrac{1}{3}\begin{pmatrix} 2 & 1 \\ -1 & 1 \end{pmatrix}\begin{pmatrix} 1 & 0 \\ 0 & 4^{20} \end{pmatrix}\begin{pmatrix} 1 & -1 \\ 1 & 2 \end{pmatrix} \\ &= \tfrac{1}{3}\begin{pmatrix} 2+4^{20} & -2+2\times 4^{20} \\ -1+4^{20} & 1+2\times 4^{20} \end{pmatrix} \end{aligned}$$

Chapter 7

Stochastic matrices and equilibrium

In the previous chapter, we introduced new tools to study (linear) dynamical systems such as dynamics of animal population. In such a model, animals die and are born, and the total number of animals can vary (and as we have seen, the asymptotic behaviour is closely related to the eigenavlues of the associated matrix.)
In this chapter, we focus on a class of dynamical systems where the total number of 'elements' under study stays the same, modelling instead a phenomenon of *redistribution*. As we will see in the following motivating problem, many such dynamical systems reach some equilibrium, and the goal of this chapter will be to understand why.

7.1 A motivating problem: Optimising a distribution of bikes

A company is planning to introduce to-rent bikes in Edinburgh. The plan is to have various spots scattered in four main zones: Old Town (zone 1), Haymarket (zone 2), Fountainbridge (zone 3), and Newington (zone 4). During a trial period, the company monitored the displacements of 200 bikes to understand the behaviour of their possible customers. They noticed the following:

(i) On average, 50% of bikes in zone 1 are parked back in zone 1 at the beginning of the next day, 20% are parked in zone 2, 20% in zone 3, and 10% in zone 4.

(ii) On average, 20% of bikes in zone 2 are parked back at zone 2 at the beginning of the next day, 40% are parked in zone 1, 20% in zone 3, and 20% in zone 4.

(iii) On average, 40% of bikes in zone 3 are parked back in zone 3 at the beginning of the next day, 30% are parked in zone 1, 20% in zone 2, and 10% in zone 4.

(iv) On average, 50% of bikes in zone 4 are parked back in zone 4 at the beginning of the next day, 10% are parked in zone 1, 20% in zone 2, and 20% in zone 3.

A natural question (in order to decide the amount of parking slots needed) is to decide whether there exists a stable distribution of bikes, that is, so that the number of bikes parked at the beginning of every day remains constant over time. If we denote by $(x_1,\dots,x_4)^T$ the vector whose components encode the distribution of bikes in zone 1, …, 4 respectively, this amounts to solving the following equation (Check it!):

$$\begin{pmatrix} 0.5 & 0.4 & 0.3 & 0.1 \\ 0.2 & 0.2 & 0.2 & 0.2 \\ 0.2 & 0.2 & 0.4 & 0.2 \\ 0.1 & 0.2 & 0.1 & 0.5 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix}.$$

This computation was done in Tutorial 1, and we found that the unique solution to this system was the vector $(70,40,50,40)^T$.
However, the company might just have decided to start with an equal distribution of 50 bicycles at each location. In that situation, one can study the evolution over time of the distribution of bicycles. Indeed, if we denote by $\vec{x}_k$ the vector whose components encode the average distribution of bicycles in zones 1, to 4 after k days, then

$$\vec{x}_k = \begin{pmatrix} 0.5 & 0.4 & 0.3 & 0.1 \\ 0.2 & 0.2 & 0.2 & 0.2 \\ 0.2 & 0.2 & 0.4 & 0.2 \\ 0.1 & 0.2 & 0.1 & 0.5 \end{pmatrix}^k \begin{pmatrix} 50 \\ 50 \\ 50 \\ 50 \end{pmatrix}.$$

We find:

$$\vec{x}_1 = \begin{pmatrix} 65 \\ 40 \\ 50 \\ 45 \end{pmatrix}, \quad \vec{x}_2 = \begin{pmatrix} 68 \\ 40 \\ 50 \\ 42 \end{pmatrix}, \quad \vec{x}_3 = \begin{pmatrix} 69.2 \\ 40 \\ 50 \\ 40.8 \end{pmatrix}, \quad \vec{x}_4 \simeq \begin{pmatrix} 69.7 \\ 40 \\ 50 \\ 40.3 \end{pmatrix}, \ldots$$

In particular, we see that, over time, the distribution seems to converge (rather quickly) to the stable distribution we computed before. There are thus two natural questions to ask oneself:

- Does that sort of system (and what sort of system exactly are we looking at?) always have a stable distribution?
- Does the distribution always evolve over time towards the same stable distribution, regardless of the initial conditions?

The goal of this final chapter will be to answer in the affirmative these two questions, and to see further applications.

7.2 Markov diagrams and transition matrices

The dynamical systems we wish to encode correspond to the evolution over time of a set of 'elements' (bikes, people, animals, etc.) that can be in a certain number of 'states'. At every time, each element can either stay in the same state or change to a different state with a given probability. The only condition is that this process is *without memory*, that is, the probability from moving to a state s_j to a state s_i depends only on the states s_i, s_j, and not of the states the process was at at previous times. Such processes, heavily studied in probability theory and statistics, are called *finite Markov chains*.

Example 7.1. We give here some examples to illustrate the type of processes we are trying to model:

(i) The study of bikes mentioned in the introduction can be modelled in this way: the set of elements correspond to the 200 bikes considered, the states correspond to the four possible zones, and the averaged probabilities measured in the introduction give the probability of a bike to move from one state to an other after one day.

(ii) We flip a fair coin and record the occurrences of heads and tails. Here, the set of elements considered consists of the coin itself, and the two possible states are 'heads' and 'tails'. Since the coin is fair, the probability from moving from one state to the other is always $\frac{1}{2}$.

Example 7.2. Here is an example of a process that cannot be modelled in this way. We play a game by flipping a fair coin as follows: We start with 10 pounds. Every time we get heads we win 1 pound, and every we time we get tails we lose 1 pound. However, being reasonable, we decide to stop playing if we lose three times in a row. One way we could try to model this game is by taking the set of elements to consist of a single player, and the three states can be: 'win', 'lose', and 'stop playing', and we go from one state to the other depending on the outcome of the coin flips. This process however does not fit the previous description. Indeed, knowing that we just lost 1 pound (i.e. we are currently on the 'lose' state), we cannot tell the probability to move to the other states without knowledge of the prior events: if we had already lost twice before, then we will stop playing and move to state 'stop playing' with probability 1. Otherwise, we will keep playing and move to state 'win' or 'lose' with probability $\frac{1}{2}$.

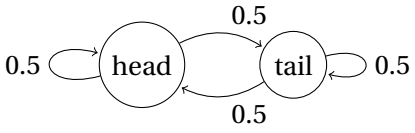
Now that we have some intuition about the type of processes we are trying to model, we will present an algebraic framework to study them. The key thing is to encode the various probabilities of moving from one state to another in a workable way. This is done as follows, using a *Markov diagram*.

Definition 7.3. A *Markov diagram* is a finite labelled oriented graph:

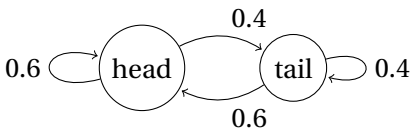
- for each pair of vertices v_i, v_j, there is at most one oriented edge from v_j to v_i, and the label of that edge is an element $a_{ij} \in (0,1]$.
- for each vertex v_j, we have:
$$\sum_{i=1}^{n} a_{ij} = 1.$$

Intuitively, a Markov diagram encodes (in a diagrammatic way) the types of processes we are interested: the vertices correspond to the states of our process, and an oriented edge with label a_{ij} from the vertex v_j to the vertex v_i corresponds to a probability a_{ij} of moving from state v_j to state v_i. We only indicate non-zero probabilities in the diagram, hence the requirement that $a_{ij} \in (0,1]$. Moreover, the second condition simply means that the diagram encodes all the possible transitions from one state to the other: if we are at a state v_j at a given time, with probability 1 ($= \sum_{i=1}^{n} a_{ij}$) we will move to one of the other states indicated in the diagram by following an oriented edge starting at v_j.

Example 7.4. Let us consider again the example of the coin flip, and the coin is assumed to be fair: heads and tails happen with probability $\frac{1}{2}$. The Markov diagram for this process is the following:



If instead we considered an unfair coin giving a head with a probability of 60%, then the Markov diagram would be:



Another way to encode a this data is by means of a matrix, called a *transition matrix*, which is particularly useful to understand the evolution over time of the process.

Definition 7.5. The *transition matrix* associated to a Markov diagram with n states is the $n \times n$ matrix defined by:

$$a_{ij} = \text{probablity of going from state } s_j \text{ at time } n \text{ to state } s_i \text{ at time } n+1.$$

Example 7.6. (i) The transition matrix associated to the fair flip coin above is:

$$A = \begin{pmatrix} 0.5 & 0.5 \\ 0.5 & 0.5 \end{pmatrix},$$

while the transition matrix associated to the unfair flip coin above is:

$$A = \begin{pmatrix} 0.6 & 0.6 \\ 0.4 & 0.4 \end{pmatrix},$$

(ii) The introductory example involving bikes can be modelled using a Markov diagram with four states $s_1, \ldots, s_4$ corresponding to the four zones $1, \ldots, 4$ respectively. The associated transition matrix is:

$$A = \begin{pmatrix} 0.5 & 0.4 & 0.3 & 0.1 \\ 0.2 & 0.2 & 0.2 & 0.2 \\ 0.2 & 0.2 & 0.4 & 0.2 \\ 0.1 & 0.2 & 0.1 & 0.5 \end{pmatrix}.$$

Note that a transition matrix is a particular example of a stochastic matrix, as defined below:

Definition 7.7. An $n \times n$ matrix is called a *stochastic matrix* if the following holds:

(S1) $a_{ij} \geq 0$ for every $1 \leq i, j \leq n$,

(S2) for every column C_j of A, the sum of its entries is 1.

The transition matrix can be used to model the evolution over time of the distribution of a given number of elements taking at each time one of the possible states. Suppose that we start originally with x_1 elements in the state s_1, x_2 elements in the state s_2, etc. We encode this by a vector $\vec{x}_0 = (x_1, \ldots, x_n)^T$, whose coordinates record the initial distribution of objects under study. At each iteration of the process, each object moves from one state to another with following the probabilities encoded in the Markov diagram (or equivalently, in the transition matrix). At a given time, we have the following:

Theorem 7.8. Let $\vec{x}_0 = (x_1, \ldots, x_n)^T$ be the vector whose coordinates record the initial distribution of elements in states $s_1, \ldots, s_n$. Let us denote by $\vec{x}_k$ the vector whose components record the expected distribution after k iterations of the process of elements in states $s_1, \ldots, s_n$ respectively. We have for every $k \geq 0$:

$$\vec{x}_{k+1} = A\vec{x}_k \text{ for every } k \geq 0.$$

In particular, we get by induction:

$$\vec{x}_k = A^k \vec{x}_0.$$

Definition 7.9. A vector $\vec{x}$ such that $A\vec{x} = \vec{x}$ is called a *stationary distribution*. In other words, it is an eigenvector of A for the eigenvalue 1.

A stationary distribution represents the distribution over the possible states when the system is at equilibrium, i.e. does not change over time.

Example 7.10. In the Markov chain modelling the distribution of bikes mentioned in the introduction, the vector $(70, 40, 50, 40)^T$ is a stationary distribution.

7.3 The Perron-Frobenius Theorem for stochastic matrices

Definition 7.11. Let A be an $n \times n$ matrix.

- We say that A is *non-negative* if $a_{ij} \geq 0$ for every i, j, and *positive* if $a_{i,j} > 0$ for every i, j.
- We say that a non-negative matrix A is *regular* if A^k is positive for some integer $k \geq 1$.

The transition matrix associated to a Markov diagram is non-negative, but not necessarily positive as there may be a probability 0 of going from some state s_i to some other state s_j.

Remark 7.12. The integer k in the definition above depends on the matrix. In particular, for every k it is possible to construct a regular matrix A such that A^k is positive but $A, A^2, \ldots, A^{k-1}$ are not positive.

Proposition 7.13. The transition matrix associated to a Markov diagram is regular if and only if there exists an integer $n \geq 1$ such that one can go from any vertex of its Markov diagram to any other by a sequence of exactly n oriented edges.

Proof. The (i, j) term of A^k is given by:

$$(A^k)_{i,j} = \sum_{1 \leq i_1, \ldots, i_{k-1} \leq n} a_{i,i_1} a_{i_1,i_2} \cdots a_{i_{k-2},i_{k-1}} a_{i_{k-1},j}.$$

Since all the coefficients $a_{i,j}$ are non-negative, A^k is positive if and only if for every i, j, there exists $1 \leq i_1, \ldots, i_{k-1} \leq n$ such that $a_{i,i_1} a_{i_1,i_2} \cdots a_{i_{k-2},i_{k-1}} a_{i_{k-1},j}$ are all non-zero. Since a directed edge in the Markov diagram corresponds to a positive probability to move from one state to the other, this is equivalent to saying that for every i, j, there exists a sequence of n directed edges from the state i to the state j. □

Example 7.14. The following matrix is a regular stochastic matrix:

$$A := \begin{pmatrix} 1/2 & 1/2 \\ 1/2 & 1/2 \end{pmatrix}.$$

Indeed, it is clearly stochastic. Moreover, since $A = A^1$ is positive, it is also regular.

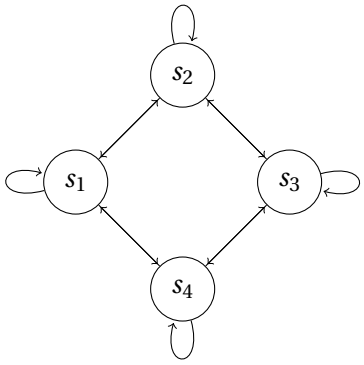
Example 7.15. The following matrix is a regular stochastic matrix:

$$A := \begin{pmatrix} 1/3 & 1/3 & 0 & 1/3 \\ 1/3 & 1/3 & 1/3 & 0 \\ 0 & 1/3 & 1/3 & 1/3 \\ 1/3 & 0 & 1/3 & 1/3 \end{pmatrix}.$$

Indeed, the fact that A is stochastic is clear, and we have:

$$A^2 = \begin{pmatrix} 1/3 & 2/9 & 2/9 & 2/9 \\ 2/9 & 1/3 & 2/9 & 2/9 \\ 2/9 & 2/9 & 1/3 & 2/9 \\ 2/9 & 2/9 & 2/9 & 1/3 \end{pmatrix},$$

which is positive. The corresponding Markov diagram is the following (where for simplicity we omit the probabilities, and we represent arrows from s_i to s_j and from s_j to s_i simply by a double arrow):



and we see that it is indeed possible to find an directed path of length 2 between any pair of states.

The key theorem of this chapter is the following:

Theorem 7.16 (Perron-Frobenius Theorem for stochastic matrices). Let A be a regular stochastic matrix. Then the following holds:

(i) There exists stationary distributions for A. In other words, 1 is an eigenvalue of A.

Moreover, there exists a stationary distribution all of whose components are strictly positive.

(ii) For every vector $\vec{x} = (x_1, \ldots, x_n)^T$ such that $\sum_i x_i \neq 0$, we have that the sequence of vectors $A^k \vec{x}$ converges to a stationary distribution of A.

(iii) $\dim E_1(A) = 1$.

(iv) For every other eigenvalue λ of A, we have $|\lambda| < 1$.

The proof of the Perron-Frobenius is rather long and technical (you can find the proof on the internet, or come and ask me after class/tutorials.). I will just present the proofs of two of the fours results, that illustrate nicely some of the concepts we have encountered in the course so far.

Proposition 7.17. We have For any square matrix, we have

$$\det(A^T) = \det(A).$$

Proposition 7.18. A square matrix and its transpose have the same eigenvalues.

Proof. Indeed, since $\det(A) = \det(A^T)$ for any square matrix by the above proposition 7.17, it follows that

$$\chi_{A^T}(\lambda) = \det(A^T - \lambda I_n) = \det((A - \lambda I_n)^T) = \det(A - \lambda I_n) = \chi_A(\lambda).$$

In particular,

$$\lambda \text{ is an eigenvalue of } A \iff \chi_A(\lambda) = 0 \iff \chi_{A^T}(\lambda) = 0 \iff \lambda \text{ is an eigenvalue of } A^T.$$

□

Proof of Theorem 7.16. (i) Notice that, since A is stochastic, the entries in each column of A add up to 1. This can be reformulated as follows:

$$A^T \begin{pmatrix} 1 \\ \vdots \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ \vdots \\ 1 \end{pmatrix}.$$

In particular, we see that 1 is an eigenvalue of A^T. But A and A^T have the same eigenvalues by the above theorem. Since 1 is an eigenvalue of A^T, it is also an eigenvalue of A.
(ii) Let us prove this statement in the case where A is diagonalisable. We choose a basis $\vec{e}_1, \ldots, \vec{e}_n$ of eigenvectors for the eigenvalues $1 = \lambda_1, \lambda_2, \ldots, \lambda_n$ respectively. Let $\vec{x} \in \mathbb{R}^n$ and let us write $\vec{x} = \alpha_1\vec{e}_1 + \ldots + \alpha_n\vec{e}_n$, for some $\alpha_1, \ldots, \alpha_n \in \mathbb{R}$. We then have for $k \geq 1$:

$$\begin{aligned} A^k\vec{x} &= \alpha_1 A^k\vec{e}_1 + \alpha_2 A^k\vec{e}_2 + \ldots + \alpha_n A^k\vec{e}_n \\ &= \alpha_1\vec{e}_1 + \alpha_2\lambda_2^k\vec{e}_2 + \ldots + \alpha_n\lambda_n^k\vec{e}_n. \end{aligned}$$

Since $|\lambda_2|, \ldots, |\lambda_n| < 1$ by (iv), it follows that $A^k\vec{x}$ converges to $\alpha_1\vec{e}_1$. To conclude that the limit is an eigenvector (for the eigenvalue $\lambda_1 = 1$), it remains to show that $\alpha_1\vec{e}_1 \neq \vec{0}$. (an eigenvector cannot be the null vector.) But since A is stochastic, the sum of the components of $A^k\vec{x}$ remains the same at every stage. Since we started from a vector $\vec{x}$ such that $\sum_i x_i \neq 0$, it follows that the sum of the components of $\alpha_1\vec{e}_1$ is non-zero, and in particular $\alpha_1\vec{e}_1 \neq \vec{0}$. □

7.4 Application: stationary distributions in dynamical systems.

Distribution of bikes. Let us go back to the problem stated in the introduction. The problem of modelling the changes in the distribution of bicycles can be modelled using a Markov diagram, whose transition matrix A was given in the introduction. This associated transition matrix is regular, as it is positive. Thus, Theorem 7.16 applies. Let us denote by $\vec{x}_0 = (50, 50, 50, 50)^T$ the vector corresponding to the initial distribution of bikes. After k days, the expected distribution of bikes is

given by $A^k\vec{x}_0$. By (iv), this sequence of vectors converges towards a stationary distribution $\vec{x}$. This stationary distribution is a vector whose components add up to 200, since each vector $A^k\vec{x}_0$ satisfies the same property (intuitively, the bikes are just redistributed, there is no gain nor loss of bike in the process). By (ii), since the eigenspace $E_1(A)$ is a line (i.e. has dimension 1), there is a unique vector of $E_1(A)$ whose components add up to 200, and this is exactly our stationary distribution $\vec{x}$. Note that this characterisatio of $\vec{x}$ does not depend on the original distribution $\vec{x}_0$. This explains in hindsight why the stationary distribution is unique, and does not depend on the original vector $\vec{x}_0$.

Speed of convergence. One natural question would be to understand the speed of convergence. When the transition A matrix is diagonalisable, then there exists a basis $\vec{e}_1,\dots,\vec{e}_n$ of $\mathbb{R}^n$ for the associated eigenvalues $1=\lambda_1>\lambda_2\geq\dots\geq\lambda_n>-1$. In particular, if we write our initial vector $\vec{x}$ as a linear combination of the $\vec{e}_i$, as $\vec{x}=\alpha_1\vec{e}_1+\dots+\alpha_n\vec{e}_n$, we have for $k\geq 1$:

$$A^k\vec{x}=\alpha_1\vec{v}+\alpha_2\lambda_2^k\vec{e}_2+\dots+\alpha_n\lambda_n^k\vec{e}_n.$$

In particular, we see that $A^k\vec{x}\to_\infty \alpha_1\vec{v}$ and that $||A^k\vec{x}-\alpha_1\vec{e}_1||$ converges to zero exponentially, and as fast as μ^k (up to a multiplicative constant), where

$$\mu=\max_{2\leq i\leq n}|\lambda_i|.$$

Rephrased using Landau's notations, we have:

$$||A^k\vec{x}-\vec{v}||=O(\mu^k)$$

If A is not diagonalisable, a slightly weaker inequality holds, namely:

$$||A^k\vec{x}-\vec{v}||=o(\mu^k)$$

for every $\max\limits_{2\leq i\leq n}|\lambda_i|<\mu<1$. We will not prove this stronger statement in this course.

7.5 Application: The PageRank algorithm

One can imagine the internet as a (very large) oriented graph. Vertices correspond to pages and oriented edges correspond to links from a page to another.
To this oriented graph, we can associate a finite Markov chain whose states correspond to the pages, by assuming that a person on a given webpage will click with equal probability on one of the links of the page. To construct the associated transition matrix, we denote by n_{ij} the number of links from page j to page i, and by d_j the total number of links on page j, then the probability a_{ij} that someone on page j clicks on a link leading to page i is:

$$a_{ij}=\frac{n_{ij}}{d_j}.$$

Let us consider a small example of a very small network consisting of four webpages $p_1,\dots,p_4$. The table representing the links between pages is given below, where the coefficient in row i and column j represents the number of links from page j to page i:

	p_1	p_2	p_3	p_4
p_1	0	0	2	2
p_2	2	0	2	1
p_3	1	1	0	2
p_4	1	1	1	0

The transition matrix associated to this Markov chain is:

$$A = \begin{pmatrix} 0 & 0 & 2/5 & 2/5 \\ 1/2 & 0 & 2/5 & 1/5 \\ 1/4 & 1/2 & 0 & 2/5 \\ 1/4 & 1/2 & 1/5 & 0 \end{pmatrix}.$$

However, this model fails to take into account that quite often people will go to a webpage not because it was linked from a previous page, but because they started thinking of something completely different, independently of the content of the current page. This can be modelled by another Markov process where at each time, the probability of going from any page to any other page is exactly $1/n$, where n is the number of pages (in our example, $n = 4$). This finite Markov chain has a transition matrix a $n \times n$ matrix of the form

$$U = \begin{pmatrix} 1/n & \cdots & 1/n \\ \vdots & \ddots & \vdots \\ 1/n & \cdots & 1/n \end{pmatrix}.$$

What Google does to take both processes into consideration is to consider that what is going on is a combination of the previous cases. There exists a constant α, called *damping factor*, such that with probability α, a person will choose the next page by clicking on one of the links, and with probability $(1-\alpha)$ that person will choose the next page completely at random. A standard value for α is $\alpha = 0.85$. This corresponds to a Markov process whose transition matrix is given by the following barycenter of A and U:

$$G := \alpha A + (1-\alpha)U.$$

Note that this transition matrix is now strictly positive, and in particular regular. We can thus apply the Perron-Frobenius Theorem, which tells us that there exists a stationary distribution with positive components and with sum 1, called the *ranking vector*. By the convergence property, this corresponds to the expected distribution of people on pages when this process has been repeated infinitely many times. Pages with a comparatively high distribution correspond to 'popular' pages, and are thus ranked highly by Google. This ranking vector is (a simplified version of) what Google uses to rank pages.

Definition 7.19. A *ranking vector* used in the Google PageRank algorithm is a stationary distribution of the 'Google matrix' $G = \alpha A + (1-\alpha)U$ with positive components. (Such a vector is unique up to multiplication by a positive constant, by the Perron-Frobenius Theorem)
The ranking of the webpages is obtained by ranking the components of that vector.

Let us go back to our simple example. Taking the damping factor to be 0.85, the new transition matrix for the process is

$$\begin{aligned} G &= 0.85 \begin{pmatrix} 0 & 0 & 2/5 & 2/5 \\ 1/2 & 0 & 2/5 & 1/5 \\ 1/4 & 1/2 & 0 & 2/5 \\ 1/4 & 1/2 & 1/5 & 0 \end{pmatrix} + 0.15 \begin{pmatrix} 1/4 & 1/4 & 1/4 & 1/4 \\ 1/4 & 1/4 & 1/4 & 1/4 \\ 1/4 & 1/4 & 1/4 & 1/4 \\ 1/4 & 1/4 & 1/4 & 1/4 \end{pmatrix} \\ &= \begin{pmatrix} 0.0375 & 0.0375 & 0.3775 & 0.3775 \\ 0.4625 & 0.0375 & 0.3775 & 0.2075 \\ 0.25 & 0.4625 & 0.0375 & 0.3775 \\ 0.25 & 0.4625 & 0.2075 & 0.0375 \end{pmatrix}. \end{aligned}$$

Finding the stationary distribution corresponds to solving the system $G\vec{x} = \vec{x}$, which we do as usual using Gaussian elimination, with a computer as the computations are too cumbersome. The unique stationary distribution $\vec{r}$ whose components are positive and add up to 1 is (with approximations to the second digit)

$$\vec{r} \simeq \begin{pmatrix} 0.21 \\ 0.26 \\ 0.28 \\ 0.24 \end{pmatrix}.$$

So the ranking of these pages, from most popular to least popular, is: page 3, page 2, page 4, page 1.

Speed of convergence (extra-curricular). We know from the Perron-Frobenius Theorem that a way to obtain a ranking vector is to look at the asymptotic behaviour of $G^k\vec{x}$ for an arbitrary vector $\vec{x}$ with $\sum x_i = 1$. We wish to find an upper bound on the norm $||G^k\vec{x} - \vec{r}||$. However,instead of looking at this norm, we will be looking at a slightly different one that is easier to use in this situation:

$$||\vec{x}||_1 = \sum_i |x_i|.$$

Note that we still have the triangle identity $||\vec{x} + \vec{y}||_1 \leq ||\vec{x}||_1 + ||\vec{y}||_1$. Moreover, for every vector $\vec{x} = (x_1, \ldots, x_n)^T$, we have:

$$||A\vec{x}||_1 = \sum_i |\sum_j a_{ij}x_j| \leq \sum_i \sum_j a_{ij}|x_j| \leq \sum_j |x_j|(\sum_i a_{ij}) \leq \sum_i |x_i| = ||\vec{x}||_1.$$

Theorem 7.20. For every $k \geq 1$ and every positive vector $\vec{x}$ whose components add up to 1, we have:

$$||G^k\vec{x} - \vec{r}||_1 \leq 2\alpha^k.$$

Proof. We have:

$$G\vec{x} = \alpha A\vec{x} + (1-\alpha)U\vec{x} = \alpha A\vec{x} + (1-\alpha)\begin{pmatrix} 1/n \\ \vdots \\ 1/n \end{pmatrix}.$$

In particular, for non-negative vectors $\vec{x}, \vec{y}$, we get

$$G\vec{x} - G\vec{y} = \alpha A(\vec{x} - \vec{y}),$$

and

$$||G\vec{x} - G\vec{y}||_1 = \alpha||A(\vec{x} - \vec{y})||_1 \leq \alpha||\vec{x} - \vec{y}||_1.$$

By induction, we get

$$||G^k\vec{x} - G^k\vec{y}||_1 \leq \alpha^k||\vec{x} - \vec{y}||_1$$

and thus, for every positive vector $\vec{x}$ whose components add up to 1, we get:

$$||G^k\vec{x} - \vec{r}||_1 = ||G^k\vec{x} - G^k\vec{r}||_1 \leq \alpha^k||\vec{x} - \vec{r}||_1 \leq \alpha^k(||\vec{x}||_1 + ||\vec{r}||_1) \leq 2\alpha^k.$$

□

Note in particular that this speed of convergence is independent on the size of the matrix. For instance, for $\alpha = 0.85$, then the power method gives an approximation of the ranking vector with error less than 10^{-6} after 90 iterations.